

Notes on Master Studies

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João, é bom ver que você aprendeu e fez tantas coisas bonitas! Continue assim, muito feliz! Brinque e aprenda muito.

Te amo muito, *Papai*

Chapter 1

Introduction

Random Matrices can be thought of in two ways: As a measurable function from a probability space into a set of matrices or, more concretely, as a matrix whose entries are random variables with some joint distribution. In some cases there is an explicit density formula for the eigenvalues of such a random matrix. When there is none for finite size matrices, one can try, with the help of some mathematical tools such as orthogonal polynomials, express the equilibrium measures for the eigenvalues asymptotically. There are many instances of “models” of random matrices, each one with its symmetries and own probability density. Those are what we call *ensembles*.

Borrowing some concepts from physics, an ensemble can be thought of as a virtual collection of all possible states that a system can be in, with more probable states appearing more often in the collection. There is an analogy to be made with ensembles in a thermodynamic setting: When considering sets of matrices with equal probability - in some sense to be more clear soon - we get the Gibbs weight for the probability distribution as we would in the physical *Canonical Ensemble*, thought of as a system of particles with constant temperature. More concretely, in the context of Random Matrix Theory, we have the definition.

Definition 1.0.1. A random matrix ensemble $\mathcal{E} = (\mathcal{S}, \mathcal{P})$ is a set of matrices $\mathcal{S}$ of fixed size equipped with a probability density function (p.d.f.) $\mathcal{P} : \mathcal{S} \rightarrow [0, \infty)$. If we say that a matrix $\hat{X}$ is drawn from the random ensemble $\mathcal{E}$, we are saying that it is a random variable assuming values in $\mathcal{S}$ with p.d.f. $\mathcal{P}$.

A commonplace thing to do when defining ensembles is to prescribe the p.d.f.'s $\mathcal{P}_{i,j}$ on the independent entries $X_{i,j}$ of a representative matrix $\hat{X} \in \mathcal{E}$. In such cases, we say that

$$\mathcal{P}(\hat{X}) = \prod_{i,j | X_{i,j} \text{ independent}} \mathcal{P}_{i,j}(X_{i,j}).$$

With this definition we can already define many ensembles of interest. For many, the set of matrices is described by applying constraints to the set of real matrices $\mathbb{M}_{m,n}(\mathbb{R})$, the set of complex matrices $\mathbb{M}_{m,n}(\mathbb{C})$ or the set of quaternionic matrices $\mathbb{M}_{m,n}(\text{He})$. Then, oftentimes, $\mathcal{S}$ is simply the set of symmetric, hermitian or skew-symmetric matrices. When it comes to determining the p.d.f. $\mathcal{P}$ on $\mathcal{S}$, there are many options that have been worked on.

The definition of ensembles is of great use when modeling other phenomena with the use of Random Matrix. Each ensemble has some symmetries and scaling properties that can better describe a chosen system. We still need to understand what it means to model such a system. This idea is illustrated in the following citation by Deif.

Citation 1.0.2. [3] “This procedure can be viewed as follows: A scientist wishes to investigate some statistical phenomenon. What he has at hand is a microscope and a handbook of matrix ensembles. The data $\{a_k\}$ are embedded on a slide which can be inserted into the microscope. The only freedom that the scientist has is to center the slide, $a_k \rightarrow a_k - A$, and then adjust the focus $a_k - A \rightarrow \tilde{a}_k = \gamma_a(a_k - A)$ so that on average one data point $\tilde{a}_k$ appears per unit length on the slide. At that point the scientist takes out his handbook, and then tries to match the statistics of the $\tilde{a}_k$'s with those of the eigenvalues of some ensemble. If the fit is good, the scientist then says that the system is well-modeled by RMT.”

Perhaps the first instance of the modeling by random matrices was in the works of Wigner on the scattering of heavy nuclei. Physicists have been doing scattering for very long: Project a small (interactive or not) particle in the direction of a nuclei of interest and change the energy of the incoming particle. You should observe what physicist

call a cross-section (an classical equivalent radius of the deflecting object). If one plots this phenomena, one obtains a graph of hundreds of peaks $E - 1 < E_2 < \dots$ and valleys $E'_1 < E'_2 < \dots$. If $E \approx E_j$ for some j , the projected particle must have been repelled from the nucleus, and if $E \approx E'_j$ for some j , then the particle goes through the nucleus, essentially unchanged. The E_j 's are the scattering resonances. If one is interested, and many physicists are, on the typical distances of theses scattering energies, one could of course write down a Schrodinger-type equation for the scattering system. This has, however, very high dimensionality and therefore offers no easy way of treatment for a solution for the E_j 's - either analytically or numerically. However, as more experiments were done on heavy nuclei, a consensus emerged that the theory of resonances may as well be statistical, and here Wigner led the way.

Citation 1.0.3. (*Eugene Wigner - On the Wigner Surmise (1956)*) “Perhaps I am now too courageous when I try to guess the distribution of the distances between successive levels (of energies of heavy nuclei). Theoretically, the situation is quite simple if one attacks the problem in a simpleminded fashion. The question is simply what are the distances of the characteristic values of a symmetric matrix with random coefficients.”

This description gave way for a whole lot of development, as Wigner also proposed that the curve must hold some universality that only depends on a few of the main properties of the nuclei - namely its symmetries. This modeling was repeated for many other highly correlated processes and was very successful as a general theory of random correlated systems. This can now be observed in the plenitude of posterior system modeled with such statistics, for example: Ulam's Problem [21], Zeros of the Riemann Zeta Function [5], Matching Random Sub-sequences [11], Tiling problems (Aztec Diamond for instance) and Interacting Random Walks, to name a few.

In this development, another point of interest became clear: There was a universal law in the edge of the distribution of the eigenvalues - and, more shockingly, it appears to insert itself in many of the other random correlated phenomena. Already at Wigner's Matrices one could note an instance of these distributions: The Tracy-Widom [19]. At some point, Majumdar and Schehr have shown for large N , the Tracy-Widom distribution describes the crossover function between stable and unstable regimes of a random dynamical system, a specific phase transition, which also occurs in the Coulomb gas, two-dimensional QCD [9], and several other systems of mathematical and physical interest. This distribution appears to be somehow related to the universality of phase transitions - such as the physical transition of water or superconductivity. This made way for a whole theory of universality and led the theory to the study of edge scaling.

On the scale of the eigenvalue spacing the fluctuation of individual eigenvalues becomes relevant. It has first been conjectured by Wigner, and subsequently formalized as the Wigner-Dyson-Mehta (WDM) conjecture, that the local statistics of eigenvalues is universal in the sense that their distribution is independent of any model specifics. The emerging distributions should be viewed as the random matrix analogue of the central limit theorem and the normal distribution. They depend only on the symmetry class of the matrix and the local singularity type of the density, but on no other model specifics. The classification indicates that for symmetric matrices, up to scaling by a constant, the density around single eigenvalues can only exhibit three different behaviors. Within the spectral bulk the density ρ is positive and real analytic and therefore, on the scale $1/N$ of individual eigenvalues, is essentially constant. Around edges and cusps the density is simply given by $x^{1/2}$ or $x^{1/3}$, indicating that the typical distance between consecutive eigenvalues is $N^{-2/3}$ and $N^{-3/4}$. The WDM conjecture roughly states that eigenvalue fluctuations depend only on the local behavior of the density, and not on far away effects or any other characteristics of the ensemble.

Conjecture 1.0.4. (*WDM conjecture for hermitian symmetry class [16]*) Assume that b , e and c are bulk, edge and cusp points of some density $\mathbf{p}$ with parameters γ_e , γ_c defined in such a way that

$$\rho(e \pm x) = \gamma_e^{3/2} x^{1/2} / \pi + \mathcal{O}(x^{1/2}), \quad \rho(c + x) = \sqrt{3} \gamma_c^{4/3} |x|^{1/3} / 2\pi + \mathcal{O}(|x|^{1/3}).$$

Then the universal correlation functions are given by

$$\begin{aligned} \frac{1}{\rho(b)^k} \mathbf{p}_k^N(b + \frac{x}{\rho(b)N}) &\approx \det \left(\frac{\sin \pi(x_i - x_j)}{\pi(x_i - x_j)} \right)_{i,j \in [k]}, & (\text{Bulk}) \\ \frac{N^{k/3}}{\gamma_e^k} \mathbf{p}_k^N(e + \frac{x}{\gamma_e N^{2/3}}) &\approx \det (K_{\text{Airy}}(x_i, x_j))_{i,j \in [k]}, & (\text{Edge}) \\ \frac{N^{k/4}}{\gamma_c^k} \mathbf{p}_k^N(c + \frac{x}{\gamma_c N^{3/4}}) &\approx \det (K_{\text{Pearcey}}(x_i, x_j))_{i,j \in [k]}, & (\text{Cusp}) \end{aligned}$$

The Airy functions (related to the Airy Kernel and Tracy-Widom function) themselves are given by integrals in which the exponents have a cubic singularity, arising from the coalescence of two saddle points in an asymptotic

analysis. However, in the scope of this work, we will be concerned with the *Pearcey Processes*. Pearcey functions are given by integrals in which the exponents have a quartic singularity, arising from the coalescence of three saddle points - these are the functions which give rise to the Pearcey Processes. The Pearcey process describes the local eigenvalue statistics of large random matrices near the points of the spectrum where the limiting mean eigenvalue density vanishes as a cubic root. This has been established rigorously for random matrix ensembles with an external source. More precisely, let $\hat{H}$ be an $n \times n$ GUE matrix (with n even), suitably scaled, and $\hat{H}_0$ a fixed Hermitian matrix with eigenvalues $\pm a$ each of multiplicity $n/2$. Let $n \rightarrow \infty$. If a is small the density of eigenvalues is supported in the limit on a single interval. If a is large then it is supported on two intervals. At the “closing of the gap” the limiting eigenvalue distribution is described by the Pearcey kernel. These models of matrices have p.d.f. given by functions

$$\mathcal{P}(M)dM = \frac{1}{Z_{N,\beta}} e^{-N(\text{Tr } V(M)+AM)} dM, \quad M \in \mathbb{S} \quad \text{and} \quad A \in \Lambda,$$

where $Z_{N,\beta}$ is a partition function, such that $\mathcal{P}$ is a probability density of the matrices. The potential $V : \mathbb{R} \rightarrow \mathbb{R}$ is a suitably regular function, acting as a confining potential on the eigenvalues $\lambda_1, \dots, \lambda_N$. The set Λ is the set of diagonal real matrices (generally hermitian, but simply diagonal) and $\mathbb{S}$ is such as the real, complex, or quaternionic sets. Especially, we can describe the model by its partition function

$$Z_{N,\beta} = \int e^{-N(\text{Tr } V(M)+AM)} dM.$$

Moreover, we say that $A = \text{diag}(a_1, \dots, a_p)$ and that each a_i has multiplicity n_p . Note that $\sum n_p = n$. This type of model has many reasons to exist and possible interpretation. One could think of a system with impurities where the Hamiltonian looked like $\mathcal{H} = \mathcal{H}_0 + V$ where V is a potential for the impurity in the system. With that, one can create critical points of some new type and study its properties by the asymptotic of its partition function.

Chapter 2

Universality

It is well known among physicists and mathematicians alike that random independent phenomena share a common characteristics when we are dealing with large quantities. This is formally expressed in terms of the central limit theorem: Take $\{x_n\}_{n \geq 1}$ i.i.d, then, if we center and scale the variables $x_n \rightarrow y_n := (x_n - \mathbb{E}(x_n))/\sqrt{\text{Var}(x_n)}$, then we have the limit

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\frac{\sum_{k=1}^n y_k}{\sqrt{n}} \leq t \right) = \int_{-\infty}^t e^{-\frac{u^2}{2}} \frac{du}{\sqrt{2\pi}}.$$

This is an amazing result as it appoints the Gaussian distribution as a universal distribution for great numbers of uncorrelated random processes. There is however, many phenomena where these assumptions can't hold. This is very natural for the many interacting physical systems or mathematical models such as roots of polynomials, where the points 'show repulsion'. For these cases another Universality result must be considered, one where the Gaussian distribution is substituted with a more appropriate distribution: The Tracy-Widow,

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\frac{\sum_{k=1}^n \left(\lambda_k - z_n^{(\beta)} \right)}{s_n^{(\beta)}} \leq t \right) = \mathbb{F}_\beta(t)$$

where $s_n^{(\beta)}$ has something to do with the class of the problem - gives the scale needed, and $F_\beta(t)$ is the cumulative distribution given by the (β) -Tracy-Widow

An amazing source for a brief introduction - and the main source used for a number of the examples - is the great overview in [3] by Deift, 2006. For the meaning of what is going to be the main subset of this chapter - modeling by a random matrix model - we cite an analogy given by this same work.

Citation 2.0.1. (Deift) *This procedure can be viewed as follows: A scientist wishes to investigate some statistical phenomenon. What s/he has at hand is a microscope and a handbook of matrix ensembles. The data $\{a_k\}$ are embedded on a slide which can be inserted into the microscope. The only freedom that the scientist has is to center the slide, $a_k \rightarrow a_k - A$, and then adjust the focus $a_k - A \rightarrow \tilde{a}_k = \gamma_a(a_k - A)$ so that on average one data point $\tilde{a}_k$ appears per unit length on the slide. At that point the scientist takes out his/her handbook, and then tries to match the statistics of the $\tilde{a}_k$'s with those of the eigenvalues of some ensemble. If the fit is good, the scientist then says that the system is well-modeled by RMT.*

2.1 Heavy Atom Nuclei

This section is based on [3].

Consider the scattering of neutrons of a heavy nucleus, say uranium U^{238} . The scattering cross-section is plotted as a function of the energy E of the incoming neutrons, and one obtains a jagged graph with many hundreds of sharp peaks $E - 1 < E_2 < \dots$ and valleys $E'_1 < E'_2 < \dots$. If $E \approx E_j$ for some j , the neutron is strongly repelled from the nucleus, and if $E \approx E'_j$ for some j , then the neutron sails through the nucleus, essentially unimpeded. The E_j 's are called scattering resonances. The challenge faced by physicists in the late 40's and early 50's was to develop an effective model to describe these resonances. One could of course write down a Schrodinger-type equation for the scattering system, but because of the high dimensionality of the problem there is clearly no hope of solving the equation for the E_j 's either analytically or numerically. However, as more experiments were done on heavy nuclei, each with hundreds of E_j 's, a consensus began to emerge that the "correct" theory of resonances was statistical,

and here Wigner led the way. Any effective theory would have to incorporate two essential features present in the data, i.e.

1. Modulo certain natural symmetry considerations, all nuclei in the same symmetry class exhibited universal behavior;
2. In all cases, the E_j 's exhibited repulsion, or, more precisely, the probability that two E_j 's would be close together was small.

Citation 2.1.1. (*Eugene Wigner - On the Wigner Surmise (1956)*) “Perhaps I am now too courageous when I try to guess the distribution of the distances between successive levels (of energies of heavy nuclei). Theoretically, the situation is quite simple if one attacks the problem in a simpleminded fashion. The question is simply what are the distances of the characteristic values of a symmetric matrix with random coefficients.”

At some point in the mid-1950's, in a striking development, Wigner suggested that the statistics of the neutron scattering resonances was governed by GOE (This is for time-reversal-invariant data, if this was not the case, this should be governed by the GUE). If one equips the space of real, symmetric matrices with a probability measure with a smooth density, the probability of a matrix $\hat{M}$ having equal eigenvalues would be zero and the eigenvalues $\lambda_1, \dots, \lambda_n$ of $\hat{M}$ would comprise a random set with repulsion built in. So Wigner had a model, or more precisely, a class of models, which satisfied constraint (ii). But why choose GOE? This is where the universality constraint (i) comes into play. We quote from: Wigner) “Let me say only one more word. It is very likely that the curve in Figure 1 is a universal function. In other words, it doesn't depend on the details of the model with which you are working. There is one particular model in which the probability of the energy levels can be written down exactly. I mentioned this distribution already in Gatlinburg. It is called the Wishart distribution. Consider a set... .” So in this way Wigner introduced GOE into theoretical physics: It provided a model with repulsion (and time-reversal) built in. Furthermore, the energy level distribution could be computed explicitly. By universality, it should do the trick!

2.2 Random Walkers

In his investigation of wetting and melting phenomena, Fisher introduced various “vicious” walker models. Here we will consider the so-called random turns vicious walker model. In this model, the walks take place on the integer lattice $\mathbb{Z}$ and initially the walkers are located at $0, 1, 2, \dots$. The rules for a walk are as follows:

1. at each tick of the clock, precisely one walker makes a step to the left
2. no two walkers can occupy the same site (hence “vicious walkers”)

For example, consider the following walk from time $t = 0$ to time $t = 6$:

```

      · × · · × × × · ×
      · · × · × × × · ×
      · · × · × × · × ×
      · · × · × · × × ×
      · · · × × · × × ×
      · · · × · × × × ×
      · · · · × × × × ×

```

At $t = 0$, clearly only the walker at 0 can move. At time $t = 1$, either the walker at -1 or at $+1$ can move, and so on. Let dN be the distance traveled by the walker starting from 0. In the above example, $d_6 = 3$. For any time N , there are clearly only a finite number of possible walks of duration $t = N$. Suppose that all such walks are equally likely.

It has been proved that, as $N \rightarrow \infty$, dN , the distance traveled by the walker starting from 0, behaves statistically like the largest eigenvalue of a GOE matrix. More precisely, they showed that

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\frac{dN - 2\sqrt{N}}{N^{\frac{1}{6}}} \leq t \right) = \mathbb{F}_1(t)$$

where $\mathbb{F}_1$ is the (1)-Tracy-Widow. In a variant of this model, [For3], the walkers again start at $0, 1, 2, \dots$, and move to the left for a time N ; thereafter they must move to the right, returning to their initial positions $0, 1, 2, \dots$ at

time $2N$. Let $d'N$ denote the maximum excursion of the walker starting from 0. Then Forrester shows that $d'N$ behaves statistically like the largest eigenvalue of a GUE matrix,

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\frac{dN - 2\sqrt{N}}{N^{\frac{1}{6}}} \leq t \right) = \mathbb{F}_2(t)$$

where $\mathbb{F}_2$ is the (2)-Tracy-Widow.

2.3 Aztec Diamonds

Consider tilings $\{T\}$ of the tilted square $T_n = \{(x, y) : |x| + |y| \leq n+1\}$ in $\mathbb{R}^2$ by horizontal and vertical dominos of length 2 and width 1. For each tiling the dominos must lie strictly within T_n . The tilings T are called Aztec diamonds because the boundary of T in $\{(x, y) : y > 0\}$, say, has the shape of a Mexican pyramid. It is a nontrivial theorem that for any n , the number of domino tilings of T_n is $2^{\binom{n}{2}}$. Assume that all tilings are equally likely. The first thing one might notice when plotting tilings is that the most common tilings have a quirky behaviour: They freeze the borders and stay random at a circle in the center. This is formally expressed in [15],

Theorem 2.3.1. *(The Arctic Circle Theorem) Fix $\epsilon > 0$. For each n , consider a uniformly random domino tiling of the Aztec Diamond of size n scaled by the factor $1/n$ in each axis to fit into the limiting diamond*

$$AD_\infty := \{|x| + |y| \leq 1\}$$

and let $P_n^0 \subset n^{-1}AD_n$ be the image of the polar regions of the random tiling under this scaling transformation. Then as $n \rightarrow \infty$ the event that

$$\{(x, y) \in AD_\infty : x^2 + y^2 > \frac{1}{2} + \epsilon\} \cap (n^{-1}AD_n) \subset P_n^0 \subset \{(x, y) \in AD_\infty : x^2 + y^2 > \frac{1}{2} - \epsilon\}$$

holds with probability that tends to 1.

The interesting result is in how much the border of the random region - or temperate, as some authors call it - deviates from the Arctic Circle as N grows large. For $-1 < \alpha < 1$ and $\alpha \neq 0$, let

$$(x_\alpha^+, y_\alpha^+) = \left(\frac{\alpha + \sqrt{1 - \alpha^2}}{2}, \frac{\alpha + \sqrt{1 + \alpha^2}}{2} \right) \quad \text{and} \quad (x_\alpha^-, y_\alpha^-) = \left(\frac{\alpha - \sqrt{1 - \alpha^2}}{2}, \frac{\alpha + \sqrt{1 + \alpha^2}}{2} \right)$$

denote the two points of intersection of the line $u + v = \alpha$ with $C_0 = \{u^2 + v^2 = \frac{1}{2}\}$. Then for fixed α , Johansson showed that the fluctuations of the boundary of the temperate zone along the line $u + v = \alpha$ about the points (x_α^+, y_α^+) and (x_α^-, y_α^-) were described by the Tracy-Widom distribution F_2 .

2.4 Ulam's Problem

This section is based on the Notes of the MasterClass given in Angers, France (2024) by Mattia Cafasso and Sofia Tarricone.

Let us consider the symmetric group S_N of the permutations of the finite set $\{1, 2, \dots, N\}$.

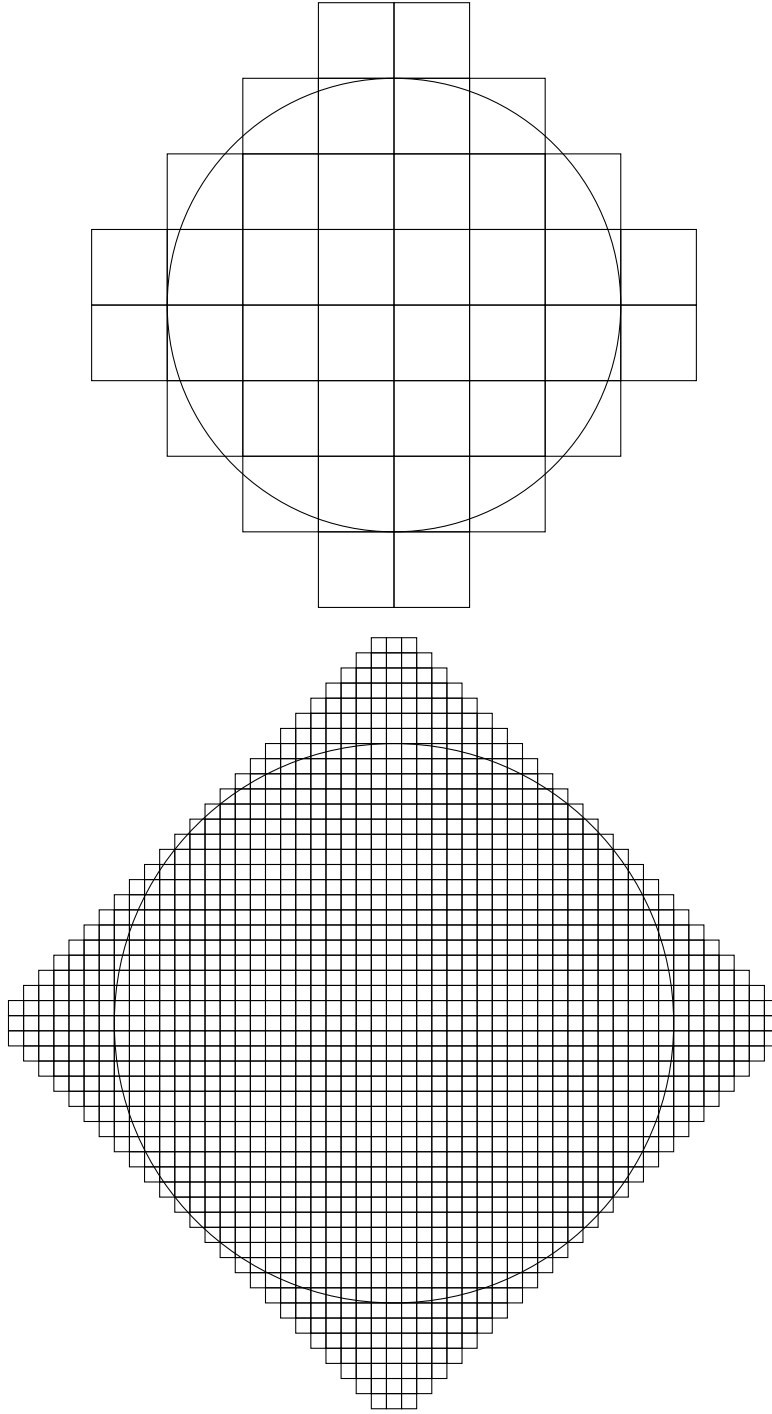
Definition 2.4.1. *Let $\pi \in S_n$ be a permutation. Given k integers*

$$1 \leq i_1 \leq i_2 \leq \dots \leq i_k \leq N$$

we say that $\pi(i_1), \dots, \pi(i_k)$ is an increasing sub-sequence of π of length k if π preserves the order of the integers i_j , $j = 1, \dots, k$, i.e.

$$\pi(i_1) < \pi(i_2) < \dots < \pi(i_k).$$

We denote $l_N(\pi)$ the maximal length of all the increasing sub-sequences of π .



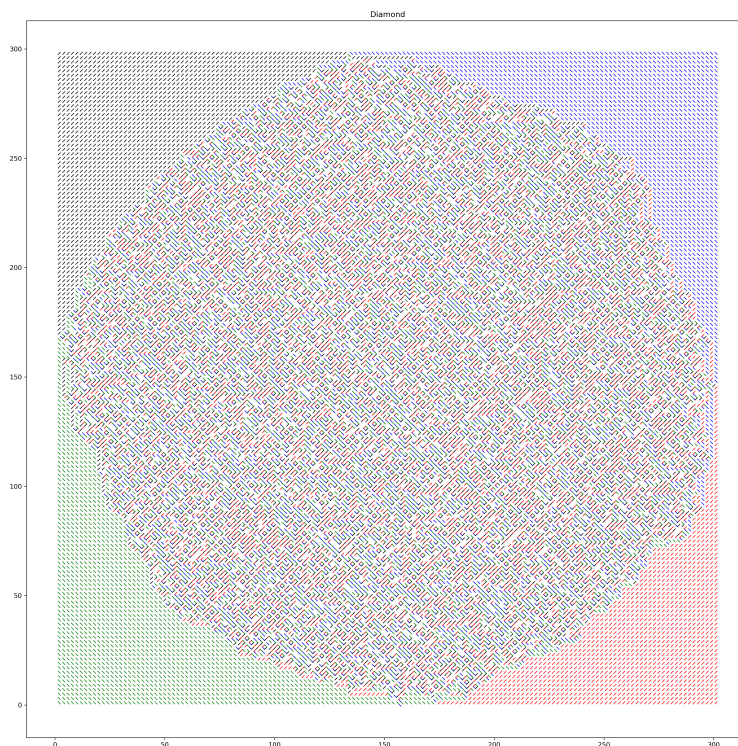
Example 2.4.2. Let $\pi = 54172386$. Then (by direct inspection), $l_N(\pi) = 4$. The maximal length sequences are $(1, 2, 3, 6)$ and $(1, 2, 3, 8)$.

Some history on the problem can be enunciated as

1. Stanislaw Ulam, in an article published in 1961, discussed the idea of using Monte Carlo approaches to study the asymptotic behavior of $\mathbb{E}[l_N]$, when $N \rightarrow \infty$. Based on his simulations, Ulam conjectured that it exists a constant c such that

$$\lim_{N \rightarrow \infty} \frac{\mathbb{E}[l_N]}{\sqrt{N}} = c.$$

The term Ulam's problem, originally, referred to the proof of this conjecture and the identification of the constant c .



2. Hammersley, in 1972, introduced a Poissonized version of this problem and proved that the constant c exists.
3. In 1977, Logan and Shepp and, independently, Vershik and Kherov showed that $c = 2$.
4. Odlyzko and Rains, in the 90's, performed analytical simulations indicating that

$$\lim_{N \rightarrow \infty} \frac{\mathbb{V}ar[l_N]}{N^{\frac{1}{3}}} = c_0 \equiv 0,819 \dots$$

and also that

$$\lim_{N \rightarrow \infty} \frac{\mathbb{E}[l_N] - 2\sqrt{N}}{N^{\frac{1}{6}}} = c_1 \equiv -1.748 \dots$$

Finally, Ulam's problem was solved by Baik, Deift and Johansson in 1999, who studied the convergence in distribution of the re-scaled variable

$$\chi_N := \frac{l_N - 2\sqrt{N}}{N^{\frac{1}{6}}}$$

given by

Theorem 2.4.3. (Baik-Deift-Johansson, 1999). Let χ be a random variable with distribution function $F_{GUE}(x)$. Then χ_N , defined as before, converges in distribution to χ :

$$\lim_{N \rightarrow \infty} \mathbb{P} \left\{ \frac{l_N - 2\sqrt{N}}{N^{\frac{1}{6}}} \leq x \right\} \rightarrow F_{(2)}(x)$$

we have, moreover, convergence of moments

$$\lim_{N \rightarrow \infty} \mathbb{E}[\chi_N^m] = \mathbb{E}[\chi^m], \quad \forall m \geq 1.$$

To study, there is a nice representation of the permuted sequences that is worth exploring.

2.4.1 Young's Diagram - and Tableaux

Consider the following process. Start with a deck of N well shuffled cards (with uniform measure) - numbered $1, 2, \dots, N$ for convenience. Now place the first card of the shuffled deck on the table. The second card will be placed on top of the first, if it is labeled with a smaller number, and on its side if it is labeled with a larger number. Now, if the third card is smaller than the two cards, it is again placed on top of both cards. If it is in-between it is placed on the side of the second card. If it is bigger than both, it is placed on the side of the first card. The fourth card follows the same process. This gives rise to what we call a *Young's Diagram*.

Definition 2.4.4. A partition $\lambda = (\lambda_1, \dots, \lambda_r)$ of N is a set of positive integers such that $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_r$ and $|\lambda| := \sum_i \lambda_i = N$. We call $|\lambda|$ the size of the partition, and r its length. We will sometimes write $\lambda \vdash N$ to indicate that λ has size N .

A convenient graphic representation of integer partitions is given by Young diagrams, which are constructing stacking r rows of boxes, each of length λ_i , in decreasing order. For instance, the partition $(5, 5, 3, 3, 2, 1, 1) \vdash 20$ corresponds to the Young diagram below: The set of Young diagrams of size N will be denoted YN , and the set of

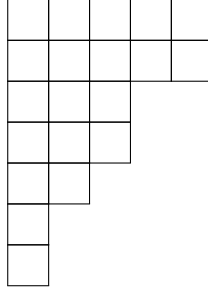


Figure 2.1: Young Diagram

all Young diagrams Y . From now on, we will implicitly identify a partition and the corresponding Young diagram.

Definition 2.4.5. A Young tableau of size N is a Young diagram $(\lambda_1, \dots, \lambda_r) \vdash N$ together with a bijection between its boxes and the integers $\{1, \dots, N\}$. A standard Young tableau is a Young tableau whose integers increase along the lines and the columns. We will denote with SYT_N the set of standard Young tableau of size N . The Young diagram of a given Young tableau P , denoted $sh(P)$, is called the shape of the tableau.

Example 2.4.6. The two Young tableaux below are of shape $(5, 5, 4, 2, 2, 1) \vdash 19$, the first one is standard and the second one is not.

1	3	4	8	10		5	3	4	8	10
2	5	9	13	14		2	1	12	13	14
6	11	12	16			6	11	9	16	
7	15					15	7			
17	18					17	18			
19						19				

Figure 2.2: Young Tableau

Theorem 2.4.7. *If we define*

$$F_\lambda := \{P \in SYT_N : sh(P) = \lambda\}$$

we have the following (amazing) identity

$$\sum_{\lambda \vdash N} F_\lambda^2 = N!$$

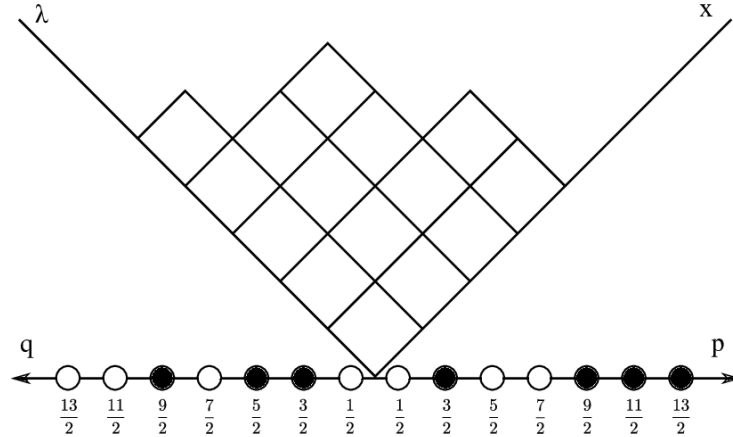
derived using the so called Robinson-Schensted correspondence between Young's Tableau and permutations,

$$RS : S_N \rightarrow \{(P, Q) \in SYT_N \times SYT_N : sh(P) = sh(Q)\}$$

The interesting part we want to discuss is when we actually change the convention a little for the Young Tableau. Namely, geometrically, we will rotate the diagram forty five degrees and centralize the bottom corner to be at the origin of a Real line. Now, we associate the YT with the following set of numbers: If we have the $YT - (\lambda_1, \dots, \lambda_n)$, then the set is given by

$$S(YT) = \left\{ \frac{\lambda_1 - 1}{2}, \frac{\lambda_2 - 3}{2}, \frac{\lambda_3 - 5}{2}, \dots \right\}$$

as depicted in the figure. When we take the size of the YT to be going to infinity we have a limiting shape



represented below - this shape is well described. More interesting, we also have a limit distribution of associated points $S(YT)$. This is excitedly the same distribution of point we get for the eigenvalues of a matrix ensemble called Wigner Matrices.

2.5 Matching Sequences

This section is based on the work on [11].

A very common - and wildly important - method used in evolutionary biology, and man correlated areas, is the alignment of sequences. The goal of this method is to search for similarities in patters in different sequences. A classic, and commonly studied alignment problem is the so called 'longest common subsequence' (LCS). The setup

is of two n - sequence and m - sequence, composed of n and m letters, respectively, of the same c -alphabet (a set of c letters). This set could be, for example, the set of basis of the DNA. Consider the sequences α, β given by

$$\begin{array}{rcl} \alpha & : & A \ C \ G \ C \ T \ A \ C \\ \beta & : & C \ T \ G \ A \ C \end{array}$$

A subsequence of α is an ordered sublist of α (entries of which need not be consecutive in α), e.g., $\{C, G, C, T\}$, but not $\{T, G, C\}$. A common subsequence of two sequences α and β is a subsequence of both of them. For example, the subsequence $\{C, G, A, C\}$ is a common subsequence of both α and β . The aim of the LCS problem is to find the longest of such common subsequences between two fixed sequences α and β .

A particularly important application of the LCS problem is to quantify the closeness between two DNA sequences. In evolutionary biology, the genes responsible for building specific proteins evolve with time and by finding the LCS of the same gene in different species, one can learn what has been conserved in time. Also, when a new DNA molecule is sequenced in vitro, it is important to know whether it is really new or it already exists. This is achieved quantitatively by measuring the LCS of the new molecule with another existing already in the database.

For a pair of fixed sequences of length i and j respectively, the length $L_{i,j}$ of their LCS is just a number. However, in the stochastic version of the LCS problem one compares two random sequences drawn from c -alphabets and hence the length $L_{i,j}$ is a random variable. A major challenge over the last three decades has been to determine the statistics of $L_{i,j}$. For equally long sequences ($i = j = n$), it has been proved that $\langle L_{n,n} \rangle \approx \gamma_c n$ for $n \gg 1$, where the averaging is performed over all realizations of the random sequences. The constant γ_c is known as the Chvátal-Sankoff constant which, to date, remains undetermined though there exists several bounds, a conjecture due to Steele that $\gamma_c = 2/(1 + \sqrt{n})$ and a recent proof that $\gamma_c \rightarrow 2/\sqrt{c}$ as $c \rightarrow \infty$. Unfortunately, no exact results are available for the finite size corrections to the leading behavior of the average $\langle L_{n,n} \rangle$, for the variance, and also for the full probability distribution of $L_{n,n}$. Thus, despite tremendous analytical and numerical efforts, exact solution of the random LCS problem has, so far, remained elusive. Therefore it is important to find other variants of this LCS problem that may be analytically tractable.

Computationally, the easiest way to determine the length $L_{i,j}$ of the LCS of two arbitrary sequences of lengths i and j (in polynomial time $O(ij)$) is via using the recursive algorithm

$$L_{i,j} = \max[L_{i-1,j}, L_{i,j-1}, L_{i-1,j-1} + \eta_{i,j}]$$

subject to the initial conditions $L_{0,j} = L_{i,0} = L_{0,0} = 0$. The variable $\eta_{i,j}$ is either 1 when the characters at the positions i (in the sequence α) and j (in the sequence β) match each other, or 0 if they do not. Note that the variables $\eta_{i,j}$'s are not independent of each other. These residual correlations between the $\eta_{i,j}$ variables make the LCS problem rather complicated. Note however that for two random sequences drawn from c alphabets, these correlations between the $\eta_{i,j}$ variables vanish in the $c \rightarrow \infty$ limit.

A natural question is how important are these correlations between the $\eta_{i,j}$ variables, e.g., do they affect the asymptotic statistics of $L_{i,j}$'s for large i and j ? Is the problem solvable if one ignores these correlations? These questions naturally lead to the Bernoulli matching (BM) model which is a simpler variant of the original LCS problem where one ignores the correlations between $\eta_{i,j}$'s for all c . The length $LBM_{i,j}$ of the BM model satisfies the same recursion relation as before, except that $\eta_{i,j}$'s are now independent and each drawn from the bimodal distribution: $p(\eta) = (1/c)\delta_{\eta,1} + (1 - 1/c)\delta_{\eta,0}$. This approximation is expected to be exact only in the appropriately taken $c \rightarrow \infty$ limit. Nevertheless, for finite c , the results on the BM model can serve as a useful benchmark for the original LCS model to decide if indeed the correlations between $\eta_{i,j}$'s are important or not. Unfortunately, even in the absence of correlations, the exact asymptotic distribution of $LBM_{i,j}$ in the BM model has so far remained elusive, mainly due to the nonlinear nature of the recursion relation.

The reference presents an exact asymptotic formula for the distribution of the length $LBM_{n,n}$ in the BM model for all c . So far, only the leading asymptotic behavior of the average length in the BM model was known using the 'cavity' method of spin glass physics,

$$\langle L_{n,n}^{(BM)} \rangle \approx \gamma_c^{(BM)} n$$

where $\gamma_c^{(BM)} = 2/(1 + \sqrt{c})$, the same as the conjectured value of the Chvátal-Sankoff constant γ_c for the original LCS model. However, other properties such as the variance or the distribution of $LBM_{n,n}$ remained untractable even in the BM model. We have shown, as illustrated below, that for large n ,

$$LBM_{n,n} \rightarrow \gamma_c^{(BM)} n + f(c)n^{1/3}\chi$$

where χ is a random variable with a n -independent distribution, $p(\chi \leq x) = F_2(x)$, which is precisely the Tracy-Widom distribution.

Thus, the Tracy-Widom distribution also describes the asymptotic distribution of the optimal matching length in the BM model, for all c . Given that the correlations in the original LCS model become negligible in the $c \rightarrow \infty$ limit, it is likely that the BM asymptotics would also hold for the original LCS model in the $c \rightarrow \infty$. An important open problem is to determine whether the Tracy-Widom distribution also appears in the LCS problem for finite c . The distribution obtained for all c in the BM model can serve as a useful benchmark to which future simulations of the LCS problem can be compared.

2.6 Public Transportation - Humam made

The city of Cuernavaca in Mexico (population about 500.000) has an extensive bus system, but there is no municipal transit authority to control the city transport. In particular there is no timetable, which gives rise to Poisson-like phenomena, with bunching and long waits between buses. Typically, the buses are owned by drivers as individual entrepreneurs, and all too often a bus arrives at a stop just as another bus is loading up. The driver then has to move on to the next stop to find his fares. In order to remedy the situation the drivers in Cuernavaca came up with a novel solution: they introduced “recorders” at specific locations along the bus routes in the city. The recorders kept track of when buses passed their locations, and then sold this information to the next driver, who could then speed up or slow down in order to optimize the distance to the preceding bus. The upshot of this ingenious scheme is that the drivers do not lose out on fares and the citizens of Cuernavaca now have a reliable and regular bus service. In the late 1990’s two Czech physicists with interest in transportation problems, M. Krbalek and P. Seba, heard about the buses in Cuernavaca and went down to Mexico to investigate. For about a month they studied the statistics of bus arrivals on Line 4 close to the city center. In particular, they studied the bus spacing distribution, and also the bus number variance measuring the fluctuations of the total number of buses arriving at a fixed location during a time interval T .

Krbalek and Seba found that both the bus spacing distribution and the number variance are well modeled by GUE. In order to provide a plausible explanation of the observations in, the authors in introduced a microscopic model for the bus linethat leads simply and directly to GUE.

The main features of the bus system in Cuernavaca are

1. the stop-start nature of the motion of the buses,
2. the “repulsion” of the buses due to the presence of recorders.

To capture these features, the authors introduced a model for the buses consisting of n (number of buses) independent, rate 1 Poisson processes moving from the bus depot at time $t = 0$ to the final terminus at time T , and conditioned not to intersect for $0 \leq t \leq T$. The authors then showed that at any observation point x along the route of length $N > n$, the probability distribution for the (rescaled) arrival times of the buses, $y_j = \frac{2t_j}{T} - 1 \in [-1, 1]$ for $i < j < n$ is given by

$$c \prod_{j=1}^n w_J(y_j) \prod_{1 \leq i < j \leq n} (y_i - y_j)^2 dy_1 \cdots dy_n$$

where

$$w_J(y) = (1 + y)^{x-1} (1 - y)^{N-x-n+1}, \quad -1 < y < 1$$

This formula is precisely the eigenvalue distribution for the so-called Jacobi Unitary Ensemble . In the appropriate scaling limit, GUE then emerges by universality. The authors also compute the distributions of the positions $x_1, \dots, x_n$ of the buses at any time $t \in (0, T)$. Again the statistics of the x_j ’s are described by a Unitary Ensemble, but now w_J is the same, but replaced by the weight for the Krawtchouk polynomials: by universality, GUE again emerges in the appropriate scaling limit.

In an intriguing recent paper, Abul-Magd [Abu] noted that drivers have a tendency “to park their cars near to each other and at the same time keep a distance sufficient for manoeuvring.” He then analyzed data measuring the gaps between parked cars on four streets in central London and showed quite remarkably that the gap size distribution was well represented by the spacing distribution of GUE. It is an interesting challenge to develop a microscopic model for the parking problem, analogous to the model for the bus problem.

2.7 Riemann zeta function - An anecdote

Here we consider the work of H. Montgomery in the early 1970's on the zeros of the Riemann zeta function $\zeta(s)$. Assuming the Riemann hypothesis, Montgomery rescaled the imaginary parts $\gamma_1 \leq \gamma_2 \leq \dots$ of the (nontrivial) zeros $\{1/2 + i\gamma_j\}$ of $\zeta(s)$,

$$\gamma_j \rightarrow \tilde{\gamma}_j = \frac{\gamma_j \log \gamma_j}{2\pi}$$

to have mean spacing 1 as $T \rightarrow \infty$, i.e.

$$\lim_{T \rightarrow \infty} \frac{\#\{j \geq 1 : \tilde{\gamma}_j \leq T\}}{T} = 1$$

For any $a < b$, he then computed the two-point correlation function for the $\tilde{\gamma}_j$'s

$$\#\{\text{ordered pairs}(j_1, j_2), j_1 \neq j_2 : 1 \leq j_1, j_2 \leq N, \tilde{\gamma}_1 - \tilde{\gamma}_2 \in (a, b)\}$$

and showed, modulo certain technical restrictions, that

$$R(a, b) \equiv \lim_{N \rightarrow \infty} \frac{1}{N} \#\{\text{ordered pairs}(j_1, j_2), j_1 \neq j_2 : 1 \leq j_1, j_2 \leq N, \tilde{\gamma}_1 - \tilde{\gamma}_2 \in (a, b)\}$$

exists and is given by a certain explicit formula. Soon after completing his work on the scaling limit (40) of the two-point correlation function for the zeros of zeta, Montgomery was visiting the Institute for Advanced Study in Princeton and it was suggested that he show his result to Dyson. What happened is a celebrated, and oft repeated, story in the lore of the Institute: before Montgomery could describe his hard won result to Dyson, Dyson took out a pen, wrote down a formula, and asked Montgomery “And did you get this?”

$$R(a, b) = \int_a^b 1 - \left(\frac{\sin(\pi r)}{\pi r} \right)^2 dr$$

Montgomery was stunned: this was exactly the formula he had obtained. Dyson explained: “If the zeros of the zeta function behaved like the eigenvalues of a random GUE matrix, then would be exactly the formula for the two-point correlation function!”

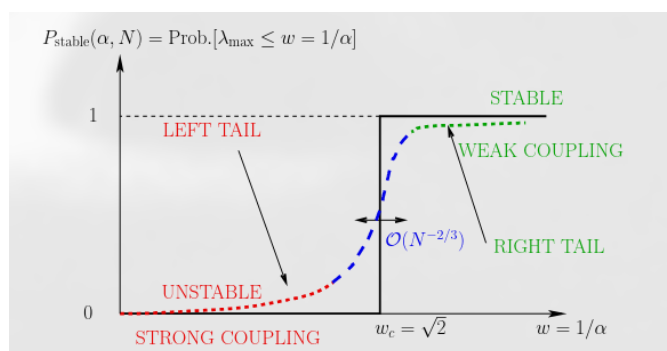
2.8 But what is the Tracy-Widow

First of all, the expected value of this distribution is actually negative. Second of all, it is highly asymmetric, with the tail on the left decaying as $\exp(-|x|^3/12)$ and the tail on the right decaying as $\exp(-4x^{3/2}/3)$. It is also worth noting that the range of validity of this distribution goes to zero as $N \rightarrow \infty$, since we scaled by $N^{1/6}$. We thus need a different tool to examine the part of the distribution where $|\lambda_{max} - \sqrt{2N}| > \mathcal{O}(N^{-1/6})$. These tails for large but finite N have a dependence that goes as e^{-N^2} on the left side and e^{-N} on the right.

Recently, Majumdar and Schehr have shown that N is large but finite, the Tracy-Widom distribution describes the crossover function between stable and unstable regimes of a random dynamical system, and furthermore that there is a third order phase transition between these regimes, which also occurs in the Coulomb gas, two-dimensional QCD, and several other systems of physical interest. The universality of the Tracy-Widom distribution could be related to the universality of phase transitions — events such as water freezing into ice, graphite becoming diamond and ordinary metals transforming into strange superconductors.

In the thermodynamic equations describing water, the curve that represents the water's energy as a function of temperature has a kink at 100 degrees Celsius, the point at which the liquid becomes steam. The water's energy slowly increases up to this point, suddenly jumps to a new level and then slowly increases again along a different curve, in the form of steam. Crucially, where the energy curve has a kink, the “first derivative” of the curve has a peak.

Similarly, the physicists realized, the energy curves of certain strongly correlated systems have a kink at $\sqrt{2N}$. The associated peak for these systems is the Tracy-Widom distribution, which appears in the third derivative of the energy curve — that is, the rate of change of the rate of change of the energy's rate of change. This makes the Tracy-Widom distribution a “third-order” phase transition.



Chapter 3

Random Matrix Theory

Random Matrices can be thought of in two ways: As a measurable function from a probability space into a set of matrices or, more concretely, as a matrix whose entries are random variables with some joint distribution. In some cases there is an explicit density formula for the eigenvalues of such a random matrix. With the help of some mathematical tools such as orthogonal polynomials one can express the equilibrium measures for the eigenvalues asymptotically.

3.1 Random Matrices

The chapter that follows is heavily based (to be very generous with my writing) on the incredibly organized and written Thesis [14]. To the author, my best wishes. The idea of this chapter is to motivate and introduce the concepts we will touch on later for the main part of the work. Also, if it is of any worth, this chapter serves as a good recap for any reader and a disclaimer of all the notation and conventions we will use in this work. What follows is my best attempt in introducing the part of Random Matrices that are pertinent to my scope of work and are of general interest, albeit inside my limitations.

3.1.1 Random Matrices Ensembles

Borrowing some concepts from physics, an ensemble can be thought of as a virtual collection of all possible states that a system can be in, with more probable states appearing more often in the collection. There is an analogy to be made with ensembles of thermodynamic: When considering sets of matrices with equal probability - in some sense to be more clear soon - we get the Gibbs weight for the probability distribution as we would in the *Canonical Ensemble* of constant temperature. More concretely, in the context of Random Matrix Theory

Definition 3.1.1. A random matrix ensemble $\mathcal{E} = (\mathcal{S}, \mathcal{P})$ is a set of matrices $\mathcal{S}$ of fixed size equipped with a probability density function (p.d.f.) $\mathcal{P} : \mathcal{S} \rightarrow [0, \infty)$. If we say that a matrix $\hat{X}$ is drawn from the random ensemble $\mathcal{E}$, we are saying that it is a random variable assuming values in $\mathcal{S}$ with p.d.f. $\mathcal{P}$.

A commonplace thing to do when defining ensembles is to prescribe the p.d.f.'s $\mathcal{P}_{i,j}$ on the independent entries $X_{i,j}$ of a representative matrix $\hat{X} \in \mathcal{E}$. In such cases, we say that

$$\mathcal{P}(\hat{X}) = \prod_{i,j | X_{i,j} \text{ independent}} \mathcal{P}_{i,j}(X_{i,j}).$$

With this definition we can already define many ensembles of interest. Common sets can be the set of symmetric matrices $\mathbb{M}_{m,n}(\mathbb{R})$, the set of complex matrices $\mathbb{M}_{m,n}(\mathbb{C})$ or the set of quaternionic matrices $\mathbb{M}_{m,n}(\text{He})$. Then, oftentimes, $\mathcal{S}$ is simply the set of symmetric, hermitian or skew-symmetric matrices. When it comes to determining the p.d.f $\mathcal{P}$ on $\mathcal{S}$, there are many options that have been worked on. A simple example is that of *Bernoulli matrices* $\hat{B}$ whose entries $B_{i,j}$ are all independently and identically distributed Bernoulli random variables ± 1 : $\mathcal{S} = \mathbb{M}_{n,n}(\mathbb{R})$ and

$$P_{i,j}(B_{i,j}) = \frac{1}{2} [\delta(B_{i,j} - 1) + \delta(B_{i,j} + 1)].$$

If we instead require $\hat{B}$ to be symmetric, its diagonal entries to be zero, and its independent entries have p.d.f.

$$P_{i,j}(B_{i,j}) = \frac{c}{n} \delta(B_{i,j} - 1) + \left(1 - \frac{c}{n} \delta(B_{i,j})\right),$$

then $\hat{B}$ belongs to the ensemble of *Erdos-Renyi*. Consider now one of the most important ensembles we will define.

Definition 3.1.2. (*Ginibre Ensemble*). The $m \times n$ real, complex, and quaternionic Ginibre Ensembles are respectively $\mathbb{M}_{m,n}(\mathbb{R}), \mathbb{M}_{m,n}(\mathbb{C}), \mathbb{M}_{m,n}(\mathbb{H})$ with p.d.f

$$\mathcal{P}^{(Gin)}(\hat{G}) = \pi^{-mn \frac{\beta}{2}} \exp\left(-\text{Tr}(\hat{G}^\dagger \hat{G})\right) = \prod_{i=1}^m \prod_{j=1}^n \pi^{\frac{\beta}{2}} \exp(-|G_{i,j}|^2),$$

defined in respect to the Lebesgue measure

$$d\hat{G} = \prod_{i=1}^m \prod_{j=1}^n \prod_{s=1}^{\beta} dG_{i,j}^{(s)}.$$

The coefficient β is called the *Dyson index* and counts the generic number of real components of each matrix entry $G_{i,j}$, so that $\beta = 1, 2, 4$ refers to real, complex and quaternionic entries. These are the Ginibre orthogonal, unitary and symplectic ensembles.

For most applications there are three ideologies that stand out. The first one is to assume little on the system and focus on universal results. This results in the study of the *Wigner matrix ensembles*: Let $\hat{X}$ be drawn from either the independently distributed real symmetric, complex hermitian or quaternionic skew-symmetric $n \times n$ matrix ensemble. Then its entries $X_{i,j}$ are independently distributed with mean zero and bounded moments. Also, its diagonal entries are equally distributed as are its upper triangular entries. Many results hold for independently and identically distributed centered random variables. To construct the general form of such matrices take the matrix $\hat{Y} = \{Y_n\}_{i,j}$ with $Y_{i,j} = Y_{j,i}^\dagger$ and such that

1. $\{Y_{i,j}\}_{1 \leq i \leq j \leq n}$ are independent;
2. $\{Y_{i,j}\}$ are identically distributed. $\{Y_{i,i}\}$ are identically distributed;
3. $\mathbb{E}(Y_{i,j}) = 0$ and $\mathbb{E}(Y_{i,j}^k) < \infty$, i.e, $r_k = \max(\mathbb{E}(Y_{i,j}^k), \mathbb{E}(Y_{i,j}^k)) < \infty$.

Definition 3.1.3. Let every $\{Y_n\}_{i,j}$ be as before and set $r_2 < \infty$ with $\mathbb{E}(Y_{1,2}^2) > 0$. Then, we say $\hat{Y}$ is a *Wigner Matrix*. Furthermore, if $\hat{X}_n = n^{-\frac{1}{2}} \hat{Y}_n$, $\hat{X}$ is what we call a *normalized Wigner Matrix*.

The scale and moment assumption taken are not as unnatural as they may seem at first. To properly understand this definition, note that if $\lambda_1^2, \lambda_2^2, \dots, \lambda_n^2$ are the ordered eigenvalues of $\{Y_n\}_{i,j}$, we can estimate that

$$\lambda_1^2 + \dots + \lambda_n^2 \leq n \max_{j=1}^n (\lambda_j^2) = n \max(\lambda_1^2, \lambda_n^2)$$

therefore, if we want the eigenvalues to be limited when we take large matrices we should impose that $1/n \cdot (\lambda_1^2 + \dots + \lambda_n^2)$ must be limited. This is the Hilbert-Schmidt or Frobenius norm for these normal matrices:

$$\|\hat{Y}\|_{HS} = \frac{1}{n} \sum_{i=1}^n \lambda_i^2 = \frac{1}{n} \sum_{i,j} Y_{i,j}^2 = \frac{1}{n} \text{Tr}(\hat{Y}^2)$$

then if we only take $\hat{X}_n = \alpha_n \hat{Y}_n$

$$\begin{aligned} \mathbb{E}(\|\hat{X}_n\|_{HS}) &= \alpha_n^2 \mathbb{E}(\|\hat{Y}_n\|_{HS}) \\ &= \frac{\alpha_n^2}{n} \sum_{i,j} \mathbb{E}(Y_{i,j}^2) \\ &= \alpha_n^2 \mathbb{E}(Y_{1,1}^2) + (n-1) \alpha_n^2 \mathbb{E}(Y_{1,2}^2) \end{aligned}$$

then it is clear that there are two cases

1. $\mathbb{E}(Y_{1,2}^2) = 0$: This is the non-interesting case where $\alpha_n \approx 1$
2. $\mathbb{E}(Y_{1,2}^2) > 0$: Here, we must take $\alpha_n \approx n^{-\frac{1}{2}}$

Now, it is likely more clear the motivation of the definition: limit the eigenvalues to a compact region. As the norm is limited, also are the eigenvalues of the scaled matrix.

The second ideology is assuming equal probability on each state of the system - this is a common assumption in thermodynamics. So this approach is to assume that every matrix is equally likely.

Observation 3.1.4. *We have focused on random matrix ensembles distributed according to p.d.f.s $\mathcal{P}$ defined with respect to the flat Lebesgue measure induced by the canonical embedding of $\mathcal{S}$ into Euclidean space. Note here, which was not obvious for me, that the measures ν we define are given in terms of the p.d.f.s $\mathcal{P}$ and the Lebesgue measure μ by Radon-Nikodim*

$$\nu(A) = \int_A \mathcal{P} d\mu,$$

or more simply, $d\nu = \mathcal{P} d\mu$.

Theorem 3.1.5. *(Haar, compact case). Given a compact group $\mathcal{S}$, let $U \in \mathcal{S}$ be a generic variable. There exists a unique (up to normalization measure) $d\mu(U)$ on $\mathcal{S}$, called the **Haar measure** such that*

$$\int_{\mathcal{S}} d\mu(U) < \infty$$

and for any fixed $U_0 \in \mathcal{S}$,

$$d\mu(U_0 U) = d\mu(U) = d\mu(U U_0),$$

that is, finite for compact sets and invariant by left and right actions. If $d\mu(U)$ integrates to unity in $\mathcal{S}$ it is referred as the **Haar probability measure**.

Definition 3.1.6. *The $n \times n$ circular unitary ensemble (CUE) is the unitary group $U(n)$ with its Haar measure. If $U \in CUE(n)$, then the symmetric unitary matrix $U^T U$ represents the $n \times n$ circular orthogonal ensemble (COE); we have $\mathcal{S} \cong U(n)/O(n)$. Note that if $U \in U(n)$ and $O \in O(n)$, we have that $U^T U = (OU)^T (OU)$, that is, U and OU are in the same co-class.*

The third ideology is that of working with algebraic combinations of matrices whose entries are independently and identically distributed. This will not be as much discussed.

3.1.2 Some quantities

To study random matrices, we need to determine which quantities related to the matrix we want to measure. This may be the expectation of some function on the matrices or even on the eigenvalues and eigenvector, if they exist. Assume a matrix is such that is able to be determined by eigenvalues and eigenvectors. While eigenvectors may be of interest in some cases, eigenvalue statistics have been considerably more explored. Hence, in addition to the p.d.f.s $\mathcal{P}(X)$ of, say, $n \times n$ random matrices $\hat{X}$, we are also interested in the eigenvalue j.p.d.f. $p(\lambda_1, \lambda_2, \dots, \lambda_n)$ of the eigenvalues $\{\lambda_i\}_{i=1}^n$ of $\hat{X}$. The eigenvalue j.p.d.f. is defined to be such that

$$\int_{[a_1, b_1] \times \dots \times [a_n, b_n]} p(\lambda_1, \lambda_2, \dots, \lambda_n) d\lambda_1 \dots d\lambda_n$$

is the probability that $a_i < \lambda_i < b_i$ for every i .

Example 3.1.7. *The eigenvalue j.p.d.f. of the $n \times n$ circular ensembles is*

$$p^{(C)}(e^{i\theta_1}, \dots, e^{i\theta_n}; \beta) = \frac{1}{\mathcal{Z}_{n,\beta}^{(C)}} |\Delta_n(e^{i\theta})|^\beta$$

where $Z_{n,\beta}^{(C)}$ is a normalization constant and

$$\Delta_n(\lambda) := \det \left[\lambda_i^{j-1} \right]_{i,j=1}^n = \prod_{1 \leq i < j \leq n} (\lambda_i - \lambda_j)$$

is the Vandermonde determinant, and $\beta = 1, 2, 4$ correspond to the COE, CUE and CSE, respectively.

Related to the eigenvalue j.p.d.f. is the univariate eigenvalue density defined by

$$\mathbf{p}(\lambda_1; n) := n \int_{\mathbb{R}^{n-1}} \mathbf{p}(\lambda_1, \lambda_2, \dots, \lambda_n) d\lambda_2 \cdots d\lambda_n,$$

which is normalized such that the expect number of eigenvalues lying between a and b is given by $\int_a^b \mathbf{p}(\lambda) d\lambda$. Note that this is not a p.d.f. if $n \neq 1$. Consider now the linear statistic of the eigenvalue density given by

$$\mathbf{p}(\lambda) = \sum_{i=1}^n \langle \delta(\lambda - \lambda_i) \rangle$$

where the brackets are understood as acting on an operator as

$$\langle \mathcal{O} \rangle = \int_{\mathbb{R}^n} \{ \mathcal{O} \mathbf{p}(\lambda_1, \dots, \lambda_n) \} d\lambda_1 \cdots d\lambda_n.$$

Many times, eigenvalues density are fully characterized by their *spectral moments*

$$m_k := \int_{\mathbb{R}} \lambda^k \mathbf{p}(\lambda) d\lambda, \quad k \in \mathbb{N}.$$

Indeed, if we are interested in a linear statistic $\sum_{i=1}^n f(\lambda_i)$ of the eigenvalues such that $f(\lambda)$ is analytic at zero with Taylor expansion $f(\lambda) = \sum_{k=0}^{\infty} f_k \lambda^k$, we can compute its expectation to be

$$\left\langle \sum_{i=1}^n f(\lambda_i) \right\rangle = \int_{\mathbb{R}} f(\lambda) \mathbf{p}(\lambda) d\lambda = \sum_{k=0}^{\infty} f_k m_k$$

The spectral moments can additionally be expressed as an average with respect to the eigenvalue j.p.d.f. $\mathbf{p}(\lambda_1, \lambda_2, \dots, \lambda_n)$ and the p.d.f. $\mathcal{P}(X)$ of the random matrix $\hat{X}$ according to

$$m_k = \left\langle \sum_{i=1}^n \lambda_i^k \right\rangle = \left\langle \text{Tr}(\hat{X}^k) \right\rangle_{\mathcal{P}(\hat{X})}$$

where we define

$$\langle \mathcal{O} \rangle_{\mathcal{P}(\hat{X})} := \int_S \{ \mathcal{O} \mathcal{P}(\hat{X}) \} d\hat{X}$$

Moment Generating Function

We can also define an useful tool for studying the spectral moments: The *Moment Generating Function* given by

$$W_1(x) := \sum_{k=0}^{\infty} \frac{m_k}{x^{k+1}},$$

which is also known as the *Resolvent* due to it being equal to the *Stieltjes Transform* of the eigenvalue density

$$W_1(x) = \int_{\text{supp}(\mathbf{p})} \frac{\mathbf{p}(\lambda)}{x - \lambda} d\lambda, \quad \text{for } x \in \mathbb{C} \setminus \text{supp}(\mathbf{p}).$$

Obtaining a closed form expression for the resolvent allows one to use the *Sokhotski-Plemelj* inversion formula to recover the eigenvalue density

$$\mathbf{p}(\lambda) = \lim_{\epsilon \rightarrow 0} \frac{W_1(\lambda - i\epsilon) - W_1(\lambda + i\epsilon)}{2\pi i}$$

as long as $\mathbf{p}(\lambda)$ is known to be continuous on its support. As we did before, we can re-express the resolvent as an ensemble average w.r.t. to the eigenvalue j.p.d.f. $\mathbf{p}(\lambda_1, \lambda_2, \dots, \lambda_n)$ and the p.d.f. $\mathcal{P}(X)$ on $\mathcal{S}$:

$$W_1(x) = \left\langle \sum_{i=1}^n \frac{1}{x - \lambda_i} \right\rangle = \left\langle \text{Tr} \frac{1}{x - \hat{X}} \right\rangle_{\mathcal{P}(\hat{X})}$$

These three expressions facilitate the dealing of the eigenvalue densities. In fact, dealing with the resolvent will be many times preferred as it presents an asymptotic expansion the densities does not.

Theorem 3.1.8. (*Borot-Guionnet*) Assuming certain hypothesis on the eigenvalues j.p.d.f. $\mathbf{p}(\lambda_1, \lambda_2, \dots, \lambda_n)$, the resolvent $W_1(x)$ and, more generally, the undefined connected m -point correlator $W_m(x_1, x_2, \dots, x_m)$, have large n expansions of the form

$$W_m(x_1, x_2, \dots, x_m, n) = n^{2-m} \sum_{l=0}^{\infty} \frac{W_m^l(x_1, x_2, \dots, x_m)}{n^l},$$

where the correlator expansion coefficients $W_m^l(x_1, x_2, \dots, x_m)$ have no dependence on n .

Some of the conditions of the theorem above guide us to rescale the eigenvalues of certain ensembles so that they are contained in a compact connected set in the large limit $n \rightarrow \infty$ regime. In these cases, we study the *global scaled eigenvalue density* $\tilde{\mathbf{p}}(\lambda)$ and the *scaled resolvent* $\tilde{W}_1(x)$ given by

$$\tilde{\mathbf{p}}(\lambda) := \frac{c_n}{n} \mathbf{p}(c_n \lambda)$$

$$\tilde{W}_1(x) := c_n W_1(c_n x) = n \int_{\text{supp}(\tilde{\mathbf{p}})} \frac{\tilde{\mathbf{p}}(\lambda)}{x - \lambda} d\lambda \quad \text{for } x \in \mathbb{C} \setminus \text{supp}(\tilde{\mathbf{p}}).$$

where c_n is a scaling parameter that differs from ensemble to ensemble. Now, assuming the theorem holds, one has that $\tilde{W}_1 = \sum_{l=1}^{\infty} W_1^l(x) n^{1-l}$. Using the Sokhotski-Plemelj inversion formula we define

$$\mathbf{p}^l(\lambda) := \lim_{\epsilon \rightarrow 0} \frac{W_1^l(\lambda - i\epsilon) - W_1^l(\lambda + i\epsilon)}{2\pi i}.$$

Then, $\tilde{\mathbf{p}}(\lambda)$ and $\sum_{l=0}^{\infty} \mathbf{p}^l(\lambda) n^{-l}$ do not agree at finite n , as one might expect. Instead, the latter sum is referred to as the *smoothed eigenvalue density*, as it lacks the oscillatory terms that are known to be present in the true scaled eigenvalue density $\tilde{\mathbf{p}}(\lambda)$.

3.1.3 Classical Ensembles

The denomination *Classical Ensemble* classically denotes the Gaussian, Laguerre and Jacobi ensembles because of their - future discussed - relation to the orthogonal polynomials. We can define them as below.

Definition 3.1.9. We define the following classical ensembles

- Let $\hat{G}$ be drawn from the $n \times n$ real, complex or quaternionic Ginibre Ensemble. Then, the random matrix $\hat{H} = \frac{1}{2}(\hat{G}^\dagger + \hat{G})$ represents the corresponding Gaussian ensemble (also known as Hermite).
- Let $\hat{G}$ be drawn from the $n \times m$ real, complex or quaternionic Ginibre Ensemble. Then, the random matrix $\hat{W} = (\hat{G}^\dagger \hat{G})$ represents the corresponding (m, n) Laguerre ensemble (also known as Wishart).
- We say $\hat{Y}$ an element of the (m_1, m_2, n) real, complex and quaternionic Jacobi Ensemble if it is of the form $\hat{Y} = \hat{W}_1(\hat{W}_1 + \hat{W}_2)^{-1}$ where each $\hat{W}_i$ belong to the corresponding (m_i, n) Laguerre ensembles with $m_i \geq n$.

Note that all ensembles presented are self-adjoint and therefore have all n eigenvalues. Gaussian ensembles are also examples of Wigner Matrices.

Proposition 3.1.10. *Let $k := \beta/2$, $a = k(m_1 - n + 1) - 1$ and $b = k(m_2 - n + 1) - 1$. From the p.d.f. of the Ginibre ensembles, we can change variables to obtain the p.d.f.s of the $n \times n$ Gaussian, (m_1, n) Laguerre, and (m_1, m_2, n) Jacobi ensembles, respectively*

$$\begin{aligned} p^{(G)}(\hat{H}) &= \frac{1}{\mathcal{Z}_{n,\beta}^{(G)}} \exp(-\text{Tr}(\hat{H}^2)) \\ p^{(L)}(\hat{W}) &= \frac{1}{\mathcal{Z}_{n,a}^{(L)}} (\det(\hat{W}))^a \exp(-\text{Tr}(\hat{W})) \\ p^{(J)}(\hat{Y}) &= \frac{1}{\mathcal{Z}_{n,a,b}^{(J)}} (\det(\hat{Y}))^a (\det(\hat{I}_n - \hat{Y}))^b \end{aligned}$$

With $\beta = 1, 2, 4$ corresponding to the choice of $\mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{H}$, the random matrix ensembles introduced in the definition are defined with the latter p.d.f.s onto the respective matrix sets

$$\begin{aligned} \mathcal{S}^{(G)} &= \{\hat{H} \in M_{n,n}(\mathbb{F}) | \hat{H} = \hat{H}^\dagger\} \\ \mathcal{S}^{(L)} &= \{\hat{W} \in \mathcal{S}^{(G)} | \hat{W} \text{ positive definite}\} \\ \mathcal{S}^{(J)} &= \{\hat{Y} \in \mathcal{S}^{(L)} | \hat{I}_n - \hat{Y} \text{ positive definite}\} \end{aligned}$$

The contemporary definition of classical ensembles introduces another ensemble into the family.

Definition 3.1.11. *A random matrix ensemble is said to be a classical matrix ensemble if its eigenvalue j.p.d.f. is of the form*

$$p^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_n; \beta) = \frac{1}{\mathcal{N}_{n,\beta}^{(w)}} \prod_{i=1}^n w(\lambda_i) |\Delta_n(\lambda)|^\beta, \quad \beta = 1, 2, 4,$$

where $\mathcal{N}_{n,\beta}^{(w)}$ is the normalization constant, $\Delta_n(\lambda)$ is the Vandermonde determinant and the weight function $w(\lambda)$ is such that

$$\frac{d}{d\lambda} \log(w(\lambda)) = \frac{w'(\lambda)}{w(\lambda)} = \frac{f(\lambda)}{g(\lambda)}$$

with f and g polynomials of degree at most one and two, respectively.

Up to fractional linear transformations $\lambda \mapsto (a_{11}\lambda + a_{12})/(a_{21}\lambda + a_{22})$ with $a_{ij} \in GL_2(\mathbb{R})$, there are precisely four weight functions that satisfies the condition. These are

$$w(\lambda) = \begin{cases} e^{-\lambda^2} & \text{Gaussian,} \\ \lambda^a e^{-\lambda} \chi_{\{\lambda > 0\}} & \text{Laguerre,} \\ \lambda^a (1 - \lambda)^b \chi_{\{0 < \lambda < 1\}} & \text{Jacobi,} \\ (1 - i\lambda)^\eta (1 + i\lambda)^{\bar{\eta}} & \text{Generalized Cauchy.} \end{cases}$$

where χ is the indicator function.

Definition 3.1.12. Taking $k = \beta/2$ and $c_n = \sqrt{kn}, kn, 1$ respectively for the Gaussian, Laguerre and Jacobi ensembles. The global scaled eigenvalue densities of the classical matrix ensembles are

$$\begin{aligned}\tilde{\mathbf{p}}^{(G)}(\lambda) &= \sqrt{\frac{k}{n}} \mathbf{p}^{(G)}(\sqrt{kn}\lambda) \\ \tilde{\mathbf{p}}^{(L)}(\lambda) &= k \mathbf{p}^{(L)}(kn\lambda) \\ \tilde{\mathbf{p}}^{(J)}(\lambda) &= \frac{1}{n} \mathbf{p}^{(J)}(\lambda) \\ \tilde{\mathbf{p}}^{(Cy)}(\lambda) &= \frac{1}{n} \mathbf{p}^{(J)}(\lambda)\end{aligned}$$

3.2 Invariant Measure

Invariant ensembles of random matrices are probability spaces which are invariant under conjugation of the appropriate matrix group - these ensembles only depend on their eigenvalue distributions while their eigenvectors become independent and are Haar distributed. To properly understand this concept, let's make precise the definition of three infamous ensembles: Unitary, Orthogonal and Symplectic ensembles. Other ensembles many times can be of interest, but their theory may be less straightforward.

1. *Orthogonal Ensembles* consist of $n \times n$ real symmetric matrices $\hat{M} = \overline{\hat{M}} = \hat{M}^T$ together with a probability distribution that is invariant under orthogonal conjugation $\hat{M} \mapsto \hat{O}\hat{M}\hat{O}^T$, $\hat{O}$ orthogonal, i.e., $\hat{O}\hat{O}^T = \hat{O}^T\hat{O} = \hat{I}$.
2. *Unitary Ensembles* consists of $n \times n$ Hermitian matrices $\hat{M} = \hat{M}^*$ together with a probability distribution that is invariant under unitary conjugation $\hat{M} \mapsto \hat{U}\hat{M}\hat{U}^*$, $\hat{U}$ unitary, i.e., $\hat{U}\hat{U}^* = \hat{U}^*\hat{U} = \hat{I}$.
3. *Symplectic Ensembles* consists of $2m \times 2m$ Hermitian self-dual matrices $\hat{M} = \hat{M}^* = \hat{J}\hat{M}^T\hat{J}^T$ where $\hat{J} = \text{diag}(\sigma, \dots, \sigma)$ with

$$\sigma = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

together with a probability distribution that is invariant under unitary-symplectic conjugation $\hat{M} \mapsto \hat{U}\hat{M}\hat{U}$, with $\hat{U}$ unitary and symplectic, i.e. $\hat{U}\hat{U}^* = \hat{U}^*\hat{U} = \hat{I}$ and $\hat{U}\hat{J}\hat{U}^T = \hat{J}$.

Examples of such invariant ensembles are the Gaussian Orthogonal, Unitary and Symplectic ensembles - *GOE*, *GUE*, *GSE*. These take Gaussian distributed and independent entries. We may also choose to include the Ginibre Ensembles in the types of invariance we will work with. The three types of Ginibre Ensembles are: Ginibre Orthogonal, Unitary and Symplectic are matrices with independent, identically distributed standard complex Gaussian entries. In this case, the eigenvalues are distributed in the complex plane.

Let's make it precise. Given a group $\mathcal{G}$ and a group action $\mathcal{A} : \mathcal{G} \times \mathcal{S} \rightarrow \mathcal{S}$, a random matrix ensemble $\mathcal{E} = (\mathcal{S}, \mathcal{P})$ is said to be $\mathcal{A}$ -invariant if the probability measure $d\mathcal{P}(\hat{X}) := \mathcal{P}(\hat{X})d\hat{X}$ satisfies

$$d\mathcal{P}(\mathcal{A}(g, \hat{X})) = d\mathcal{P}(\hat{X})$$

for every $g \in \mathcal{G}$. The term *invariant matrix ensemble* usually refers to invariance under the mapping $\hat{X} \mapsto g\hat{X}g^{-1}$ for g drawn from the appropriate compact group $\mathcal{G}$; when a matrix ensemble is considered to be irreducible and its underlying set of matrix is $S \subseteq \mathbb{M}_{n,n}(\mathbb{R}), \mathbb{M}_{n,n}(\mathbb{C}), \mathbb{M}_{n,n}(\mathbb{H})$, the group $\mathcal{G}$ must be one of $O(n), U(n), Sp(2n)$. Then, we say the ensemble is orthogonal, unitary or symplectic.

We can also determine the probability measure $d\mathcal{P}(\hat{X})$ to be invariant under the adjoint action of a group $\mathcal{G}$ if we establish that both the p.d.f. $\mathcal{P}(\hat{X})$ and the Lebesgue measure $d\hat{X}$ are invariant under this action. We can say directly using 3.1.2 and 3.1.10 that the p.d.f.s for Ginibre, Gaussian, Laguerre and Jacobi are invariant under the appropriate adjoint actions.

Proof. (p.d.f.s for Ginibre, Gaussian, Laguerre and Jacobi are invariant under the appropriate adjoint actions.) $\square$

The invariance of the Lebesgue measure is more involved and depends on whether or not the associated matrix set $\mathcal{S}$ consists of purely of self-adjoint matrices. Because of that, we start with the case of the Gaussian, Laguerre and Jacobi ensembles.

Proposition 3.2.1. *Given an $n \times n$ real ($\beta = 1$), complex ($\beta = 2$) or quaternionic ($\beta = 4$) self-adjoint matrix variable $\hat{X}$, diagonalize it as $\hat{X} = \hat{U} \hat{\Lambda} \hat{U}^\dagger$ where $\hat{\Lambda} = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$ is a diagonal matrix of eigenvalues, and $\hat{U}$ is corresponding matrix of eigenvectors; the spectral theorem dictates that $\hat{U}$ is an element of $O(n)$ in the real case, $U(n)$ in the complex case, and $Sp(2n)$ in the quaternionic case. To make the mapping $\hat{X} \mapsto \hat{U} \hat{\Lambda} \hat{U}^\dagger$ bijective, we insist that the eigenvalues are listed in increasing order and the first row of $\hat{U}$ is non-negative real. The canonical Lebesgue measure*

$$d\hat{X} = \prod_{i=1}^n dX_{ii} \prod_{1 \leq j < k \leq n} \prod_{s=1}^{\beta} dX_{jk}^s$$

on the set of adjoint matrices decomposes as

$$d\hat{X} = |\Delta_n|^\beta d\hat{\Lambda} d\mu_{Haar}(\hat{U})$$

where $d\hat{\Lambda} = \prod_{i=1}^n d\lambda_i$ is the Lebesgue measure on the Weyl Chamber $\lambda_1 < \dots < \lambda_n \subset \mathbb{R}^n$ and

$$d\mu_{Haar}(\hat{U}) = \bigwedge_{1 \leq l < k \leq n} \bigwedge_{s=1}^{\beta} \left(\hat{U}^\dagger [dU_{ij}]_{i,j=1}^n \right)_{lk}^{(s)}$$

is the unnormalized Haar measure on the quotient space $O(n)/O(1)^n$ ($\beta = 1$), $U(n)/U(1)^n$ ($\beta = 2$), or $Sp(2n)/Sp(2)^n$ ($\beta = 4$), given by the exterior product of the independent real components of the entries of the matrix of differentials $\hat{U}^\dagger [dU_{ij}]_{i,j=1}^n$.

At this point, one can check (albeit with some non-trivial computations) that the Lebesgue measure is indeed invariant in these cases for the appropriate group action.

The situation concerning the $n \times n$ Ginibre Ensemble is more complicated. In the 1965 work, Ginibre extended the proposition above and its arguments starting with the diagonalization formula $\hat{G} = \hat{P} \hat{\Lambda} \hat{P}^{-1}$ - note that, with $\hat{\Lambda}$ diagonal of eigenvalues and $\hat{P}$ of eigenvectors, this formula is valid with probability one since the subset of matrices with repeated eigenvalues has measure zero. However, $\hat{P}$, diagonalizing matrix, is not necessarily orthogonal, unitary, nor symplectic. Nonetheless, QR decomposition can be used to write $\hat{P} = \hat{U} \hat{T}$ where $\hat{U}$ is as before and $\hat{T}$ is an upper triangular matrix with real positive diagonal entries. Then,

$$d\hat{G} = J(\lambda) d\hat{\Lambda} d\mu_{Tri}(\hat{T}) d\mu_{Haar}(\hat{U})$$

where $d\hat{\Lambda}$ is the Lebesgue measure on the space housing the eigenvalues, $d\mu_{Haar}(\hat{U})$ is the Haar measure defined before, $J(\lambda)$ is a Jacobian consisting of products of Vandermonde determinants involving the eigenvalues, and $\mu_{Tri}(\hat{T})$ is some measure dependent on the entries of $\hat{T}$. From here, it follows that the Lebesgue measure on $\hat{G}$ is invariant under the desired conjugation action.

3.2.1 Eigenvalues j.p.d.f.s

To obtain the eigenvalue j.p.d.f. $p(\lambda_1, \lambda_2, \dots, \lambda_n)$ from the probability measure $d\mathcal{P}(\hat{X})$, one changes variable to the eigenvalues and a set of auxiliary variables, then integrates over the latter. This will be something like this

sketch

$$\begin{aligned}
\mathcal{P}(\hat{X})d\hat{X} &= \mathcal{P}(X_{1,1}, \dots, X_{n,n}) \prod_{i,j} dX_{i,j} \\
&= \mathcal{P}\left(X_{1,1}(\hat{\Lambda}, \hat{U}), \dots, X_{n,n}(\hat{\Lambda}, \hat{U}) | J(\hat{X} \mapsto \{\hat{\Lambda}, \hat{U}\})\right) d\hat{\Lambda} d\hat{U}. \\
\Rightarrow \mathbf{p}(\lambda_1, \lambda_2, \dots, \lambda_n) &= \int_{\mathcal{G}(n)/\mathcal{G}(1)^n} \mathcal{P}(X_{1,1}(\hat{\Lambda}, \hat{U}), \dots, X_{n,n}(\hat{\Lambda}, \hat{U}) | J(\hat{X} \mapsto \{\hat{\Lambda}, \hat{U}\})) d\hat{\Lambda} d\hat{U} \\
&\stackrel{\text{invariance}}{=} \mathcal{P}(X_{1,1}(\hat{\Lambda}), \dots, X_{n,n}(\hat{\Lambda}) | J(\hat{X} \mapsto \{\hat{\Lambda}\})) d\hat{\Lambda} \int_{\mathcal{G}(n)/\mathcal{G}(1)^n} d\hat{U} \\
&= \mathcal{P}(X_{1,1}(\hat{\Lambda}), \dots, X_{n,n}(\hat{\Lambda})) |\Delta_n(\lambda)|^\beta \text{Vol}(\mathcal{G}(n)/\mathcal{G}(1)^n) d\hat{\Lambda}
\end{aligned}$$

A nice observation is that we know $\mathbf{p}(\lambda_1, \lambda_2, \dots, \lambda_n)$ to be a multivariate symmetric function - this implies that $\mathcal{P}(X_{1,1}(\hat{\Lambda}), \dots, X_{n,n}(\hat{\Lambda})) := \mathcal{P}(\hat{\Lambda})$ to be a function of $\{\text{Tr}(\hat{\Lambda}^k)\}_{k=0}^n$, hence $\{\text{Tr}(\hat{X}^k)\}_{k=0}^n$. For the self-adjoint cases, these auxiliary variables were the quotient spaces of orthogonal, unitary and symplectic matrices. For Ginibre, is the upper triangular entries. Leaving the details of Ginibre behind, we focus on the Gaussian, Laguerre and Jacobi ensembles. Set $\mathcal{P}(\hat{X})$ the p.d.f. for one of the ensembles, fix $(\beta, \mathcal{G}(n)) = (1, O(n)), (2, U(n)), (4, Sp(2n))$. The propositions cited implies that

$$\int_{\mathcal{G}(n)/\mathcal{G}(1)^n} d\mathcal{P}(\hat{X}) = \mathcal{P}(\Lambda) |\Delta_n(\lambda)|^\beta d\hat{\Lambda} \int_{\mathcal{G}(n)/\mathcal{G}(1)^n} d\mu_{Haar}(\hat{U})$$

As the eigenvalue and eigenvector statistics are decoupled, one reads off the eigenvalue j.p.d.f. is thus given by

$$\mathbf{p}(\lambda_1, \lambda_2, \dots, \lambda_n) = \frac{\text{Vol}(\mathcal{G}(n)/\mathcal{G}(1)^n)}{N!} \mathcal{P}(\hat{\Lambda}) |\Delta_n(\lambda)|^\beta,$$

where, of course,

$$\text{Vol}(\mathcal{G}(n)/\mathcal{G}(1)^n) = \int_{\mathcal{G}(n)/\mathcal{G}(1)^n} d\mu_{Haar}(\hat{U}).$$

Now, with $w(\lambda)$ the appropriate classical weight, we can check that

$$\mathcal{P}(\hat{\Lambda}) = \prod_{i=1}^n w(\lambda_i),$$

which confirms the consistence with the definition of the eigenvalue j.p.d.f. on the classical ensembles. Furthermore, note that

$$\mathcal{Z}_{n,\beta} = \frac{\text{Vol}(\mathcal{G}(n)/\mathcal{G}(1)^n)}{N!} \mathcal{N}_{n,\beta}$$

as long as the volume is known.

3.2.2 Gaussian Ensembles

Let's make it explicit the probability distribution on Gaussian Ensembles. We define $\hat{M} = n^{-\frac{1}{2}} \hat{M}' \in \mathcal{M}_{n,n}(\mathcal{F})$ to be of the Gaussian Ensemble if the entries of $\hat{M}'$ are independent and distributed as follows:

$$M'_{i,j} \sim \begin{cases} \mathcal{N}_{\mathcal{F}}(0, 1/2) & \text{for } i \neq j, \\ \mathcal{N}_{\mathcal{F}}(0, 1) & \text{for } i = j, \end{cases} \quad \text{if } i \leq j.$$

where $\mathcal{F}$ is such as the real line, complex plane or quaternion space. Note that this turns $\hat{M}$ into a Scaled Wigner Matrix. Now, take for example $\mathcal{F} = \mathbb{R}$ and

$$\mathcal{P}(\hat{M}) = \prod_{i=1}^n \frac{\exp\left(-\frac{M_{i,i}^2}{2}\right)}{\sqrt{2\pi}} \prod_{i < j} \frac{\exp\left(-M_{i,i}^2\right)}{\sqrt{\pi}} = 2^{-n/2} \pi^{-n(n+1)/4} \exp\left(-\frac{1}{2} \text{Tr } \hat{M}^2\right).$$

Then, we know the distribution of eigenvectors and eigenvalues is the product of the distribution of entries times the Jacobian of the diagonalization, that is

$$\mathcal{P}(\hat{M}) = \mathcal{P}(\hat{\Lambda}, \hat{O}) \prod_{i < j} |\lambda_j - \lambda_i|$$

Now, for the distribution on the eigenvalues, we integrate on $\mathbb{V}_n$ as

$$p(\lambda_1, \lambda_2, \dots, \lambda_n) = \int_{\mathbb{V}_n} \mathcal{P}(\hat{\Lambda}) \prod_{i < j} |\lambda_j - \lambda_i| d\hat{O} = \mathcal{P}(\hat{\Lambda}) \prod_{i < j} |\lambda_j - \lambda_i| \text{Vol}(\mathbb{V}_n)$$

which lead us to write

$$p(\lambda_1, \lambda_2, \dots, \lambda_n) = \frac{1}{\mathcal{Z}_{n,\beta=1}} \exp \left\{ -\frac{1}{2} \sum_{i=1}^n \lambda_i^2 \right\} \prod_{i < j} |\lambda_i - \lambda_j|.$$

with $\mathcal{Z}_{n,\beta}$ the partition function, or normalizer of the expression. We could also do the same calculation for the GUE and GSE ensembles, which would result in a similar expression

$$\begin{aligned} p(\lambda_1, \lambda_2, \dots, \lambda_n) &= \frac{1}{n! \mathcal{Z}_{n,\beta}^{(ord)}} \exp \left\{ - \left(\sum_{i=1}^n \frac{\lambda_i^2}{2} - \sum_{i < j} \log |\lambda_i - \lambda_j|^\beta \right) \right\}, \\ &= \frac{1}{\mathcal{Z}_{n,\beta}^{(ord)}} e^{-\beta_n \mathcal{H}_n(\vec{\lambda})}, \end{aligned}$$

a very familiar expression for a physicist, as it is a Gibbs measure.

3.2.3 Empirical Measure

In many application one is usually concerned with the limiting behavior of the spectra as the size of the matrix tends to infinity, either to properly understand large matrices or to approximate infinite-dimensional operators. A usual tool to describe the eigenvalues of a random matrix is the empirical measure: If $\hat{M}$ has eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$, then its empirical spectral measure μ_M is the random probability measure putting equal mass at each of the eigenvalues:

$$\mu_M = \frac{1}{n} \sum_{i=1}^n \delta_{\lambda_i}$$

We want to say that this measure converges, in some sense, for some other measure. A common way of doing this is the concept of *convergence in expectation*. Define $\mathbb{E}(\mu_M)$ to be as follows: for a test function f , define

$$\int f d\mathbb{E}(\mu_M) := \mathbb{E} \left(\int f d\mu_M \right)$$

If we have a sequence of matrices $\{\hat{M}_n\}$ we say that their spectral measure converges in expectation to a limiting measure μ if the sequence $\{\mathbb{E}(\mu_{M_n})\}$ converges weakly to μ , that is, if

$$\int f d\mathbb{E}(\mu_{M_n}) \xrightarrow{n \rightarrow \infty} \int f d\mu$$

for bounded continuous test functions f . A natural class of test functions are polynomials. Note that if $\lambda_1, \lambda_2, \dots, \lambda_n$ are the random eigenvalues of $\hat{M}$ random matrix we have that

$$\int (z^k + c_{k-1} z^{k-1} + \dots + c_0) d\mu_M = \frac{1}{n} \sum_{i=1}^n (\lambda_i^k + c_{k-1} \lambda_i^{k-1} + \dots + c_0) = \frac{1}{n} \text{Tr}(V(\hat{M}))$$

In a way, the potential V somehow influences the convergence.

3.3 Orthogonal Polynomial Ensembles

The subclass of invariant matrix ensembles this section refers to are those self-adjoint matrices $\hat{X}$ with p.d.f.s $\mathcal{P}(\hat{X})$ such that one has the decomposition

$$\mathcal{P}(\hat{X}) = \prod_{i=1}^n w(\lambda_i),$$

where $\{\lambda_i\}$ are the indistinguishable eigenvalues of $\hat{X}$ and $w(\lambda)$ is some continuous weight function that is real valued on $\mathbb{R}$ and decays fast enough as $|\lambda| \rightarrow \infty$. This is the same as removing the constrain on the weight function of the classical ensembles. We are interested in the probability distribution of the eigenvalues $\lambda_1, \dots, \lambda_n$ of $\hat{X}$. Because the matrices are self-adjoint, there exists $\hat{U} \in U(n)$ such that

$$\hat{U}^* \hat{X} \hat{U} = \hat{\Lambda} = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n).$$

We make such a transformation

$$\hat{X} \mapsto (\hat{U}, \hat{\Lambda})$$

and then integrate the eigenvector element away, so that we are left with the part depending on the eigenvalues. This can be done using Weyls formula.

Theorem 3.3.1. (Weyl) Suppose f is an integrable class function on $\mathbb{S}_n$, i.e., $f(\hat{U} \hat{X} \hat{U}^*) = f(M)$ for all $\hat{U} \in U(n)$. Then we have

$$\int_{\text{He}_n} f(\hat{X}) d\hat{X} = c_n \int_{\mathbb{R}^n} f(\lambda_1, \lambda_2, \dots, \lambda_n) \prod_{i < j} (\lambda_j - \lambda_i)^\beta d\lambda_1 \cdots d\lambda_n,$$

where $\mathbb{S}_n$ are the self-adjoint matrices of order n and

$$c_n = \frac{\pi^{n(n-1)/2}}{\prod_{j=1}^n j!}.$$

More generally,

$$dM = c_n \prod_{i < j} (\lambda_i - \lambda_j)^\beta d\lambda_1 \cdots d\lambda_n d\hat{U},$$

where $d\hat{U}$ is the Haar measure on $U(n)$

Because of this, we arrive at the following definition.

Definition 3.3.2. Let $\beta = 1, 2, 4$, fix $n \in \mathbb{N}$ and let $w(\lambda)$ be a continuous, non-negative, real-valued function such that $w(\lambda) = o(\lambda^{-\beta(n-1)-1})$. Then, the j.p.d.f.

$$\mathbf{p}^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_n; \beta) = \frac{1}{\mathcal{N}_{n,\beta}^{(w)}} \prod_{i=1}^n w(\lambda_i) |\Delta_n(\lambda)|^\beta$$

describes the n -sized orthogonal polynomial ensemble (OPE) when $\beta = 2$, skew-orthogonal polynomial ensemble of real type (R-SOPE) when $\beta = 1$, and the skew-orthogonal polynomial ensemble of quaternion type (H-SOPE) when $\beta = 4$.

The condition on the weight function guarantees that the j.p.d.f. has a well-defined normalization constant. To properly understand the nomenclature used, one must discuss the Vandermonde determinant. Check 3.4 for this.

If one wishes to integrate the determinantal and Pfaffian structures given by the Jacobian against the product of weight functions, it is beneficial to use monic polynomials $\{q_j(\lambda)\}_{j=0}^\infty$ that are skew-orthogonal with respect to the weight $w(\lambda)$.

Definition 3.3.3. With $w(\lambda)$ a suitable function, define the following inner products on polynomial space:

$$\begin{aligned}\langle f, g \rangle^{(1)} &:= \frac{1}{2} \int_{\mathbb{R}^2} f(x)g(y) \operatorname{sign}(y-x)w(x)w(y)dx dy \\ \langle f, g \rangle^{(2)} &:= \int_{\mathbb{R}} f(x)g(x)w(x)dx \\ \langle f, g \rangle^{(4)} &:= \frac{1}{2} \int_{\mathbb{R}} (f(x)g'(x) - f'(x)g(x))w(x)dx\end{aligned}$$

For $\beta = 1, 2, 4$, let $\{q_j^{(\beta)}\}_{j=0}^{\infty}$ be a family of monic polynomials such that each $q_j^{(\beta)}(\lambda)$ is of degree j . Impose also that exists finite positive constants $\{r_j^{(\beta)}\}_{j=0}^{\infty}$ s.t.

$$\begin{aligned}\langle q_j^{(2)}, q_k^{(2)} \rangle^{(2)} &= r_j^{(2)} \chi_{j=k} \\ \langle q_{2j}^{(\sigma)}, q_{2k+1}^{(\sigma)} \rangle^{(\sigma)} &= -\langle q_{2k+1}^{(\sigma)}, q_{2j}^{(\sigma)} \rangle^{(\sigma)} = r_j^{(\sigma)} \chi_{j=k} \\ \langle q_{2j}^{(\sigma)}, q_{2k}^{(\sigma)} \rangle^{(\sigma)} &= \langle q_{2j+1}^{(\sigma)}, q_{2k+1}^{(\sigma)} \rangle^{(\sigma)} = 0\end{aligned}$$

for $\sigma = 1, 4$ and any $j, k > 0$. Then, we say that the family $\{q_j^{(1)}(\lambda)\}_{j=0}^{\infty}$ is $\mathbb{R}$ -skew-orthogonal, $\{q_j^{(2)}(\lambda)\}_{j=0}^{\infty}$ is orthogonal, and $\{q_j^{(4)}(\lambda)\}_{j=0}^{\infty}$ is He-skew-orthogonal. All in respect to $w(\lambda)$.

We can now express the eigenvalue j.p.d.f. $p^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_n; \beta)$ as either a determinant or a Pfaffian.

Theorem 3.3.4. Let $\{q_j^{(\beta)}(x)\}_{j=0}^{\infty}, \{r_j^{(\beta)}\}_{j=0}^{\infty}$, be as in the definition above, and let $n \in \mathbb{N}$ with n even in the case for $\beta = 1$. Next, introduce the auxiliary functions

$$\begin{aligned}\phi_j(x) &:= \int_{\mathbb{R}} q_j^{(1)}(y) \operatorname{sign}(x-y)w(y)dy \\ S_j^{(1)}(x, y) &:= \phi'_{2j+1}(x)\phi_{2j}(y) - \phi'_{2j}(x)\phi_{2j+1}(y) \\ S_j^{(4)}(x, y) &:= q'_{2j+1}(x)q_{2j}(y) - q'_{2j}(x)q_{2j+1}(y) \\ D_j^{(4)}(x, y) &:= q'_{2j+1}(x)q'_{2j}(y) - q'_{2j}(x)q'_{2j+1}(y) \\ I_j^{(4)}(x, y) &:= q_{2j+1}(x)q_{2j}(y) - q_{2j}(x)q_{2j+1}(y)\end{aligned}$$

and the kernels corresponding respectively to the real, complex and quaternionic cases,

$$\begin{aligned}K_n^{(1)}(x, y) &:= \sum_{k=0}^{N/2-1} \frac{1}{r_k^{(1)}} \begin{bmatrix} \frac{1}{2} \int_{\mathbb{R}} \operatorname{sign}(x-z)S_k^{(1)}(z, y) & S_k^{(1)}(y, x) \\ -S_k^{(1)}(x, y) & \delta_y S_k^{(1)}(x, y) \end{bmatrix} - \frac{1}{2} \begin{bmatrix} \operatorname{sign}(x-y) & 0 \\ 0 & 0 \end{bmatrix}, \\ K_n^{(2)}(x, y) &:= \sqrt{w(x)w(y)} \sum_{k=0}^{N-1} \frac{1}{r_k^{(2)}} q_k^{(2)}(x)q_k^{(2)}(y), \\ K_n^{(4)}(x, y) &:= \sqrt{w(x)w(y)} \sum_{k=0}^{N-1} \frac{1}{r_k^{(4)}} \begin{bmatrix} I_k^{(4)}(x, y) & S_k^{(4)}(y, x) \\ -S_k^{(4)}(x, y) & D_k^{(4)}(x, y) \end{bmatrix}.\end{aligned}$$

Then, the eigenvalue j.p.d.f. of Definition 3.3.2 given by

$$p^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_n; \beta) = \begin{cases} \det [K_n^{(2)}(\lambda_i, \lambda_j)]_{i,j=1}^n, & \beta = 2, \\ \operatorname{Pf} [K_n^{(\beta)}(\lambda_i, \lambda_j)]_{i,j=1}^n, & \beta = 1, 4. \end{cases}$$

The significance of this theorem is that, for $\beta = 1, 2, 4$ the kernels $K_n^{(\beta)}(x, y)$ have *Reproducing properties* such as

$$\begin{aligned} \int_{\mathbb{R}} \det \left[K_n^{(2)}(\lambda_i, \lambda_j) \right]_{i,j=1}^{k+1} d\lambda_{k+1} &= (n-k) \det \left[K_n^{(2)}(\lambda_i, \lambda_j) \right]_{i,j=1}^k \\ \int_{\mathbb{R}} \text{Pf} \left[K_n^{(\beta)}(\lambda_i, \lambda_j) \right]_{i,j=1}^{k+1} d\lambda_{k+1} &= (n-k) \text{Pf} \left[K_n^{(\beta)}(\lambda_i, \lambda_j) \right]_{i,j=1}^k \end{aligned} \quad (3.3.1)$$

Thus, one may integrate the j.p.d.f. over $d\lambda_{k+1} \cdots d\lambda_n$ to obtain the *k-point correlation function*

$$\mathbf{p}_k^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_k; n; \beta) := \frac{n!}{(n-k)!} \int_{\mathbb{R}^{n-k}} \mathbf{p}^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_n; \beta) d\lambda_{k+1} \cdots d\lambda_n = \begin{cases} \det \left[K_n^{(2)}(\lambda_i, \lambda_j) \right]_{i,j=1}^k, & \beta = 2, \\ \text{Pf} \left[K_n^{(\beta)}(\lambda_i, \lambda_j) \right]_{i,j=1}^k, & \beta = 1, 4. \end{cases}$$

This implies, that the eigenvalue density $p^{(w)} = p_1^{(w)}(\lambda)$ is given by the formula

$$p^{(w)}(\lambda; \beta) = \begin{cases} K_n^{(2)}(\lambda, \lambda), & \beta = 2, \\ \text{Pf} \left[K_n^{(\beta)}(\lambda, \lambda) \right], & \beta = 1, 4. \end{cases}$$

Most properties of a determinantal point process - such as the eigenvalues of a random matrix from a unitary ensemble - follow from properties of the Kernel.

Theorem 3.3.5. *If*

$$\lim_{n \rightarrow \infty} \frac{1}{n} K_n(x, x) = \phi_V(x), \quad x \in \mathbb{R},$$

then the asymptotic distribution of the points in the point process (eigenvalues of $\hat{M}_n$) is given by the measure with density ϕ_V :

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^n f(x_j) = \int f(x) \phi_V(x) dx.$$

Moreover

Theorem 3.3.6. *Suppose ϕ_V to be supported on one interval $[a, b]$, then for $x^* \in (a, b)$ with $\phi_V(x^*) > 0$, one has*

$$\lim_{n \rightarrow \infty} \frac{1}{\phi_V(x^*)n} K_n \left(x^* + \frac{x}{\phi_V(x^*)n}, x^* + \frac{y}{\phi_V(x^*)n} \right) = \frac{\sin \pi(x-y)}{\pi(x-y)}$$

there is a similar result for the borders.

Theorem 3.3.7. *Suppose ϕ_V to be supported on one interval $[a, b]$ and near the endpoint b*

$$\phi_V(x) = \frac{c}{\pi} (b-x)^{\frac{1}{2}} (1 + o(1)), \quad x \rightarrow b.$$

Then

$$\lim_{n \rightarrow \infty} \frac{1}{(cn)^{\frac{2}{3}}} K_n \left(b + \frac{x}{(cn)^{\frac{2}{3}}}, b + \frac{y}{(cn)^{\frac{2}{3}}} \right) = \frac{Ai(x)Ai'(y) - Ai'(x)Ai(y)}{x-y},$$

where Ai is the Airy function.

3.3.1 The case for $\beta = 2$

We start illustrating the results above for the case $\beta = 2$. Begin with the following characterization of the Jacobian as a Vandermonde determinant illustrated in 3.4

$$\prod_{i < j}^n |\lambda_i - \lambda_j|^2 = \det(\lambda_i^{j-1})_{i,j=1, \dots, n}^2.$$

Now, based on the property 3.4.1, we can rewrite the entries using some class of appropriate polynomial. For this case, we will use $\{q_j^{(2)}(\lambda)\}_{j=0}^{\infty}$ orthogonal polynomials. Then,

$$\prod_{i < j}^n |\lambda_i - \lambda_j|^2 = \det \left(q_{j-1}^{(2)}(\lambda_i) \right)_{n \times n}^2$$

Using the fact that $(\det(A))^2 = \det(\sum_{j=1}^n A_{ji} A_{jk})$ we can rewrite

$$\prod_{i < j}^n |\lambda_i - \lambda_j|^2 = \det \left(\sum_{j=1}^n q_{j-1}^{(2)}(\lambda_i) q_{j-1}^{(2)}(\lambda_k) \right)_{n \times n}$$

Now, remember that

$$\mathbf{p}^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_n; 2) \propto \prod_{i=1}^n w(\lambda_i) |\Delta_n(\lambda)|^2,$$

then we will put the weight inside the determinant in the following manner: Take $w(\lambda_i) = \sqrt{w(\lambda_i)} \sqrt{w(\lambda_i)}$ and multiply one term by the i -th column and the other by the i -th line of the matrix. Now, we should get

$$\mathbf{p}^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_n; 2) \propto \det \left(\sqrt{w(\lambda_i) w(\lambda_k)} \sum_{j=1}^n q_{j-1}^{(2)}(\lambda_i) q_{j-1}^{(2)}(\lambda_k) \right)_{i,k=1}^n$$

and therefore

$$\mathbf{p}^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_n; 2) \propto \det \left[K_n^{(2)}(\lambda_i, \lambda_k) \right]_{i,k=1}^n.$$

The general formula can also be further simplified but this computation is easier with $\beta = 2$, so that's the case we will deal with for now. First, it is known that the orthogonal polynomials $\{q_j^{(2)}(\lambda)\}_{j=0}^{\infty}$ satisfy the three-term recurrence

$$q_j^{(2)}(\lambda) = (\lambda + a_j) q_{j-1}^{(2)}(\lambda) - \frac{r_{j-1}^{(2)}}{r_{j-2}^{(2)}} q_{j-2}^{(2)}(\lambda)$$

where the coefficients depend on the choice of weight. We use this recurrence to write the terms in the summation of $K_n^{(2)}$ so that it gives rise to the *Christoffel-Darboux* formula:

Proposition 3.3.8. *The $\beta = 2$ kernel defined before simplifies as*

$$K_n^{(2)}(x, y) = \frac{\sqrt{w(x)w(y)}}{r_{n-1}^{(2)}} \frac{q_n^{(2)}(x)q_{n-1}^{(2)}(y) - q_{n-1}^{(2)}(x)q_n^{(2)}(y)}{x - y}$$

Taking the limit $y \rightarrow x$ using L'Hopital's rule shows that for finite n the density has the form

$$\mathbf{p}^{(w)}(\lambda) = K_n^{(2)}(\lambda, \lambda) = \frac{w(\lambda)}{r_{n-1}^{(2)}} \left(q_{n-1}^{(2)}(\lambda) \partial_\lambda q_n^{(2)}(\lambda) - q_n^{(2)}(\lambda) \partial_\lambda q_{n-1}^{(2)}(\lambda) \right)$$

All that remains to make $\mathbf{p}^{(w)}(\lambda)$ fully explicit is knowledge of the orthogonal polynomials $\{q_j^{(2)}(\lambda)\}_{j=0}^{\infty}$. In the Gaussian, Laguerre and Jacobi cases, there are Hermite, Laguerre and Jacobi orthogonal polynomials, respectively. Another important identity we should clarify is the following

Proposition 3.3.9.

$$\mathcal{Z}_{n,2} = n! \prod_{k=0}^{n-1} r_k^{-1}$$

where r_k is the normalizing constant as in $\langle q_j^{(2)}, q_k^{(2)} \rangle^{(2)} = r_j^{(2)} \chi_{j=k}$.

Proof. We start stating the following equality which follows from 3.3.4 that

$$\det \left[K_n^{(2)}(\lambda_i, \lambda_j) \right]_{i,j=1}^n = p^{(w)}(\lambda_1, \lambda_2, \dots, \lambda_n; \beta),$$

$$\prod_{k=1}^n r_k^{-1} \det \left[\sqrt{w(x)w(y)} \sum_{k=0}^{N-1} q_k^{(2)}(x) q_k^{(2)}(y) \right]_{i,j=1}^n = \prod_{1 \leq i \leq j \leq n} (\lambda_j - \lambda_i)^2 \prod_{k=0}^n w(\lambda_k)$$

therefore

$$n! \det \left[\tilde{K}_n^{(2)}(\lambda_i, \lambda_j) \right]_{i,j=1}^n = n! \prod_{k=1}^n r_k \prod_{1 \leq i \leq j \leq n} (\lambda_j - \lambda_i)^2 \prod_{k=0}^n w(\lambda_k)$$

where

$$\mathcal{Z}_{n,2} = n! \prod_{k=1}^n r_k$$

□

3.4 Vandermonde

We have said before that the change of variables performed for the invariant ensembles give rise to an Jacobian of the form

$$\prod_{i < j}^n |\lambda_j - \lambda_i|^\beta,$$

lets try to properly understand this object. The first thing to note is that this is a determinant. This is done in two steps: First, we show that for a given matrix form, the determinant has exactly this form, secondly we show that the Jacobian is exactly this matrix. For the first part, consider a matrix V given by

$$\hat{V} = \begin{bmatrix} 1 & \alpha_1 & \alpha_1^2 & \dots & \alpha_1^{n-1} \\ 1 & \alpha_2 & \alpha_2^2 & \dots & \alpha_2^{n-1} \\ 1 & \alpha_3 & \alpha_3^2 & \dots & \alpha_3^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha_n & \alpha_n^2 & \dots & \alpha_n^{n-1} \end{bmatrix}$$

We can use of the properties of the determinant, namely that it does not change if we add a multiple of one row to another row to write

$$|\hat{V}| = \begin{vmatrix} 1 & 0 & 0 & \dots & 0 \\ 1 & \alpha_2 - \alpha_1 & \alpha_2(\alpha_2 - \alpha_1) & \dots & \alpha_2^{n-2}(\alpha_2 - \alpha_1) \\ 1 & \alpha_3 - \alpha_1 & \alpha_3(\alpha_3 - \alpha_1) & \dots & \alpha_3^{n-2}(\alpha_3 - \alpha_1) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha_n - \alpha_1 & \alpha_n(\alpha_n - \alpha_1) & \dots & \alpha_n^{n-2}(\alpha_n - \alpha_1) \end{vmatrix} = \prod_{i=2}^n (\alpha_i - \alpha_1) \begin{vmatrix} 1 & 0 & 0 & \dots & 0 \\ 1 & 1 & \alpha_2 & \dots & \alpha_2^{n-2} \\ 1 & 1 & \alpha_3 & \dots & \alpha_3^{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \alpha_n & \dots & \alpha_n^{n-2} \end{vmatrix}$$

Using now of Laplace Theorem we now that there is only one cofactor that matters. So we write

$$|\hat{V}| = \prod_{i=2}^n (\alpha_i - \alpha_1) \begin{vmatrix} 1 & \alpha_2 & \dots & \alpha_2^{n-2} \\ 1 & \alpha_3 & \dots & \alpha_3^{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha_n & \dots & \alpha_n^{n-2} \end{vmatrix}$$

Repeating this many times we of course have the expression we started with, that is

$$|\hat{V}| = \prod_{i < j}^n |\alpha_j - \alpha_i|.$$

We can now use of the properties of the Vandermonde determinant to rewrite the Jacobian as we need. For reasons that will become clearer, we will use the following properties

Affirmation 3.4.1. *Let $\hat{V}$ be a matrix with entries $\{V_{i,j}\} = v_i^{j-1}$ such that its determinant is given by $\prod_{i < j}^n |v_j - v_i|$. Then, if we define $\hat{V}'$ with $\{V_{i,j}\} = p(v_i)$ with p a monomial with degree j .*

3.5 Equilibrium Measure

Remember in 3.3.1 when we expressed the equilibrium measure in terms of the Kernel of the point process - given in 3.3.1. It was noted then that we would only need to understand the asymptotics of the orthogonal polynomial associated with the weight function to properly recover the equilibrium measure. Now, we have such an asymptotic. Remember the expression

$$\mathbf{p}^{(w)}(x) = K_n^{(2)}(x, x) = \frac{w(x)}{r_{n-1}^{(2)}} (\text{He}_{n-1}(x) \partial_x \text{He}_n(x) - \text{He}_n(x) \partial_x \text{He}_{n-1}(x))$$

We will make use of the following relation

Definition 3.5.1. *The Hermite polynomial satisfies the following propertie*

$$\partial_x \text{He}_n(x) = n \text{He}_{n-1}(x)$$

Proof.

$$\begin{aligned} \frac{d}{dx} \text{He}_n(x) &= \frac{d}{dx} \left((-1)^{(n)} e^{\frac{1}{2}x^2} \frac{d^n}{dx^n} e^{-\frac{1}{2}x^2} \right) \\ &= (-1)^{(n)} \left(x e^{\frac{1}{2}x^2} \frac{d^n}{dx^n} e^{-\frac{1}{2}x^2} + e^{\frac{1}{2}x^2} \frac{d^{n+1}}{dx^{n+1}} e^{-\frac{1}{2}x^2} \right) \\ &= x \left((-1)^{(n)} e^{\frac{1}{2}x^2} \frac{d^n}{dx^n} e^{-\frac{1}{2}x^2} \right) - (-1)^{(n+1)} \left(e^{\frac{1}{2}x^2} \frac{d^{n+1}}{dx^{n+1}} e^{-\frac{1}{2}x^2} \right) \\ &= x \text{He}_n(x) - \text{He}_{n+1}(x) \\ &= x \text{He}_n(x) - (x \text{He}_n(x) - n \text{He}_{n-1}(x)) \\ &= n \text{He}_{n-1}(x) \end{aligned}$$

□

using this result, one can write that

$$\begin{aligned} \mathbf{p}^{(w)}(x) &= K_n^{(2)}(x, x) = \frac{w(x)}{r_{n-1}^{(2)}} (He_{n-1}(x) n \text{He}_{n-1}(x) - \text{He}_n(x) (n-1) \text{He}_{n-2}(x)) \\ &= \frac{w(x)}{r_{n-1}^{(2)}} (n \text{He}_{n-1}^2(x) - (n-1) \text{He}_n(x) \text{He}_{n-2}(x)) \\ &= \frac{w(x)}{r_{n-1}^{(2)}} ((N-1) \text{He}_{n-1}^2(x) - (N-2) \text{He}_n(x) \text{He}_{n-2}(x)) \end{aligned}$$

and now it suffices to have the expression for $\text{He}_{n-2}(x)$.

$$\begin{aligned} \text{He}_n(2\sqrt{N} \cos(\gamma)) &= \sqrt{\frac{2}{\sin(\gamma)}} N^{\frac{1}{2}(N-1)} e^{N(\cos^2(\gamma) - \frac{1}{2})} \sin \left(\theta(\beta_c^+) - N \left(\frac{\sin(2\gamma)}{2} - \gamma \right) \right) (1 + \mathcal{O}(N^{-1})) \\ \text{He}_{n-1}(2\sqrt{N} \cos(\gamma)) &= \sqrt{\frac{2}{\sin(\gamma)}} N^{\frac{1}{2}(N-2)} \frac{e^{N(\cos^2(\gamma) - \frac{1}{2})}}{e^{\cos^2(\gamma)}} \sin \left(\theta(\beta_c^+) - (N-1) \left(\frac{\sin(2\gamma)}{2} - \gamma \right) \right) (1 + \mathcal{O}(N^{-1})) \\ \text{He}_{n-2}(2\sqrt{N} \cos(\gamma)) &= \sqrt{\frac{2}{\sin(\gamma)}} N^{\frac{1}{2}(N-3)} \frac{e^{N(\cos^2(\gamma) - \frac{1}{2})}}{e^{2\cos^2(\gamma)}} \sin \left(\theta(\beta_c^+) - (N-2) \left(\frac{\sin(2\gamma)}{2} - \gamma \right) \right) (1 + \mathcal{O}(N^{-1})) \end{aligned}$$

Therefore

$$\begin{aligned} \mathbf{p}^{(w)}(x) &= \Pi_N [(N-1) \sin^2(\theta(\beta_c^+) - (N-1)\Theta) - (N-2) \sin(\theta(\beta_c^+) - (N-2)\Theta) \sin(\theta(\beta_c^+) - N\Theta)] \\ &= \Pi_N N [\sin^2(\theta(\beta_c^+) - (N-1)\Theta) - \sin(\theta(\beta_c^+) - (N-2)\Theta) \sin(\theta(\beta_c^+) - N\Theta)] (1 + \mathcal{O}(1)) \end{aligned}$$

with

$$\begin{aligned}\Pi_N &= \frac{e^{-\frac{1}{2}(2\cos\gamma\sqrt{N})^2}}{(N-2)!} \left(\frac{2}{\sin(\gamma)} \frac{e^{2N(\cos^2(\gamma)-\frac{1}{2})}}{e^{2\cos^2(\gamma)}} N^{N-2} \right) \\ &= \frac{2e^{-2\cos^2(\gamma)}}{\sin(\gamma)\sqrt{2\pi N}}\end{aligned}$$

and

$$\Theta = \left(\frac{\sin(2\gamma)}{2} + \gamma \right)$$

Finally, using the identity $\cos^2(a+b) - \cos(a)\cos(a+2b) = \sin^2(b)$,

$$\begin{aligned}\mathfrak{p}^{(w)}(\alpha) &= \frac{2e^{-2\cos^2(\gamma)}}{\sin(\gamma)\sqrt{2\pi N}} \left[\sin^2(\theta(\beta_c^+) - (N-1)\Theta) - \sin(\theta(\beta_c^+) - (N-2)\Theta) \sin(\theta(\beta_c^+) - N\Theta) \right] (1 + \mathcal{O}(1)) \\ &= \frac{2e^{-2\cos^2(\gamma)}}{\sin(\gamma)\sqrt{2\pi N}} \left[\cos^2(a+\gamma) - \cos(a)\cos(a+2\gamma) \right] (1 + \mathcal{O}(1)) \\ &= \frac{2e^{-2\cos^2(\gamma)}}{\sin(\gamma)\sqrt{2\pi N}} \sin^2(\gamma) (1 + \mathcal{O}(1)) \\ &= \frac{2e^{-2\cos^2(\gamma)}}{\sqrt{2\pi N}} \sin(\gamma) (1 + \mathcal{O}(1)) \\ &= \frac{e^{-\frac{1}{2}\alpha^2}}{\sqrt{2\pi N}} \sqrt{4 - \alpha^2}\end{aligned}$$

using that

$$a - \frac{\pi}{2} = \theta(\beta_c^+) - N\Theta - \frac{\sin(2\gamma)}{2}$$

Chapter 4

Determinantal Point Processes

The definitions given in this section are a direct adaptation from [7], [17], [4].

4.1 Point Processes

In general, point processes are models to describe random numbers of occurrences of events in a given time-frame or, alternatively, the number of points in regions of some space. For this work, we mainly focus on the latter application: density of points in regions of space. To illustrate such processes, suppose there are m points at the random locations $\chi_m = \{x_1, x_2, \dots, x_m\}$ in some region Λ of the Euclidean space $\mathbb{R}^n$. We can define a quantity $\xi(A)$ for any sub-region of $A \subset \Lambda$ that measures the amount of points ($\#\{\chi_m \cap A\}$) in this region in the following way:

$$\xi(A) = \sum_{k=1}^m \mathbb{1}(x_k \in A).$$

Let $\mathcal{I}$ denote all intervals or rectangles in Λ , $\mathcal{E}$ denote the class of Borel sets and $\mathcal{B}$ the class of bounded Borel sets. One can notice that this ξ defines a *random counting measure*, i.e., a mapping $\mu : \mathcal{E} \rightarrow \{0, 1, \dots, \infty\}$ such that

$$\mu(\emptyset) = 0, \quad \mu(B) < \infty, \text{ for } B \in \mathcal{B},$$

and, for disjoint $B_1, B_2, \dots$ in $\mathcal{B}$,

$$\mu\left(\bigcup_k B_k\right) = \sum_k \mu(B_k).$$

We say that ξ is boundedly finite. Let $\mathcal{N}(\Lambda)$ denote the space of all counting measures ξ on Λ . If $B \in \mathcal{B}$, then $\xi(B)$ is finite and we can write

$$\xi|_B = \sum_{i=1}^{\xi(B)} \delta_{x_i},$$

for some $\chi = \{x_1, x_2, \dots, x_{\xi(B)}\} \subset \Lambda$. Finally, we can explicitly define point processes.

Definition 4.1.1. (*Point Processes*) Let Λ be a complete separable metric space. A **point process** $\mathbb{P}$ on Λ is a probability measure on the space of all configuration of points χ or counting measures $\mathcal{N}(\Lambda)$. We say $\mathbb{P}$ to be a n -point process if $\mathbb{P}(\#\chi = n) = 1$ or, equivalently, $\mathbb{P}(\xi(\Lambda) = n) = 1$. We say that ξ is simple if $\xi(\{x\}) \leq 1$ for all $x \in \Lambda$. The point process is simple if $\mathbb{P}(\xi \text{ is simple}) = 1$.

An important note is that ξ is the same for any ordering of the subscripts on $x_1, x_2, \dots, x_n$. This lead us the the conclusion that if $u_n(x_1, x_2, \dots, x_n)$ is a probability function on $\mathbb{R}^n$ such that it is invariant under permutations, i.e.

$$u_n(x_{\sigma(1)}, \dots, x_{\sigma(n)}) = u_n(x_1, x_2, \dots, x_n), \quad \forall \sigma \in S_n,$$

then u_n defines naturally a point process.

The mapping $\mathcal{E} \ni A \mapsto \mathbb{E}[\#(\chi \cap A)]$, that assigns to the Borel set A the expected value of the number of points in itself under the configuration χ , is a measure on $\mathbb{R}$. The mean measure is defined by

$$\mu(A) = \mathbb{E}[\xi(A)] = \mathbb{E}[\#(\chi \cap A)], \quad A \in \mathcal{E}.$$

This μ is a measure on Λ in the standard sense that $\mu : \rightarrow [0, \infty]$ satisfies $\mu(\emptyset) = 0$ and, for disjoint $\Lambda_1, \Lambda_2, \dots$ in $\mathcal{E}$,

$$\mu\left(\bigcup_k \Lambda_k\right) = \sum_k \mu(\Lambda_k).$$

Assuming there exists a density $\rho_1 : \mathcal{E} \ni A \rightarrow \mathbb{R}_+$ with respect to the Lebesgue measure (**1-point correlation function**), there is also a way to define the mean measure as

$$\mu(A) = \int_A \rho_1(x) dx$$

This can be called the rate or intensity of ξ and is the infinitesimal probability of a point at x , i.e. $\mathbb{P}(\xi(dx) = 1) = \rho_1(x)dx$. In this case μ is written as $\mu(dx) = \rho_1(x)dx$. This mean measure shows up in expressions of expectations for functionals of ξ . More generally, the **k-th moment** μ_k of ξ , written

$$\mu_k(A_1 \times A_2 \times \dots \times A_k) = \mathbb{E}[\xi_1(A)\xi_2(A)\dots\xi_k(A)], \quad A_1, A_2, \dots, A_k \dots \in \mathcal{E},$$

can be expressed in respect to the **k-th correlation function** $\rho_k : A^k \rightarrow \mathbb{R}_+$ in the following way

$$\mu_k(A_1 \times A_2 \times \dots \times A_k) = \mathbb{E}\left[\prod_{j=1}^k (\#\chi \cap A_j)\right] = \int_{A_1} \dots \int_{A_k} \rho_k(x_1, \dots, x_k) dx_1 \dots dx_k. \quad (4.1.1)$$

That is, the expected number of k -tuples $\chi_k = \{x_1, x_2, \dots, x_k\} \in \chi^k$ such that $x_j \in A_j \forall j$. For example, if $u_n(x_1, x_2, \dots, x_n)$ is a probability density function on $\mathbb{R}^n$, invariant under permutation of coordinates, then we can build an n -point process on $\mathbb{R}$ with correlation functions

$$\rho_k(x_1, x_2, \dots, x_k) := \frac{n!}{(n-k)!} \int_{\mathbb{R}^{n-k}} u_n(x_1, x_2, \dots, x_n) dx_{k+1} \dots dx_n. \quad (4.1.2)$$

The mean measure also arises in expressions of expectations of functionals of ξ . For example, the Campbell formula that, for $E \in \mathcal{E}$, states that the integral

$$\int_E f(x) \xi(dx) = \sum_{k=1}^m f(x_k),$$

where $f : E \rightarrow \mathbb{R}_+$, has expectation

$$\mathbb{E}\left[\int f(x) \xi(dx)\right] = \int f \mu(dx)$$

if the integral $\int_E |f(x)| \mu(dx)$ is finite.

For simple points processes on $\mathbb{R}$ there is a direct way to understand correlation functions. Let $A_i = [x_i, x_i + \Delta x_i]$, $1 \leq i \leq k$, be disjoint intervals. If the Δx_i are small enough, then we should expect either one or zero particles in each interval A_i . Hence, the product $\xi(A_1) \dots \xi(A_k)$ is typically either 1 or 0 depending on whether we have one particle in each interval. From (4.1.1), we should expect

$$\rho_k(\lambda_1, \lambda_2, \dots, \lambda_k) = \lim_{\Delta x_i \rightarrow 0} \frac{\mathbb{P}(\text{one particle in each } A_i)}{\Delta x_1 \Delta x_2 \dots \Delta x_k}.$$

This means that $\rho_1(\lambda)$ is the density of particles at λ , but since we have many particles, the events of finding particles at their respective points are not disjoint events, even for distinct points. It follows from the definition that if we have a simple point process, then $\rho_k(x_1, x_2, \dots, x_k)$ is exactly the probability of finding particles at $x_1, x_2, \dots, x_n$.

Lets explore an important example for our work. Let $\mathcal{H}_N$ be the space of all $N \times N$ Hermitian matrices. This space can be identified naturally with $\mathbb{R}^{N^2}$. If μ_N is a probability measure on $\mathcal{H}_N$ and $\{\lambda_1(M), \dots, \lambda_N(M)\}$ denotes the set of eigenvalues of $M \in \mathcal{H}_N$, then

$$M \rightarrow \sum_{j=1}^N \delta_{\lambda_j(M)}$$

maps μ_N to a finite point process P on $\mathbb{R}$. If dM is the Lebesgue measure on $\mathcal{H}_N$, then

$$d\mu_N(M) = \frac{1}{Z_N} e^{-\text{Tr } M^2} dM$$

is a Gaussian probability measure on $\mathcal{H}_N$, called the *GUE*. It can be shown that for any symmetric, continuous function f on $\mathbb{R}^N$ with compact support

$$\int_{\mathcal{H}_N} f(\lambda_1(M), \dots, \lambda_N(M)) d\mu_N(M) = \int_{\mathbb{R}^N} f(x_1, x_2, \dots, x_N) u_N(x_1, x_2, \dots, x_N) d^N x$$

where

$$u_N(x_1, x_2, \dots, x_N) = \frac{1}{Z_N N!} \prod_{1 \leq i < j \leq N} (x_i - x_j)^2 \prod_{j=1}^N e^{-x_j^2}$$

is the induced eigenvalue measure on $\mathbb{R}^N$. Hence the point process P on $\mathbb{R}$ defined by this induced measure has correlation functions given by (4.1.2).

Most, if not all, of the processes discussed on this work will have an rightmost or last particle. We say that a point process ξ has a last particle if there is a $t \in \mathbb{R}$ such that $\xi(t, \infty) < \infty$. If $x_1 \leq x_2 \leq \dots \leq x_{n(\epsilon)}$ are finitely many particles in (t, ∞) , we define $x^* = x_{n(\epsilon)}$, the position of the last particle. The distribution $\mathbb{P}[x^* \leq t]$ is called the last particle distribution.

Proposition 4.1.2. *Take ξ a point process on $\mathbb{R}$, or a subset of, all whose correlation functions ρ_k exist, and assume that*

$$\sum_{n=0}^{\infty} \frac{1}{n!} \int_{(t, \infty)^n} \rho_n(x_1, x_2, \dots, x_n) d\mu_n(x) < \infty \quad (4.1.3)$$

for each $t \in \mathbb{R}$. then the process ξ has a last particle and

$$\mathbb{P}[x^* \leq t] = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \int_{(t, \infty)^n} \rho_n(x_1, x_2, \dots, x_n) d\mu_n(x). \quad (4.1.4)$$

Proof. We take $t < s$, then

$$\mathcal{P}[\text{no particle in } (t, s)] = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \int_{(t, s)^n} \rho_n(x_1, x_2, \dots, x_n) d\mu_n(x).$$

We see from (4.1.3) and the dominated convergence theorem that we can let $s \rightarrow \infty$ and obtain (4.1.4). $\square$

4.2 Determinantal Point Processes

Repulsive random processes are, in general, intractable, that is, they are usually difficult or even impossible to, for example, compute marginals and conditional probabilities. In these cases, their theoretical analysis is a major challenge. However, there is a class of processes, repulsive ones, that are an rare exception to this rule: Determinantal Point Processes (DDP), or Fermionic point processes in the physics literature, are random processes that have the property of being both repulsive and tractable [20]. This property comes from the fact that their correlation function can be expressed as a certain determinantal form. Let's make this precise.

Definition 4.2.1. Consider a point process P on a complete separable metric space Λ , with reference measure μ , all of whose correlation functions ρ_k exist. If there is a function $K : \Lambda \times \Lambda \rightarrow \mathbb{C}$ such that

$$\rho_k(x_1, x_2, \dots, x_k) = \det(K(x_i, x_j))_{i,j=1}^k$$

for all $x_1, x_2, \dots, x_k \in \Lambda$, $k \geq 1$, then we say that P is a **determinantal point process**. We call K the **correlation kernel** of the process.

Important examples of DDP's are constructed by using the following result

Theorem 4.2.2. (Construction for DDP [4]) Consider a kernel $K : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}_+$ with the following properties:

- Trace-class: $\text{Tr}(K) = \int_{\mathbb{R}} K(x, x) dx = n < +\infty$;
- Positivity: $\det[K(x_i, x_j)]_{i,j=1}^n$ is non-negative for every $x_1, x_2, \dots, x_n \in \mathbb{R}$;
- Reproducing property: $\forall x, y \in \mathbb{R}$

$$K(x, y) = \int_{\mathbb{R}} K(x, s) K(s, y) ds.$$

Then,

$$u_n(x_1, x_2, \dots, x_n) := \frac{1}{n!} \det[K(x_i, x_j)]_{i,j=1}^n$$

is a probability measure on $\mathbb{R}^n$, invariant under coordinated permutations. The associated point process is a determinantal point process with K as a correlation kernel.

All important information is contained in the correlation kernel for these processes and, therefore, all quantities of interest can be expressed in terms of the same K . The kernel K can be seen as an integral kernel of an operator K on $L^2(\Lambda, \lambda)$,

$$Kf(x) = \int_{\Lambda} K(x, y) f(y) d\lambda(y) \quad (4.2.1)$$

as long as the right hand side is well defined.

4.2.1 Gap probabilities

Consider a point process on $\mathbb{R}$ with correlation function ρ_k and let $A \in \mathcal{E}$ be a Borel set such that, with probability 1, there are only finitely many points in A (for example, $A \in \mathcal{B}$). Denote by $\rho_A(n)$ the probability that there are exactly n points in A . If there are n points in A , then the number of ordered k -tuples in A is $\binom{n}{k}$. Also, remember by (4.1.1), that

$$\mathbb{E} \left[\prod_{j=1}^k (\# \chi \cap A_j) \right] = \frac{1}{k!} \int_A \cdots \int_A \rho_k(x_1, \dots, x_k) dx_1 \cdots dx_k$$

should be the expected number of ordered k -tuples $(x_1, x_2, \dots, x_k)$ such that $x_1 < x_2 < \cdots < x_k$ and $x_j \in A$ for every $j = 1, \dots, k$. Therefore it must hold that

$$\frac{1}{k!} \int_{A^k} \rho_k(x_1, x_2, \dots, x_k) dx_1 \cdots dx_k = \sum_{n=k}^{\infty} \binom{n}{k} \rho_A(n).$$

Assume the following alternating series is absolutely convergent, then

$$\begin{aligned} \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \int_{A^k} \rho_k(x_1, x_2, \dots, x_k) dx_1 \cdots dx_k &= \sum_{k=0}^{\infty} \sum_{n=k}^{\infty} (-1)^k \binom{n}{k} \rho_A(n), \\ &= \sum_{n=0}^{\infty} \left(\sum_{k=0}^n (-1)^k \binom{n}{k} \right) \rho_A(n). \end{aligned}$$

On the other hand, $\sum_{k=0}^{\infty} (-1)^k \binom{n}{k}$ vanishes unless n is zero. In conclusion,

$$\rho_A(0) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \int_{A^k} \rho_k(x_1, x_2, \dots, x_k) dx_1 \cdots dx_k,$$

where we call $\rho_A(0)$ the **gap probability**, i.e. the probability of finding 0 particles in A . In particular, if the point process is determinantal, we have

$$\rho_A(0) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \int_{A^k} \det[K(x_i, x_j)]_{i,j=1}^k dx_1 \cdots dx_k.$$

Let $d(z) = \det(I + zA)$ be the Fredholm determinant given by

$$d(z) = \sum_{k=0}^{\infty} \frac{z^k}{k!} \int_{[a,b]^k} \det[K(x_i, x_j)]_{i,j=1}^k dx_1 \cdots dx_k.$$

then it is clear that $\rho_0(A)$ is the Fredholm determinant $d(-1) = \det(I - \mathbf{K}|_A)$ of the integral operator

$$\mathbf{K}|_A : L^2(A) \rightarrow L^2(A)$$

$$f(x) \mapsto \mathbf{K}[f](x) = \int_{\mathbb{R}} K(x, y) f(y) dy.$$

as defined in (4.2.1). A more general result can be posed as follows:

Theorem 4.2.3. [4] *Consider a determinantal point process with kernel K . For any finite Borel sets $A_j \subset \mathbb{R}, j = 1, \dots, n$, the generating function of the probability distribution of the occupation number $\#_{A_j} := \#\{x_i \in A_j\}$ is given by*

$$G(z_1, z_2, \dots, z_n) = \mathbb{E} \left(\prod_{i=1}^n z_i^{\#_{A_i}} \right) = \det \left(I + \sum_{j=1}^n (z_j - 1) K|_{A_j} \right)$$

From the properties of generating functions, i.e. $G(z) = \mathbb{E}(z^X) = \sum_{x=0}^{\infty} p(x) z^x$, it is true that

$$p(k) = \mathbb{P}(X = k) = \frac{G^{(k)}(0)}{k!}.$$

Then, for example, the probability of having k points in A is

$$\mathbb{P}(\chi \cap A = k) = \frac{G^{(k)}(0)}{k!} = \frac{1}{k!} \frac{d^k}{dz^k} \det(I + (z - 1)K|_A)_{z=0}.$$

4.2.2 Outermost Particle

Proposition 4.2.4. *Consider a determinantal point process ξ on a subset Λ of $\mathbb{R}$ with a hermitian correlation kernel $K(x, y)$, i.e., $K(y, x) = \overline{K(x, y)}$. Assume that $K(x, y)$ defines a trace class operator K on $L^2(t, \infty)$ for each $t \in \mathbb{R}$, and that*

$$\text{Tr } K = \int_t^{\infty} K(x, x) d\lambda(x) < \infty.$$

Then ξ has a last particle almost surely and

$$\mathcal{P}[\lambda^* \leq t] = \det(I - K)_{L^2(t, \infty)}$$

Proof. Since correlation functions are non negative and $K(y, x) = K(\bar{x}, y)$, the matrix $K(x_i, x_j)$ is positive definite. In thta case Hadamard's inequality says that

$$\det (K(x_i, x_j))_{1 \leq i, j \leq n} \leq \prod_{j=1}^n K(x_j, x_j).$$

Hence,

$$\sum_{n=0}^{\infty} \frac{C^n}{n!} \int_{(t, \infty)^n} \det (K(x_i, x_j))_{1 \leq i, j \leq n} d^n \lambda(x) \leq \sum_{n=0}^{\infty} \frac{C^n}{n!} \left(\int_t^{\infty} K(x, x) d\lambda(x) \right) \exp \left(C \int_t^{\infty} K(x, x) d\lambda(x) \right).$$

□

Consider for example the *Airy Kernel*,

$$A(x, y) = \int_0^{\infty} Ai(x+t) Ai(y+t) dt$$

This kernel is Hermitian and has

$$\int_t^{\infty} A(x, x) dx < \infty, \quad \forall t \in \mathbb{R}.$$

It can be shown that $A(x, y)_-(t, \infty)$ is the kernel of a trace class operator and there is a point process ξ on $\mathbb{R}$, the *Airy kernel point process*, which is determinantal with kernel $A(x, y)$. We have

$$F_{TW}(t) := \mathcal{P}[\lambda^*(\xi) \leq t] = \det(I - A)_{L^2(t, \infty)}.$$

The distribution function $F_{TW}(t)$ for the last particle in the Airy kernel point process is a natural scaling limit of certain finite determinantal point processes. We call it the *Tracy-Widow distribution*.

4.2.3 Biorthogonal Ensemble

Let $p_t(x, y)$ be the transition probability density from point x to point y at time t of a one-dimensional strong Markov process with continuous sample paths. The following infamous theorem give us a determinantal formula for the probability that a number of paths with given starting and ending positions fall in certain sets at some time without intersecting in the mean time.

Theorem 4.2.5. (Karlin, McGregor [8]) Consider n independent copies $X_1(t), \dots, X_n(t)$ of a one-dimensional strong Markov process with continuous sample paths, conditioned so that

$$X_j(0) = a_j, \quad a_1 < a_2 < \dots < a_n \in \mathbb{R}.$$

Let $p_t(x, y)$ be the transition probability function of the Markov process and let $E_1, \dots, E_n \subset \mathbb{R}$ be disjoint Borel sets (we assume $\sup E_j < \inf E_{j+1}$). Then,

$$\frac{1}{Z_n} \int_{E_1} \dots \int_{E_n} \det[p_t(a_i, x_j)]_{i,j=1}^n dx_1 \dots dx_n$$

is equal to the probability that each path X_j belongs to the set E_j at time t , without any intersection between paths in the time interval $[0, t]$, for some normalizing constant Z_n .

However, this is not a determinantal process, since the correlation functions are not expressible in terms of the determinant of a kernel. If we additionally condition the paths to end at time $T > 0$ at some given points $b_1 < \dots < b_n$, without any intersection between the paths along the whole time interval $[0, T]$, then it can be shown (using an argument again based on the Markov property) that the random positions of the n paths at a given time $t \in [0, T]$ have the joint probability density function

$$\frac{1}{Z_n} \det[p_t(a_i, x_j)]_{i,j=1}^n \det[p_{T-t}(x_i, b_j)]_{i,j=1}^n = \det[K_n(x_i, x_j)]_{i,j=1}^n \quad (4.2.2)$$

with a suitable normalizing constant $\mathcal{Z}_n$ and kernel

$$K_n(x, y) := \sum_{j=1}^n \phi_j(x) \psi_j(y),$$

$$\phi_j \in \text{span}\{p_t(a_i, x)\}, \quad \psi_k \in \text{span}\{p_{T-t}(x, b_i)\},$$

$$\int_{\mathbb{R}} \phi_j(x) \psi_k(x) dx = \delta_{jk}.$$

A interesting case is the confluent regime when two or more starting (or ending) points collapse together. For example, in the confluent limit as $a_j \rightarrow a$ and $b_j \rightarrow b$, for all j 's, applying L'Hopital rule to (4.2.2) gives

$$\frac{1}{\tilde{\mathcal{Z}}} \det \left[\frac{d^{i-1}}{da^{i-1}} p_t(a_i, x_j) \right]_{i,j=1}^n \det \left[\frac{d^{i-1}}{db^{i-1}} p_{T-t}(x_i, b_j) \right]_{i,j=1}^n$$

which is still a determinantal point process with kernel derived along the same method as before.

4.3 Limit and Transformations of Determinantal Point Processes

Let $f \in C^1(\mathbb{R})$ be a bijection, with continuously differentiable inverse $f^{-1} = g \in C^1(\mathbb{R})$. Then, f maps configurations on $\mathbb{R}$ to configuration on $\mathbb{R}$. The push-forward of a point process $\mathbb{P}$ under such mapping is $f(\mathbb{P})$ and it is again a point process.

Proposition 4.3.1. [4] *If K is the kernel of a DDP $\mathbb{P}$, then $f(\mathbb{P})$ is also a DDP with kernel*

$$\hat{K}(x, y) = \sqrt{g'(x)g'(y)} K(g(x), g(y)).$$

where $f^{-1} = g$.

In particular, if we consider a linear function $f(x) = \alpha(x - x^*)$, for $\alpha > 0$ and $x^* \in \mathbb{R}$, then the transformed point process $f(\mathbb{P})$ is a scaled and translated version of the original point process $\mathbb{P}$. The new correlation kernel is

$$\hat{K}(x, y) = \frac{1}{\alpha} K\left(x^* + \frac{x}{\alpha}, x^* + \frac{y}{\alpha}\right).$$

Others transformations are also possible. Consider what happens when one fixes $x^* \in \mathbb{R}$ and, given that $x^* \in \chi$, removes x^* from the configuration. The resulting process $\chi \setminus \{x^*\}$ is a new point process.

Proposition 4.3.2. [4] *Let K be the correlation kernel of a DDP and let x^* be a point such that $0 < K(x^*, x^*) < +\infty$. Then, the point process obtained by removing x^* is determinantal with kernel*

$$\hat{K}(x, y) = K(x, y) - \frac{K(x, x^*)K(x^*, y)}{K(x^*, x^*)}.$$

We now want to consider sequences of finite determinantal point processes $\mathbb{P}_n$ with correlation kernel K_n and understand how the limit behaves.

Proposition 4.3.3. [4] *Let $\mathbb{P}$ and $\mathbb{P}_n$ be determinantal point processes with kernels K and K_n respectively. Let K_n converge pointwise to K*

$$\lim_{n \rightarrow \infty} K_n(x, y) = K(x, y)$$

uniformly in x, y over compact subsets of $\mathbb{R}$. Then, the point processes $\mathbb{P}_n$ converge to $\mathbb{P}$ weakly.

A very important case studied here is of the limit of an scaled and centered determinantal point process. Suppose a sequence of kernels K_n and fixed point x^* of interest. We first perform a centering and rescaling of the form

$$x \mapsto Cn^\alpha(x - x^*)$$

for some suitable value of $C, \gamma > 0$. Then the scaled kernels have a limit

$$\lim_{n \rightarrow \infty} \frac{1}{Cn^\alpha} K_n \left(x^* + \frac{x}{Cn^\alpha}, x^* + \frac{y}{Cn^\alpha} \right) = K(x, y)$$

Therefore, the scaling limit K is a kernel that corresponds to a determinantal point process with an infinite number of points. Interestingly, in many situations, the same scaling limit K may appear. The phenomenon is known as **universality** in RMT. Examples of such kernels are the sine kernel

$$K_{\text{sin}} = \frac{\sin \pi(x - y)}{x - y}$$

and the Airy kernel

$$K_{\text{Ai}} = \frac{\text{Ai}(x)\text{Ai}'(y) - \text{Ai}'(x)\text{Ai}(y)}{x - y}$$

where Ai is the Airy function.

Chapter 5

Orthogonal Polynomials

Orthogonal Polynomials are of great importance for the theory of random matrices, as their properties allow us to perform operations that would otherwise be very convoluted - see (3.3.1) for example. We start our discussion with $\beta = 2$ - Hermite Polynomials when we use the Gaussian weight function, for its relatively simplicity.

5.1 Connection to RMT

Consider the weight $w(x) = e^{-V(x)}$ and the corresponding orthogonal polynomials

$$\int_{-\infty}^{\infty} P_n(x) P_m(x) e^{V(x)} dx = 0 \quad \Longleftrightarrow \quad m \neq n.$$

An important property of the weight function is its moments defined by

$$w_k = \int_{-\infty}^{\infty} x^k e^{-V(x)} dx, \quad k \in \mathbb{N}.$$

Using the moments, one can define the *Hankel determinant* D_n defined as

$$D_n = \begin{vmatrix} w_0 & w_1 & w_2 & \cdots & w_{n-1} \\ w_1 & w_2 & w_3 & \cdots & w_n \\ w_2 & w_3 & w_4 & \cdots & w_{n+1} \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ w_{n-1} & w_n & w_{n+1} & \cdots & w_{2n-2} \end{vmatrix},$$

to which there is a integral expression given by

$$D_n = \frac{1}{n!} \int_{\mathbb{R}^n} \prod_{i < j} (x_j - x_i)^2 \prod_{j=1}^n e^{-V(x)} dx_1 \cdots dx_n.$$

The intuition for this formula is that you can extract the integrals for each of the rows of the determinant and then the Vandermonde can be extracted from what remains. Interestingly, the monic polynomial $P_n(x) = x^n + \cdots$ is given by

$$P_n(x) = \frac{1}{D_n} \begin{vmatrix} w_0 & w_1 & w_2 & \cdots & w_n \\ w_1 & w_2 & w_3 & \cdots & w_{n+1} \\ w_2 & w_3 & w_4 & \cdots & w_{n+2} \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ w_{n-1} & w_n & w_{n+1} & \cdots & w_{2n-1} \\ 1 & x & x^2 & \cdots & x^n \end{vmatrix},$$

From the same intuition as before,

$$P_n(x) = \frac{1}{n! D_n} \int_{\mathbb{R}^n} \prod_{j=1}^n (x - x_j) \prod_{i < j} (x_j - x_i)^2 \prod_{j=1}^n e^{-V(x)} dx_1 \cdots dx_n.$$

which can be identified as the expected characteristic polynomial over the density we began with. This is a way to show that the orthogonal polynomial is the expected characteristic polynomial.

5.2 Hermite Polynomials

Consider $q_n^{(2)}$ when $w^{(G)}(x) := (\sqrt{2\pi})^{-1} e^{-\frac{1}{2}x^2}$. We make standard the notation

$$\text{He}_n(x) := q_n^{(2, w^{(G)})}.$$

This means that *Hermite Polynomials* are such that

$$\int_{-\infty}^{\infty} \text{He}_n(x) \text{He}_m(x) w^{(G)}(x) dx = \begin{cases} 0, & \text{if } n \neq m \\ n!, & \text{if } n = m. \end{cases}$$

There are some other representation for the polynomial - one of the most common is their derivative formula

$$\text{He}_n(x) = (-1)^{(n)} e^{\frac{1}{2}x^2} \frac{d}{dx^n} e^{-\frac{1}{2}x^2}$$

the recursion formula

$$\text{He}_{n+1}(x) = x \text{He}_n(x) - n \text{He}_{n-1}(x).$$

or even the differential equation

$$\text{He}_n''(x) - 2x \text{He}_n'(x) = -2n \text{He}_n(x)$$

One can also write the explicit representation (or *Standard*),

$$\text{He}_n(x) = n! \sum_{l=0}^{\lfloor \frac{n}{2} \rfloor} (-1)^l \frac{(x)^{n-2l}}{2^l (n-2l)! l!}.$$

This is all good, but can be a little inconvenient when doing computation. A nice representation is presented by [12]. The Gaussian integral formula for the Hermite Polynomials. This development will be based on this reference.

Definition 5.2.1. *The Gaussian integral Formula, or equivalently the Gaussian Expectation formula, for the Hermite polynomials He_n is given by*

$$\text{He}_n(x) = \mathbb{E}_{Z \sim \mathcal{N}(0,1)} [(x + iZ)^n] = \int_{-\infty}^{\infty} (x + iz)^n e^{-\frac{1}{2}z^2} dz$$

This deeply simplifies some computation. For example, note that someone can easily compute the identities

$$\text{He}_n(0) = \mathbb{E}[(0 + iZ)^n] = \begin{cases} (-1)^{n/2} (n-1)!!, & \text{for } n \text{ even} \\ 0, & \text{for } n \text{ odd} \end{cases},$$

$$\text{He}_n(x + y) = \mathbb{E}[(x + y + iZ)^n] = \mathbb{E} \left[\sum_{k=0}^n \binom{n}{k} y^k (x + iZ)^{n-k} \right] = \sum_{k=0}^n \binom{n}{k} y^k \text{He}_{n-k}(x).$$

And with that one can get a explicit expression for the polynomial setting $x = 0$ in the second expression and using the first one to calculate the values.

5.2.1 Average characteristic polynomial

Hermite Polynomials are the average characteristic polynomials of the Gaussian Ensemble. That is, if $\hat{M}$ is an order n matrix which is hermitian and with joint density given by

$$\frac{1}{\mathcal{Z}} e^{-\text{Tr } \hat{M}^2} dM$$

then the average characteristic polynomial

$$P_n(x) = \mathbb{E} \left[\det \left(x \hat{I}_n - \hat{M}_n \right) \right]$$

satisfies

$$P_n(x) = xP_{n-1}(x) - \frac{n-1}{2}P_{n-2}(x).$$

If one notices that $P_0(x) = 1$ and $P_1(x) = x$, then one can also see that the recurrence equation give us the Hermite Polynomials.

Proof. We begin by calculating de characteristic determinant

$$\begin{aligned} \det \left(\lambda \hat{I}_n - \hat{M}_n \right) &= \begin{vmatrix} \lambda - m_{1,1} & -m_{1,2} & \cdots & -m_{1,n} \\ -m_{2,1} & \lambda - m_{2,2} & \cdots & -m_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ -m_{n,1} & -m_{n,2} & \cdots & \lambda - m_{n,n} \end{vmatrix} \\ &= (\lambda - m_{n,n}) \det \left(\lambda \hat{I}_{n-1} - \hat{M}_{n-1} \right) + \sum_{k=1}^{n-1} (-1)^{n+k} (-m_{n,k}) \det \left(\lambda \hat{I}_n - \hat{M}_n \right)_{[n,k]} \end{aligned}$$

where

$$\det \left(\lambda \hat{I}_n - \hat{M}_n \right)_{[n,k]}$$

is the determinant of the matrix obtained from $\lambda \hat{I}_n - \hat{M}_n$ by deleting row n and column k . Moreover,

$$\det \left(\lambda \hat{I}_n - \hat{M}_n \right)_{[n,n]} = \det \left(\lambda \hat{I}_{n-1} - \hat{M}_{n-1} \right).$$

Where we still have some dependence between term in the summation because of the index n . To solve for this, once more expand in cofactors,

$$\begin{aligned} &= (\lambda - m_{n,n}) \det \left(\lambda \hat{I}_{n-1} - \hat{M}_{n-1} \right) + \sum_{k=1}^{n-1} (-1)^{n+k} (-m_{n,k}) \det \left(\lambda \hat{I}_n - \hat{M}_n \right)_{[n,k]} \\ &= (\lambda - m_{n,n}) \det \left(\lambda \hat{I}_{n-1} - \hat{M}_{n-1} \right) + \sum_{k=1}^{n-1} \sum_{j=1}^{n-1} (-1)^{n+k} (-1)^{n-1+j} (m_{n,k} m_{j,n}) \det \left(\lambda \hat{I}_n - \hat{M}_n \right)_{[n,k],[j,n]}. \end{aligned}$$

Finally, because

$$\begin{aligned} \mathbb{E} [\lambda - m_{n,n}] &= \lambda, \\ \mathbb{E} \left[\det \left(\lambda \hat{I}_n - \hat{M}_n \right) \right] &= P_{n-1}, \\ \mathbb{E} [m_{n,k} m_{j,n}] &= \begin{cases} 0 & \text{if } j \neq k \\ \mathbb{E} [\Re(m_{n,n})^2 + \Im(m_{n,n})^2] & \text{if } j = k \end{cases} \\ \mathbb{E} \left[\det \left(\lambda \hat{I}_n - \hat{M}_n \right)_{[n,k],[j,n]} \right] &= P_{n-2} \end{aligned}$$

then we get the recurrence expression

$$P_n(x) = xP_{n-1} - \frac{n-1}{2}P_{n-2}$$

which give rise to the Hermite polynomials. $\square$

This is also true for other ensembles and give rise to Laguerre and Jacobi polynomials. Moreover, using Weyls formula, one can say that

$$\mathbb{E} \left[\det \left(x \hat{I}_n - \hat{M} \right) \right] = \frac{1}{Z_n} \int_{\mathbb{R}^n} \prod_{j=1}^n (x - x_j) \prod_{i < j} (x_j - x_i)^2 \prod_{j=1}^n e^{-V(x)} dx_1 \cdots dx_n.$$

5.2.2 Arbitrary Variance

We can use this formula to work with Hermite Polynomials of arbitrary variance. Assume one adds a temporal dependence parameter t . Denote this polynomial $\tilde{H}_n(x; t)$ defined as

Definition 5.2.2.

$$\tilde{H}_n(x; t) := \mathbb{E}_{B_t \sim \mathcal{N}(0, t)} [(x + iB_t)^n] = \mathbb{E}_{Z \sim \mathcal{N}(0, 1)} [(x + i\sqrt{t}Z)^n]$$

The notation has the tilde because historically there is a preference for the derivative formula instead of the integral formula. The notation H_n is reserved for the dual of this polynomial, defined as below

Definition 5.2.3.

$$H_n(x; t) := \mathbb{E}_{Z \sim \mathcal{N}(0, \frac{1}{t})} \left[\left(\frac{x}{t} + iZ \right)^n \right] = \mathbb{E}_{Z \sim \mathcal{N}(0, 1)} \left[\left(\frac{x}{t} + i \frac{Z}{\sqrt{t}} \right)^n \right].$$

Which is dual in the sense that

$$\int_{-\infty}^{\infty} \tilde{H}_n(x; t) H_m(x; t) (\sqrt{2\pi t})^{-1} e^{-\frac{1}{2} \frac{x^2}{t}} dx = \begin{cases} 0, & \text{if } n \neq m \\ n!, & \text{if } n = m. \end{cases}$$

Finally, note that

$$\tilde{H}_n(x; t) = t^{n/2} \tilde{H} \left(\frac{x}{\sqrt{t}}; 1 \right) = t^{n/2} \text{He} \left(\frac{x}{\sqrt{t}} \right), \quad H_n(x; t) = t^{-n/2} \tilde{H} \left(\frac{x}{\sqrt{t}}; 1 \right) = t^{-n/2} \text{He} \left(\frac{x}{\sqrt{t}} \right).$$

5.2.3 Combinatorial Arguments

Our attention will be on the Asymptotic of the Hermite Polynomials, to get there, we will work our way with some combinatorial arguments. We start with the *Wick Formula/Isserlis' Theorem* for Gaussian expectations.

Theorem 5.2.4. (*Wick Formula/Isserlis' Theorem*) If $Z_1, Z_2, \dots, Z_n$ are a collection of mean zero, jointly Gaussian random variables, then the expectation of the product $Z_1 \cdots Z_n$ can be written in terms of the individual correlations $\mathbb{E}[Z_a Z_b]$ in the following way

$$\mathbb{E}[Z_1 \cdots Z_n] = \sum_{\pi \in P(\{1, \dots, n\})} \prod_{\{a, b\} \in \pi} \mathbb{E}[Z_a Z_b]$$

where $P(\{1, \dots, n\})$ are the set of pairings (or pair partitions) on the set $S = \{1, \dots, n\}$. That is, $P(S)$ denotes the way of dividing S into disjoint pairs.

If one applies this result for example, for $\mathbb{E}[Z^n]$, the expectation of a n -repeated Gaussian Variable, this result gives the moments of the normal distribution

$$\mathbb{E}[Z^n] = |P(\{1, \dots, n\})| \cdot \mathbb{E}[Z^2]^{n/2} = \begin{cases} (n-1)!! \cdot \text{Var}[Z]^{n/2}, & \text{if } n \text{ is even,} \\ 0, & \text{if } n \text{ is odd.} \end{cases}$$

If we want to apply this combinatorial argument for the polynomials $\tilde{H}_n(x; t) = \mathbb{E}_{Z \sim \mathcal{N}(0, 1)} [(x + i\sqrt{t}Z)^n]$, we must expand the product in one of the terms on each of the brackets - which yields a sum over 2^n monomials. Then, taking the expectation over Z amounts to adding up the pairings for each term as in 5.2.4. The number of pairings depends on the number of times m that $i\sqrt{t}Z$ was chosen: there are $(m-1)!!$ such pairings if m is even and 0 pairings if m is odd. Therefore, the only contributions come from the terms with even order.

When exactly m factors $i\sqrt{t}Z$ are chosen, there must be $n-m$ copies of x chosen, and so the resulting contribution of the term is $(i\sqrt{t})^m x^{n-m}$. Since m is even we will actually have $(-t)^{m/2} x^{n-m}$. We can count $\binom{n}{m} (m-1)!!$ such

terms - m choices of the term and the subsequent $(m-1)!!$ ways of pairing it up. This can be formalized as

$$\begin{aligned}
\tilde{H}_n(x; t) &= \sum_{S \subset \{1, \dots, n\}, \pi \in P(S)} (i\sqrt{t})^{|\pi|} x^{n-|S|} \\
&= \sum_{\{m=0 \mid m \text{ is even}\}} \binom{n}{m} (m-1)!! (-t)^{m/2} x^{n-m} \\
&= \sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} \binom{n}{2k} (2k-1)!! (-t)^k x^{n-2k} \\
&= x^n - \binom{n}{2} 1!! \cdot t x^{n-2} + \binom{n}{4} 3!! \cdot t^2 x^{n-4} - \dots
\end{aligned}$$

This property can also be used to prove the orthogonality and recursion of the polynomials - for the proof, check [12].

5.2.4 Asymptotic - $x = \mathcal{O}(n)$

We can study the asymptotic of this polynomial for $x = \mathcal{O}(n)$ only using the combinatorial argument given in this chapter.

Proposition 5.2.5. *Let $c \in \mathbb{R}$ be fixed, and set $x = cn$. We have the following asymptotic formula as $n \rightarrow \infty$:*

$$\text{He}_n(cn) = (cn)^n e^{-\frac{1}{2c^2}} (1 + o(1))$$

that is,

$$\text{He}_n(cn) \approx x^n e^{-\frac{n^2}{2x^2}}$$

for $x = \mathcal{O}(n)$.

Proof. Using the integral formula 5.2.1 we can write

$$\text{He}_n(cn) = \mathbb{E}[(cn + iZ)^n] = (cn)^n \mathbb{E} \left[\left(1 + i \frac{Z}{cn} \right)^n \right].$$

Now, note that

$$\lim_{n \rightarrow \infty} \left(1 + i \frac{Z}{cn} \right)^n = e^{i \frac{Z}{c}}$$

almost surely in Z , and in light of the expectation $\mathbb{E}[e^{\beta Z}] = e^{\frac{1}{2}\beta^2}$ we find that

$$\mathbb{E} \left[\left(1 + i \frac{Z}{cn} \right)^n \right] = e^{-\frac{1}{2c^2}} (1 + o(1))$$

from where the results follows. □

Other regimes for the asymptotic will have to use some more sophisticated techniques.

5.2.5 Integral Representation

Follow [22, Proposition 2.1]. A standard method of asymptotic of orthogonal polynomials is the so called *Deift-Zhou* steepest descent method, which considers to asymptotically solve a Riemann-Hilbert problem associated to the orthogonal polynomials. However, the asymptotic may be simpler to compute if the polynomial has an integral representation by direct inspection. The Hermite polynomials have such representation.

Proposition 5.2.6. *The Hermite polynomial $\text{He}_n(x)$ has the following ntegral representation*

$$\text{He}_n(x) = \frac{n!}{2\pi i} \int_{|t|=1} \frac{e^{tx - \frac{t^2}{2}}}{t^{n+1}} dt$$

Proof. We will explicitly arrive at the *Standart* notation for the polynomial using the integral representation

$$\text{He}_n(x) = \frac{n!}{2\pi i} \int_{|t|=1} \frac{e^{tx - \frac{t^2}{2}}}{t^{n+1}} dt$$

Take the Taylor expansion for the exponential

$$\begin{aligned} e^{tx} &= \sum_{i=0}^{\infty} \frac{(x)^i}{i!} t^i \\ e^{-\frac{t^2}{2}} &= \sum_{j=0}^{\infty} \frac{(-1)^j}{2^j j!} t^{2j} \end{aligned}$$

and explicitly substitute the expressions into the integral form

$$\text{He}_n(x) = \frac{n!}{2\pi i} \sum_{i=0}^{\infty} \frac{(x)^i}{i!} \sum_{j=0}^{\infty} \frac{(-1)^j}{2^j j!} \int_{|t|=1} t^{-n-1+i+2j} dt$$

and note that this integral is valued zero except when $2j - (i - n) = 0$. Then, we can write, for every even $i - n$,

$$\begin{aligned} \text{He}_n(x) &= n! \sum_{i=0 | i-n \text{ even}}^{\infty} (-1)^{\frac{n-i}{2}} \frac{x^i}{2^{\frac{n-i}{2}} i! \left(\frac{n-i}{2}\right)!} \\ &= n! \sum_{l=0}^{\lfloor \frac{n}{2} \rfloor} (-1)^l \frac{x^{n-2l}}{2^l l! (n-2l)!} \end{aligned}$$

which is exactly the expression given by 5.2. □

Proposition 5.2.7. *I believe there is a connection between the two integral representations if we start with the Gaussian expectation formula, expand the binomial and rewrite the moments with the Cauchy formula, as its done to derive this representation starting from the derivative formula. Missing proof.*

5.3 Multiple Orthogonal Polynomials

Let $\mu_1, \mu_2, \dots, \mu_r$ be r positive measures on the real line and $\vec{n} = (n_1, n_2, \dots, n_r)$ in $\mathbb{N}^r$ with length $|\vec{n}| = n_1 + \dots + n_r$.

Definition 5.3.1. *The type I multiple orthogonal polynomials $(A_{\vec{n},1}, \dots, A_{\vec{n},r})$ for the multi-index $\vec{n} = (n_1, n_2, \dots, n_r)$ are such that $\deg(A_{\vec{n},j}) \leq n_j - 1$ and*

$$\sum_{j=1}^r \int x^k A_{\vec{n},j}(x) d\mu_j(x) = 0, \quad 0 \leq k \leq |\vec{n}| - 2,$$

with normalization

$$\sum_{j=1}^r \int x^{|\vec{n}|-1} A_{\vec{n},j}(x) d\mu_j(x) = 1.$$

Or also

Definition 5.3.2. *The type II multiple orthogonal polynomials $P_{\vec{n}}$ for the multi-index $\vec{n} = (n_1, n_2, \dots, n_r)$ is the monic polynomial of degree $|\vec{n}|$ that satisfies*

$$\int x^k P_{\vec{n}} d\mu_j(x) = 0, \quad 0 \leq k \leq n_j - 1,$$

for $1 \leq j \leq r$.

For the type I polynomials, take $\mu = \mu_1 + \dots + \mu_r$, and put $w_j(x) = d\mu_j(x)/d\mu(x)$ and set

$$Q_{\vec{n}}(x) = \sum_{j=1}^r A_{\vec{n},j}(x) w_j(x)$$

such that we can set

$$\int P_{\vec{n}}(x) Q_{\vec{m}}(x) d\mu(x) = \begin{cases} 0, & \text{if } \vec{m} \leq \vec{n}, \\ 0, & \text{if } |\vec{m}| > |\vec{n}| + 1, \\ 1, & \text{if } |\vec{m}| = |\vec{n}| + 1, \end{cases}$$

The question that remains is if there is such a thing as a determinantal point process happening here. Consider the Kernel

$$K_{\vec{n}}(x, y) = \sum_{k=0}^{|\vec{n}|-1} P_{\vec{n}_k}(x) Q_{\vec{n}_{k+1}}(y)$$

where $(\vec{n}_k)_{0 \leq k \leq |\vec{n}|}$ is a path in $\mathbb{N}^r$ from which $\vec{n}_0 = \vec{0}$, $\vec{n}_{|\vec{n}|} = \vec{n}$ and $\vec{n}_{k+1} - \vec{n}_k = \vec{e}_i = (0, \dots, \delta_i, \dots, 0)$ for some $1 \leq i \leq r$. The point process with joint density

$$P(x_1, x_2, \dots, x_N) = \frac{1}{N!} (K_{\vec{n}}(x_i, x_j))_{i,j=1}^N, \quad N = |\vec{n}|$$

defines a *multiple orthogonal polynomial ensemble* whenever this density is non-negative for every $(x_1, x_2, \dots, x_N) \in \mathbb{R}^N$. The self-reproducing property is also valid in this case, the problem is to know under which conditions over $\mu_1, \mu_2, \dots, \mu_r$ we have $K_{\vec{n}}(x, x) \geq 0$. Similarly for the limit on the multi-index of the Kernel. Some cool open problems can be discussed here.

5.3.1 Multiple Hermite Polynomials

If $c_1, c_2, \dots, c_r$ are the eigenvalues of $\hat{A}$ with multiplicities $n_1, n_2, \dots, n_r$, then

$$\int_{-\infty}^{\infty} \text{He}_{\vec{n}}(x) x^k e^{-x^2 + c_j x} dx = 0, \quad 0 \leq k \leq n_j - 1$$

for $1 \leq j \leq r$. The work done by Bleher was mostly on $r = 2$.

Chapter 6

Asymptotic

We have the sense in the previous chapters that we want to study some distributions when our matrices are large enough - be that for feasibility of computations or approximation of infinite dimension operators. This is equivalent, in some sense, to study the regime of large n for some orthogonal polynomials previously discussed. For some of those, a integral representation is available, allowing us to use asymptotic methods on integrals to solve for the large n behavior.

6.1 Quadrature

We first start briefly introducing a useful tool in the development of the formula we wish to use: the Bernoulli numbers B_n , polynomials $B_n(x)$ and periodic polynomials $\tilde{B}_n(x)$. The Bernoulli numbers B_n are a sequence of rational numbers that frequently appears in the Taylor series expansion of trigonometric-type functions. These number can also be seen as special values obtained by the Bernoulli polynomials $B_n(x)$. This is because these polynomials can be written explicitly as

$$B_k(x) = \sum_{n=0}^k \binom{k}{n} B_n x^{k-n},$$

then, we must have that $B_k(0) = B_k$. More interestingly for us is the fact that these polynomials satisfy the differential equation

$$B'_n(x) = nB_{n-1}(x), \quad n = 1, 2, \dots \quad (6.1.1)$$

Finally, the periodic Bernoulli polynomial is simply the Bernoulli polynomials evaluated at the fractional part of the argument, i.e., $\tilde{B}_n(x) = B_n(x - \lfloor x \rfloor)$. Now, let B_n and $B_n(x)$ denote the n th Bernoulli number and polynomial, respectively, and $\tilde{B}_n(x)$ the n th Bernoulli periodic function $B_n(x - \lfloor x \rfloor)$. Assume that a, m and n are integers such that $n > a$, $m > 0$, and $f^{(2m)}(x)$ is absolutely integrable over $[a, n]$. Then [13], Let's start by

$$\sum_{j=a}^n f(j) = \int_a^n f(x)dx + \frac{1}{2} (f(a) + f(n)) + \sum_{s=1}^{m-1} \frac{B_{2s}}{(2s)!} \left(f^{(2s-1)}(n) - f^{(2s-1)}(a) \right) + \int_a^n \frac{B_{2m} - \tilde{B}_{2m}(x)}{(2m)!} f^{(2m)}(x)dx.$$

This formula is called the *Euler-Maclaurin* formula. This equation can be extremely useful when estimating the asymptotic of the summation on the function f when its integral can be more easily evaluate. In essence, this is a quadrature technique; a numerical integration: the approximation of an integral $\int f(x)dx$ of some function f by a discrete summation $\sum w_i f(x_i)$ over a set of points x_i with some weight function w_i . For example, in the classical Riemann sums, one approximates the integral $\int f(x)dx$ over some interval $[a, b]$ with the limit of the summation $\sum f(x_i^*)\Delta x_i$, where $x_i^* \in [x_{i-1}, x_i]$ and $\Delta x_i = x_i - x_{i-1}$. Different methods can be generated by choosing x_i^* differently.

The trapezoidal rule, for example, is a Riemann sum where x_i^* is chosen to be the mean point $x_i^* = (x_{i-1} + x_i)/2$.

In this way, we would estimate the area using trapezoids, as the name suggests. Then, one could write

$$\begin{aligned}\int_a^n f(x)dx &\approx \sum_{j=0}^n f(x_j^*) = \sum_{j=0}^n \frac{f(x_i + x_{i-1})}{2}, \\ &= \sum_{j=1}^{n-1} f(x_j^*) + \frac{1}{2} (f(n) + f(0)) = \sum_{j=0}^{n-1} f(x_j^*) + \frac{1}{2} (f(n) - f(n)).\end{aligned}$$

Of course, this has some error term. Suppose that for some (x_i, x_{i-1}) we have the same k -derivatives at both extreme points for every $k \geq 1$, i.e., the function is constant. Then, there is no error in the approximation. If however, the two extreme points differ only on the k th derivative, there is some curvature defined by this difference where the error comes from. Therefore, there is indication of some error function given in respect to higher order derivatives.

Following the computations in [1], take a function $f \in C^p(\mathbb{R})$ for $p \geq 1$. To derive the Euler-Maclaurin for the interval $[a, b]$, let $h = (b - a)/n$ and $c = a + jh$, then we can express

$$\int_a^b f(x)dx = \sum_{j=0}^{n-1} \int_{a+jh}^{a+(j+1)h} f(x)dx$$

where we can use the property (6.1.1) and integration by parts to rewrite each integral as

$$\begin{aligned}\int_c^{c+h} f(x)dx &= h \int_0^1 B_0(1-x)f(c+xh)dx \\ &= -h \frac{B_1(1-x)}{1!} f(c+xh)|_0^1 + h^2 \int_0^1 \frac{B_1(1-x)}{1!} f'(c+xh)dx, \\ &\vdots \\ &= -\sum_{r=0}^{p-1} \frac{h^{r+1} B_{r+1}(1-x)}{(r+1)!} f^{(r)}(c+xh)|_0^1 + h^{p+1} \int_0^1 \frac{h^{r+1} B_p(1-x)}{(p)!} f^{(p)}(c+xh)dx, \\ &= h \frac{f(c) + f(c+h)}{2} - \sum_{r=0}^{p-1} \frac{h^{r+1} B_{r+1}(1-x)}{(r+1)!} [f^{(r)}(c+h) - f^{(r)}(c)] + h^{p+1} \int_0^1 \frac{B_p(1-x)}{(p)!} f^{(p)}(c+xh)dx.\end{aligned}$$

where one can start to notice the trapezoidal rule creeping out on the first term. Notice that it was necessary to use that $-B_1(0) = B_1(1) = 1/2$ and $B_n(0) = B_n(1) = B_n$ for $n \neq 1$. Returning to the summation, we can rearrange terms as

$$\begin{aligned}h \left[\frac{f(a)}{2} + f(a+h) + \dots + f(b-h) + \frac{f(b)}{2} \right] &= \frac{1}{2} (f(b) - f(a)) + \sum_{j=0}^{n-1} f(a+jh), \\ &= \int_a^b f(x)dx + \sum_{r=0}^{p-1} \frac{h^{r+1} B_{r+1}(1-x)}{(r+1)!} [f^{(r)}(b) - f^{(r)}(a)] - h^{p+1} \int_0^1 \frac{B_p(1-x)}{(p)!} f^{(p)}(c+xh)dx.\end{aligned}$$

This is the Euler-Maclaurin formula, somehow an extension of the trapezoidal rule including the higher order terms.

6.2 Asymptotics on Integrals

We will try to explore the asymptotic techniques on a specific type of function, as $\lambda \rightarrow \infty$, given by

$$\phi(\lambda) := \int_{\Gamma} g(s) e^{-\lambda f(s)} ds,$$

where Γ is a contour in the plane and f, g are given **suitable** functions. Suitable here, means that we impose g and f to be continuous (or analytic), f to have global maximum with zero derivative (maybe in more than one point) and strictly positive second derivative at this point(s) of maximum. There is also a more technical regularity condition on f to ensure the approximations holds.

There is some motivations for such.

- Airy Equation $\psi''(x) = x\psi(x)$: Taking the Fourier transform we have

$$-\omega^2 \Phi(x) = i\Phi'(x)$$

for which we can solve for $\Phi(x)$ and then take the inverse Fourier Transform to get

$$\phi(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i/3\omega^3 + ix\omega} d\omega.$$

The imaginary part of the argument is an odd function and therefore zero when integrated. The rest is expressed

$$\phi(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \cos\left(\frac{\omega^3}{3} + x\omega\right) d\omega.$$

But how to know if this is the only solution? One can prove that for a more general definition of a Transform

$$\tilde{F}f(w) := \frac{1}{\sqrt{2\pi i}} \int_{\Gamma} e^{-i\omega s} f(s) ds$$

holds very similar properties of the Fourier Transformation and solves the equation for any contour that gives a convergent integral. We can actually choose independent contour such that each give us a different independent solution to the equation.

6.3 Laplace Method

All approximation in this chapter will use in some way the Lemma of Laplace, or the Laplace Method for integrals of the form

$$\Phi(\lambda) := \int_a^b g(x) e^{-\lambda\phi(x)} dx \quad (6.3.1)$$

We first state a non-precise version of this theorem and discuss some ideas surrounding it.

Theorem 6.3.1. *Let Φ be as in (6.3.1), with $g(x)$ a continuous function and $f(x)$ assuming a global maximum at $x^* \in (a, b)$ with $f'(x^*) = 0$ and $f''(x^*) > 0$. Then, under some regularity conditions, the function Φ satisfies an estimate of the form*

$$\Phi(\lambda) = e^{-\lambda\phi(x^*)} \frac{\sqrt{2\pi}g(x^*)}{\sqrt{|\phi''(x^*)|\lambda^{1/2}}} (1 + o(1))$$

as $\lambda \rightarrow \infty$.

The idea of the proof is quite more revealing than the proper result. We abstain for now from giving a proper formal proof and instead give a general sketch that should be enough to illustrate the important point for this section.

Proof. (Sketch of Proof) The main idea is to approximate ϕ around its global maximum by Taylor's theorem

$$\phi(x) = \phi(x^*) - \frac{1}{2}|\phi''(x^*)|(x - x^*)^2 + \epsilon_2(x - x^*)$$

and then perform a change of variables $\frac{s}{\sqrt{\lambda}} = x - x^*$, so that

$$\int_{x^* - \frac{c}{\sqrt{\lambda}}}^{x^* + \frac{c}{\sqrt{\lambda}}} e^{-\lambda\phi(x)} d(x) = e^{-\lambda\phi(x^*)} \int_{-c}^c e^{\frac{1}{2}|\phi''(x^*)|t^2 - \lambda\epsilon_2\left(\frac{t}{\sqrt{\lambda}}\right)} \frac{dt}{\sqrt{\lambda}}.$$

Now, using the technical assumption of regularity on the function ϕ , one can argue that the error term ϵ_2 goes to zero sufficiently fast, leaving a Gaussian Integral. More that that, one need to argue that the integral outside this small neighborhood of x^* is exponentially smaller than $e^{\lambda\phi(x^*)}$ and use continuity of g to include it in the argument. $\square$

Some important points to highlight:

1. More precise error estimates can be obtained if one uses higher order terms of the Taylor Series expansion of ϕ .
2. This argument and result could also be used on curves on the plane. This is clear from the fact that every (smooth) curve is locally a straight line. This will lead us to the Method of *Steepest Descent*.
3. For maximizer x^* which have null second derivative (a double critical point), the Laplace Approximation can be used expanding to the third order terms of the Taylor Series $\phi(x) = \phi(x^*) = \frac{1}{6}\phi'''(x^*)(x-x^*)^3 + \epsilon_3(x-x^*)$ and making the change of variables $\frac{t}{\sqrt[3]{n}} = x - x^*$ instead. This is precisely one of the cases in the expansion for the Hermite Polynomial.

These next subsections are in the spirit of proving a bit more formally the expansion given by the Laplace Method.

6.3.1 Watson's Lemma

We start this study in the real line by looking at the functions $F : \mathbb{R} \rightarrow \mathbb{R}$

$$F(\lambda) = \int_0^T f(t) e^{-\lambda t} dt$$

which can be identified as the Laplace transform of a function f supported on $[0, T]$.

Theorem 6.3.2. (*Watson's Lemma*) Take $f \in L^1(0, T)$ and suppose that for some $\delta > 0$ it is of the form

$$f(t) = t^\alpha f_0(t), \quad 0 < t \leq \delta,$$

where $\alpha \in (-1, \infty)$ and f_0 is continuously differentiable on $[0, \delta]$ with $f_0(0) \neq 0$. Then the function F in 6.3.1 satisfies

$$F(\lambda) = \frac{f_0(0)\Gamma(\alpha+1)}{\lambda^{\alpha+1}}(1 + \mathcal{O}(\lambda^{-1}))$$

when $\lambda \rightarrow \infty$ and where Γ is the Gamma function.

Proof. A weak estimative of the integral values would be given by

$$\int_0^T f(t) e^{-\lambda t} dt \leq \|f\|_{L^1}$$

where we only use the fact that $f \in L^1(0, T)$. However, we have more information on the behavior of the function in the section $0 < t < \delta$. To use this information we start by decomposing the function in two parts

$$F(\lambda) = (1) + (2) = \int_0^\delta f(t) e^{-\lambda t} dt + \int_\delta^T f(t) e^{-\lambda t} dt.$$

For (2), we can use the weak estimative, leaving us with

$$F(\lambda) \leq \int_0^\delta f(t) e^{-\lambda t} dt + \|f\|_{L^1} e^{-\delta\lambda}.$$

Now, we remind of the condition of continuous differentiability and the hypothesis on $f(t)$ to write (1) as

$$(1) = \int_0^\delta t^\alpha (f_0(0) + r(t)) e^{-\lambda t} dt = f_0(0) \int_0^\delta t^\alpha e^{-\lambda t} dt + \int_0^\delta t^\alpha r(t) e^{-\lambda t} dt = f_0(0) \cdot (1.1) + (1.2)$$

For these two integrals we can try to simplify the problem by rewriting

$$(1.1) = \int_0^\infty t^\alpha e^{-\lambda t} dt - \int_\delta^\infty t^\alpha e^{-\lambda t} dt = \frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} - \int_\delta^\infty t^\alpha e^{-\lambda t} dt$$

$$|(1.2)| \leq \|f'\|_\infty \int_0^\delta t^{\alpha+1} e^{-\lambda t} dt = \|f'\|_\infty \left(\int_0^\infty t^{\alpha+1} e^{-\lambda t} dt - \int_\delta^\infty t^{\alpha+1} e^{-\lambda t} dt \right) = \|f'\|_\infty \left(\frac{\Gamma(\alpha+2)}{\lambda^{\alpha+2}} - \int_\delta^\infty t^{\alpha+1} e^{-\lambda t} dt \right).$$

We are now only missing one ingredient to get the result, the order of

$$G(\lambda, \beta) := \int_\delta^\infty t^\beta e^{-\lambda t} dt, \quad \lambda > 1, \quad \beta > -1.$$

Using Cauchy-Schwartz we can estimate

$$\begin{aligned} |G(\lambda, \beta)|^2 &\leq \left(\int_\delta^\infty e^{-\lambda t} dt \right) \left(\int_\delta^\infty t^{2\beta} e^{-\lambda t} dt \right) \\ &= \left(\frac{e^{-\lambda\delta}}{\lambda} \right) \left(\int_\delta^\infty t^{2\beta} e^{-\lambda t} dt \right) \\ &\stackrel{(u=\lambda t)}{=} \left(\frac{e^{-\lambda\delta}}{\lambda} \right) \left(\int_\delta^\infty e^{-u} u^{2\beta} \frac{1}{\lambda^{2\beta+1}} du \right) \\ &= \frac{e^{-\lambda\delta}}{\lambda^{2\beta+2}} M_\beta \end{aligned}$$

Where M_β is independent of λ . That leave us with

$$G(\lambda, \beta) = \mathcal{O} \left(\frac{e^{-\lambda\delta/2}}{\lambda^{\beta+1}} \right) = \mathcal{O} (e^{-\epsilon\lambda})$$

where $\epsilon > 0$ depends on β and δ . Finally we can rewrite

$$(1.1) = \frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} - G(\lambda, \alpha) = \frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} + \mathcal{O} (e^{-\epsilon\lambda})$$

$$|(1.2)| \leq \|f'\|_\infty \left(\frac{\Gamma(\alpha+2)}{\lambda^{\alpha+2}} - G(\lambda, \alpha+1) \right) = \|f'\|_\infty \left(\frac{\Gamma(\alpha+2)}{\lambda^{\alpha+2}} + \mathcal{O} (e^{-\epsilon\lambda}) \right) = \mathcal{O} (\lambda^{-\alpha-2})$$

and finally

$$F(\lambda) \leq \mathcal{O} (\lambda^{-\alpha-2}) + f_0(0) \left(\frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} + \mathcal{O} (e^{-\epsilon\lambda}) \right) + \|f\|_{L^1} e^{-\delta\lambda}.$$

by the dominance of the exponential error, we can finally get the result

$$F(\lambda) = \frac{f_0(0)\Gamma(\alpha+1)}{\lambda^{\alpha+1}} (1 + \mathcal{O}(\lambda^{-1}))$$

□

In the estimative the Gamma function and the factor of λ^{-1} is the result of integrating the linear exponent of e given by $-\lambda t$. If we, however, change this exponent to be of the general type $-\lambda t^d$ we can still have a similar result on the behavior of the function.

Proposition 6.3.3. *For $T, S > 0$, define the function*

$$G(\lambda) := \int_{-S}^T g(t) e^{-\lambda t^2} dt$$

and assume $g(x)$ absolutely integrable in $[-S, T]$, of class C^1 in a fixed neighborhood of the origin. Then the function satisfies

$$G(\lambda) = \frac{g(0)\Gamma(1/2)}{\lambda^{1/2}} + \mathcal{O} \left(\frac{1}{\lambda^{3/2}} \right)$$

when $\lambda \rightarrow \infty$ and where Γ is the Gamma function.

Proof. The proof is a reduction of the problem to Watson's Lemma. Assume, $T > S$ and write

$$G(\lambda) = \int_{-S}^S g(t) e^{-\lambda t^2} dt + \int_S^T g(t) e^{-\lambda t^2} dt$$

The second integral we know to be of order

$$\int_S^T g(t) e^{-\lambda t^2} dt \leq \|g\|_\infty e^{-\lambda S^2},$$

exponential on λ , so we drop it. For the first integral we rewrite

$$\begin{aligned} \int_{-S}^S g(t) e^{-\lambda t^2} dt &= \int_0^S (g(t) + g(-t)) e^{-\lambda t^2} dt \\ &\stackrel{(u=t^2)}{=} \frac{1}{2} \int_0^{S^2} \left(\frac{g(\sqrt{u}) + g(-\sqrt{u})}{\sqrt{u}} \right) e^{-\lambda u} du \end{aligned}$$

where now we can use Watson's Lemma. □

6.3.2 Laplace's Lemma

A seemingly natural extension of the type of function we were dealing with are functions of the type

$$\Phi(\lambda) := \int_a^b g(t) e^{-\lambda \phi(t)} dt$$

for $a, b \in \mathbb{R} \cup \{-\infty, \infty\}$ and $a < b$. For now, ϕ is a real-valued function. To begin the study let's make some assumptions. The first one is that a is finite, that means that we can set $a = 0$ without loss of generality. The second assumption is that ϕ is a increasing function on $[0, b]$, so that $\phi(0) < \phi(t)$ for any t in the interval. With that we note that the coefficient $e^{-\lambda \phi(t)} / e^{-\lambda \phi(0)}$ decreases exponentially, that is, the integral is accumulated as the neighborhood of 0,

$$\Phi(\lambda) \approx \int_0^\delta g(t) e^{-\lambda \phi(t)} dt.$$

As we know $\phi(t)$ to be increasing, it's first non zero derivative must be of even order, so that when we expanded the function by Taylor,

$$\phi(t) \approx \phi(0) + \phi_{2k+1} t^{2k+1}$$

for some $k \geq 0$ and $|t| < \delta$. As ϕ_{2k+1} , the $(2k+1)$ th-derivative, must be positive we also have that, as $\lambda \rightarrow \infty$,

$$\begin{aligned} \Phi(\lambda) &\approx e^{-\lambda \phi(0)} \int_0^\delta g(t) e^{-\lambda \phi_{2k+1}(t) \cdot t^{2k+1}} dt \\ &\stackrel{(u=t(\lambda \phi_{2k+1})^{1/(2k+1)})}{\approx} e^{-\lambda \phi(0)} \frac{g(0)}{(\lambda \phi_{2k+1})^{\frac{1}{2k+1}}} \int_0^\infty e^{-u^{2k+1}} du \\ &\stackrel{(s=u^{2k+1})}{\approx} e^{-\lambda \phi(0)} \frac{g(0)}{(2k+1)(\lambda \phi_{2k+1})^{\frac{1}{2k+1}}} \int_0^\infty e^{-s} ds \end{aligned}$$

Where the integral can be evaluated as an Gamma function. More than anything we need to focus on an important assumption that was made: that ϕ has a global minima at one of the endpoints of integration. Is that were not the case, for example if $a < 0 < b$ and $\phi(t)$ had a global minima at $t = 0$ we would now have to consider the expansion

$$\phi(t) \approx \phi(0) + \phi_{2k} t^{2k}$$

which would lead us to the expression

$$\Phi(\lambda) \approx e^{-\lambda \phi(0)} \frac{g(0)}{(\lambda \phi_{2k})^{\frac{1}{2k}}} \int_{-\infty}^\infty e^{-u^{2k}} du$$

which differentiates of the previously result more importantly by the limits of integration. This is closely related to physical boundaries of the problem.

Theorem 6.3.4. (*Laplace's Method, endpoint contributions*) Let Φ be as in 6.3.2, with a finite and $g \in L^1(a, b)$. Assume that ϕ has a unique global minimum at $t = a$. In addition, suppose that for some $\delta > 0$ and some $\alpha > -1$, the function g takes the form

$$g(t) = (t - a)^\alpha g_0(t)$$

for $|t - a| < \delta$ and g_0 continuously differentiable in $[a, a + \delta]$ and $g_0(a) \neq 0$, whereas ϕ is twice continuously differentiable on $[a, a + \delta]$ with $\phi'(a) > 0$. Then the function Φ satisfies an estimate of the form

$$\Phi(\lambda) = \frac{1}{\lambda^{\alpha+1}} \frac{g_0(a) \Gamma(\alpha + 1)}{\phi'(a)^{\alpha+1}} e^{-\lambda \phi(a)} (1 + \mathcal{O}(\lambda^{-1}))$$

as $\lambda \rightarrow \infty$.

Proof. Start by assuming, w.l.g, that $a = 0$ and dividing the integral in two, one in the region of interest and one in the rest of the interval

$$\Phi(\lambda) = \int_0^b g(t) e^{-\lambda \phi(t)} dt = \int_0^\delta g(t) e^{-\lambda \phi(t)} dt + \int_\delta^b g(t) e^{-\lambda \phi(t)} dt = (1) + (2).$$

Knowing that ϕ has a global minima at $t = 0$ we can assume, changing δ as needed, that $\phi(t) > \phi(\delta)$ for any $t > \delta$. That give us a simple limitation for (2):

$$(2) = \int_\delta^b g(t) e^{-\lambda \phi(t)} dt \leq e^{-\lambda \phi(\delta)} \int_\delta^b g(t) dt \leq e^{-\lambda \phi(\delta)} \|g\|_{L^1[0, b]}.$$

We now get back to (1) and start by studying the difference give by $\phi(t) - \phi(0) = s$ in a similar way that we would think of a change of variable. Define the multivariate function

$$H(t, s) := \phi(t) - \phi(0) - s$$

such that $H(0, 0) = 0$ and $\partial_t H(0, 0) \neq 0$ so that we can use the Theorem of Implicit Function to say that there exists a unique bijective and continuously differentiable function $t(s) := t$ such that

$$\begin{cases} \phi(t(s)) - \phi(0) - s = 0 \\ t'(0) = \frac{\partial_s H(0, 0)}{\partial_t H(0, 0)} = \frac{1}{\phi'(0)} \end{cases}$$

If we set $T = t^{-1}(\delta)$ we would then have

$$\int_0^\delta g(s) e^{-\lambda \phi(s)} ds = \int_0^T g(t(s)) e^{-\lambda \phi(t(s))} t'(s) ds$$

for now to apply the assumption on g . We write $g(t(s)) = t^\alpha(s) g_0(t(s))$ and because of Taylor,

$$t(s) = st'(0)(1 + r(s)) = \frac{s}{\phi'(0)}(1 + r(s)).$$

Joining both results,

$$\begin{aligned} \int_0^T g(t(s)) e^{-\lambda \phi(t(s))} t'(s) ds &= e^{\lambda \phi(0)} \int_0^T g(t(s)) e^{-\lambda(\phi(t(s)) - \phi(0))} t'(s) ds \\ &= e^{\lambda \phi(0)} \int_0^T t^\alpha(s) g_0(t(s)) e^{-\lambda s} t'(s) ds \\ &= \frac{e^{\lambda \phi(0)}}{\phi'(0)^\alpha} \int_0^T s^\alpha (1 + r(s))^\alpha g_0(t(s)) e^{-\lambda s} t'(s) ds \end{aligned}$$

where, if we set $f(s) := s^\alpha (1 + r(s))^\alpha g_0(t(s))$ we can apply Watson's Lemma. □

Theorem 6.3.5. (*Laplace's Method, interior contributions*) Let Φ be as in 6.3.2, with a finite and $g \in L^1(a, b)$. Assume that ϕ has a unique global minimum at $t = c \in (a, b)$. In addition, suppose that for some $\delta > 0$ and some $\alpha > -1$, the function g takes the form

$$g(t) = (t - a)^\alpha g_0(t)$$

for $|t - a| < \delta$ and g_0 continuously differentiable in $[c - \delta, c + \delta]$ and $g_0(a) \neq 0$, whereas ϕ is three times continuously differentiable on $[c - \delta, c + \delta]$ with $\phi'(a) > 0$. Then the function Φ satisfies an estimate of the form

$$\Phi(\lambda) = e^{-\lambda\phi(c)} \left(\frac{\sqrt{2\pi}g(c)}{\sqrt{|\phi''(c)|}\lambda^{1/2}} + \mathcal{O}(\lambda^{-3/2}) \right)$$

as $\lambda \rightarrow \infty$.

Proof. Although the proof may be similar there is an important distinction in the solution. Suppose we wanted to try the same approach. We would first assume, w.l.g., that $c = 0$. Then, we could write

$$\Phi(\lambda) = \int_a^b g(t) e^{-\lambda\phi(t)} dt = \int_a^{-\delta} g(t) e^{-\lambda\phi(t)} dt + \int_{-\delta}^{\delta} g(t) e^{-\lambda\phi(t)} dt + \int_{\delta}^b g(t) e^{-\lambda\phi(t)} dt = (1) + (2) + (3).$$

For (1) and (3) the result is as it was before

$$\begin{cases} (1) = \int_{\delta}^a g(-t) e^{-\lambda\phi(-t)} dt \leq e^{\lambda\phi(-\delta)} \|g\|_{L^1[a,b]} \\ (3) = \int_{\delta}^b g(t) e^{-\lambda\phi(t)} dt \leq e^{\lambda\phi(\delta)} \|g\|_{L^1[a,b]} \end{cases}$$

For (2) we would try to define a function

$$H(t, s) := \phi(t) - \phi(0) - s^2.$$

However we could not apply the Implicit Function Theorem here as its derivative $\partial_t H = \phi'$ vanishes at $t = 0$. A nice fact about this indetermination is that it doesn't come from the fact that there are no solutions to $H(t, s) = 0$ but from the fact that there are two. To solve this we will use a technique called *blow up*. First, we introduce a variable $v = v(t)$ such that $vs = t$. Now, we redefine our function of interested to be

$$L(s, v) := \frac{\phi(sv) - \phi(0)}{s^2} - 1.$$

for $s \in (-\delta, \delta) - \{0\}$. We would think this function is also singular but note that, using the Taylor Expansion of ϕ we can write

$$\phi(t) = \frac{\phi''(0)}{2} t^2 + t^3 R(t)$$

which leave us with

$$L(s, v) = \frac{\phi''(0)}{2} v^2 + sv^3 R(sv) - 1$$

where we can indeed apply the Implicit Function Theorem. In fact, we know $L \in C^2$ and as we want to solve for $L(s, v) = 0$, we need that the v_0 we take to satisfy $L(0, v_0) = 0$, ie,

$$v_0 = \sqrt{\frac{2}{\phi''(0)}}.$$

Now, we also know that

$$\partial_v L(0, v_0) = \phi''(0)v_0 \neq 0.$$

So we get a bijective and continuously differentiable function $v = v(s)$ such that

$$\begin{cases} L(s, v(s)) = 0 \\ v(0) = v_0 \end{cases}$$

But, $L(s, v(s))$ means exactly that

$$\phi(t) - \phi(0) = s^2$$

with $t = t(s) = sv(s)$. Getting back to our equation, we have that

$$e^{\lambda\phi(0)} \int_{-\delta}^{\delta} g(t) e^{-\lambda(\phi(t)-\phi(0))} dt = e^{\lambda\phi(0)} \int_{\alpha}^{\beta} g(sv(s))(v(s) + sv'(s)) e^{-\lambda s^2} ds$$

where we can easily calculate $\alpha = -\sqrt{\phi(-\delta) - \phi(0)}$ and $\beta = \sqrt{\phi(\delta) - \phi(0)}$. We finish the prove with Proposition 6.3.3. $\square$

Example - Gamma Function

Take the function called Gamma given by the expression

$$\Gamma(x) = \int_0^{\infty} e^{-t} t^{x-1} dt$$

We are interested in what happens to the function when we take the variable x to $+\infty$. To see what happens, let's fit it into Proposition 6.3.5. Rewrite equation 6.3.2 as

$$\int_0^{\infty} e^{-t+x \log(t) - \log(t)} dt.$$

This lead us to identify ϕ as $-\log(t)$, which would give us $\phi'(t) < 0$, that is, we would have it's minimum at $t = +\infty$. However $e^{\log(t)} = t^1$ is not integrable near the origin. Something need to change about the identification. Let's introduce a variable to rescale the integral: $xs = t$, s.t.,

$$\Gamma(x) = x^x \int_0^{\infty} e^{-\lambda(s-\log(s))} e^{-s} ds.$$

Where we have used $\lambda = x - 1$. Thus we identify $\phi(s) = s - \log(s)$ and $g(s) = e^{-s}$. in fact, now ϕ has a global minima at $s = 1$ and $\phi''(1) = 1 \neq 0$. Hence

$$\begin{aligned} \Gamma(x) &= \frac{\sqrt{2\pi} x^x e^{-x}}{(x-1)^{(-1/2)}} (1 + \mathcal{O}(x^{-1})) \\ &= \sqrt{2\pi} x^{x-1/2} e^{-x} (1 + \mathcal{O}(x^{-1})) \\ &= (x-1/2) \log(x) - x + \frac{1}{2} \log(2\pi) \end{aligned}$$

Which is called the Stirling formula.

6.4 Steepest Descent

Let's now finally explore integrals over the complex plane with complex valued functions. We retake the integral

$$\phi(\lambda) := \int_{\Gamma} g(s) e^{-\lambda f(s)} ds,$$

where Γ is a contour in the complex plane and f, g are given *complex-valued, analytic functions*.

Suppose that Γ is parametrized by $\gamma : [a, b] \rightarrow \mathbb{C}$ and let us decompose the functions we have into their real and imaginary parts such as

$$f(z) = u(z) + iv(z), \quad g(z) = p(z) + iq(z), \quad \gamma(t) = \alpha(t) + i\beta(t)$$

with all functions in the decomposition being real valued. Of course we now have

$$e^{-\lambda f(z)} = e^{-\lambda u(z)} e^{-iv(z)}$$

and when λ grows large, we have the exponentially growing/decaying term $e^{-\lambda u(z)}$ and a oscillating term coming from $e^{iv(z)}$ which is hard to control. The idea to get around this fact is to use paths in which this term is constant. If we have such a case, we would write

$$\Phi(\lambda) = e^{-i\lambda v_0} \int_a^b (p(\gamma(t))\alpha'(t) - q(\gamma(t))\beta'(t)) e^{-\lambda u(\gamma(t))} dt + i e^{-i\lambda v_0} \int_a^b (p(\gamma(t))\beta'(t) - q(\gamma(t))\alpha'(t)) e^{-\lambda u(\gamma(t))} dt$$

where each integral can now be solved by Laplace's method.

Theorem 6.4.1. *Consider Φ as in 6.4 with a smooth contour Γ . Assume that the exponent $f = u + iv$ is an analytic function on a neighborhood of Γ that satisfies*

$$v(z) \equiv v_0, \quad z \in \Gamma$$

In addition, suppose that u has a unique minimum at an interior point of Γ , say on $z_0 \in \Gamma$, with $f''(z_0) \neq 0$, and also that g is analytic on a neighborhood of Γ and satisfies

$$\int_{\Gamma} |g(z)| |dz| < \infty.$$

Then Φ has a leading asymptotic as $\lambda \rightarrow +\infty$ given by

$$\Phi(\lambda) = e^{-\lambda f(z_0)} \left(\frac{\sqrt{2\pi} g(z_0) e^{i\theta(z_0)}}{\lambda^{1/2} \sqrt{|f''(z_0)|}} + \mathcal{O}(\lambda^{-3/2}) \right), \quad \lambda \rightarrow +\infty,$$

where $\theta(z_0)$ is the angle of the oriented tangent of Γ at z_0 .

Proof. Fix a parametrization $\gamma : [a, b] \rightarrow \mathbb{C}$ of Γ with $\gamma(c) = z_0, a < c < b$, and decompose Φ as before in 6.4. By the assumptions on the functions f, g we can apply Laplace's method on both equations, and yields, as $\lambda \rightarrow +\infty$,

$$\int_a^b (p(\gamma(t))\alpha'(t) - q(\gamma(t))\beta'(t)) e^{-\lambda u(\gamma(t))} dt = \sqrt{2\pi} e^{-\lambda u(z_0)} \left(\frac{p(z_0)\alpha'(c) - q(z_0)\beta'(c)}{\sqrt{|(u \circ \gamma)''(c)|} \lambda^{1/2}} + \mathcal{O}(\lambda^{3/2}) \right)$$

and

$$\int_a^b (p(\gamma(t))\beta'(t) - q(\gamma(t))\alpha'(t)) e^{-\lambda u(\gamma(t))} dt = \sqrt{2\pi} e^{-\lambda u(z_0)} \left(\frac{p(z_0)\beta'(c) - q(z_0)\alpha'(c)}{\sqrt{|(u \circ \gamma)''(c)|} \lambda^{1/2}} + \mathcal{O}(\lambda^{3/2}) \right)$$

Using that $v_0 + u(z_0) = f(z_0)$ and also that,

$$(p(z_0)\alpha'(c) - q(z_0)\beta'(c)) + i(p(z_0)\beta'(c) - q(z_0)\alpha'(c)) = g(z_0)\gamma'(c).$$

Combining the estimative for both parts we get

$$\Phi(\lambda) = \sqrt{2\pi} e^{-\lambda f(z_0)} \left(\frac{g(z_0)\gamma'(c)}{\sqrt{|(u \circ \gamma)''(c)|} \lambda^{1/2}} + \mathcal{O}(\lambda^{3/2}) \right)$$

and it suffices now to get an expression for the second derivative in the denominator. Knowing the imaginary part is constant we can write

$$\frac{d}{dt} u(\gamma(t)) = \frac{d}{dt} f(\gamma(t)) = \gamma'(t) f'(\gamma(t))$$

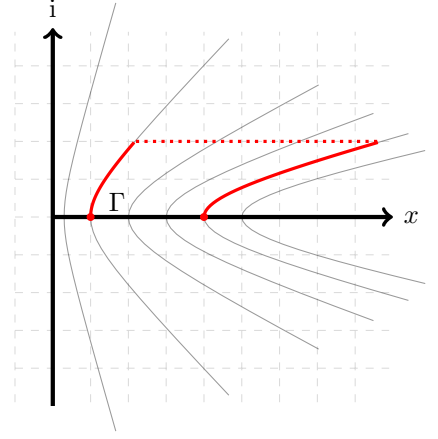
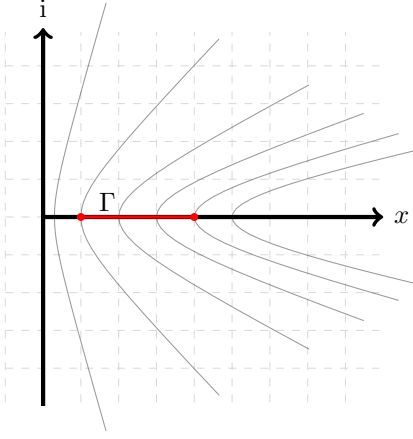
and

$$\frac{d^2}{dt} u(\gamma(t)) = \frac{d^2}{dt} f(\gamma(t)) = (\gamma'(t))^2 f''(\gamma(t)) + \gamma''(t) f'(\gamma(t)).$$

Hence,

$$\frac{\gamma'(c)}{\sqrt{|(u \circ \gamma)''(c)|}} = \frac{1}{\sqrt{|f''(z_0)|}} \frac{\gamma'(c)}{|\gamma'(c)|}$$

which concludes the proof. $\square$



We now need to discuss why this assumption of constant value of $\text{Im}(f(z))$ is not an assumption as strong as one might assume at first. Suppose we want to integrate in a random contour Γ of the plane. Of course, our assumption that the functions were analytic permit us to deform the contour keeping the value of the integral. We are going to use this to find an contour $\tilde{\Gamma}$ such that sections of the contour are such that $\text{Im}(f(z))$ is constant. in the figure above we plot such a scenario by using the level lines of the imaginary part of f as curves on the plane.

The name given by the method is also intuitive. We already know that being analytic means satisfying the *Cauchy-Riemann* equations,

$$u_x(z) = v_y(z), \text{ and } u_y(z) = -v_x(z).$$

An immediate result from these equation is that the gradients of u and v must be normal. Indeed, if $\langle \cdot, \cdot \rangle$ is the inner product on $\mathbb{R}^2$,

$$\langle \nabla u(z), \nabla v(z) \rangle = u_x(z)v_x(z) + u_y(z)v_y(z) = u_x(z)v_x(z) - v_x(z)u_x(z) = 0.$$

As we're following the curves where $\text{Im}(f(z))$ is constant, we're also following the path of *steepest descent* of the real part of the integral. As we also are looking to use Laplace's method on the integrals, we may want to have our paths going through the critical points of the function.

6.5 Case Study

6.5.1 Airy function

First, we define the function we call Airy.

Definition 6.5.1. (*Airy function*) Let γ be the union of the line segments $\{te^{i\pi/3} : t \geq 0\}$ and $\{te^{-i\pi/3} : t \geq 0\}$, that is the union of the two infinite rays in the complex plane emanating from the origin that make angles $\pi/3$ and $-\pi/3$ with the real axis. For $u \in \mathbb{C}$ the Airy function is defined by the complex contour integral

$$Ai(u) = \frac{1}{2\pi i} \int_{\gamma} \exp\left(\frac{t^3}{3} - ut\right) dt.$$

Of course, by Cauchy Theorem, we know that the definition is independent of the contour as long as the contour goes from $e^{-2\pi i/3} \infty$ to $e^{2\pi i/3} \infty$. Now, we need to get things in order to apply Laplace method on the integral. We could start defining

$$g(z) = e^{-\frac{1}{3}z^3}, \quad f(z) = z$$

but this is not a good approach as such an f won't have any critical points and won't help us calculate the integral. We then try a change of variable to get a function with those critical points. Define $x := \lambda e^{i\theta} = \lambda\omega$ so that we can

send x to infinite by sending λ in a constant direction θ . Now, introduce a variable s such that $s\lambda^\eta = z$ for some $\eta \in \mathbb{R}$. Now, we want to extract λ from the new exponent

$$-\frac{1}{3}\lambda^{3\eta}s^3 + \lambda^{1+\eta}\omega s$$

so we set $\eta = 1/2$ and have that

$$Ai(\lambda, \theta) = \frac{\lambda^{\frac{1}{2}}}{2\pi i} \int_{\Gamma} e^{-\lambda^{\frac{3}{2}}(s^3/3 - \omega s)} ds$$

$$\implies Ai(\lambda, \theta) = \frac{\lambda^{\frac{1}{2}}}{2\pi i} \Phi(\lambda^{\frac{3}{2}}), \quad \Phi(\lambda) = \int_{\Gamma} e^{-\lambda f(s)} ds, \quad f(s) = f_{\omega} := \frac{s^3}{3} - \omega s$$

Note that we have f with two critical points, $s_c^{\pm} = \pm\sqrt{\omega}$. We now need to study the steepest descent path. For reasons that will be clear soon, we divide the analysis into the cases

1. $\theta = 0$
2. $0 < \theta < \frac{2}{3}\pi$
3. $\theta = \frac{2}{3}\pi$
4. $\frac{2}{3}\pi < \theta < \pi$
5. $\theta = \pi$

the rest of the cases should follow from the same arguments based on the symmetry of the critical points. For each case we want to deform Γ to a new contour $\tilde{\Gamma}$ such that

- (a) $\tilde{\Gamma}$ is homotopically equivalent to Γ , in the sense that the integral in the definition of $\Phi(\lambda)$ is invariant under the deformation from Γ to $\tilde{\Gamma}$.
- (b) The contour $\tilde{\Gamma}$ is a level line of $\text{Im}(f)$ that passes through one of the critical points of f .

For (a) to be true we need to guarantee that the deformed path $\tilde{\Gamma}$ should start at $\infty e^{-i\theta}$ and end at $\infty e^{i\theta}$ for some $\theta \in [\pi/2, 5\pi/6]$. Now we need to discuss how to satisfy (b) for every one of the cases cited. For all cases, we refer to Image ?? for illustration of the scenario and intuition on the contour lines we take.

For the first case, (1), we can see that the integration path can be taken as the union of the lines emerging from s_c^- with angle $\pi/2$ and $3\pi/2$. For the second case we note that something happens when $\theta = 2/3\pi$. This is because $f(s_c^+) = f(s_c^-)$ when $\theta = 2/3\pi$. But until this point the topology is equivalent. For this interval of values then we take the path of integration to be basically the same as before, the union of the lines emerging from s_c^- with angle $\pi/2$ and $3\pi/2$.

For both of these cases we have the integration of Γ to be given by the contribution of the critical point s_c^- by the asymptotic of Laplace.

For $\theta = 2/3\pi$, case (3), something change. We now need to take a path that passes through both critical points. We would think that now we have to consider the asymptotic from both saddle points but, in reality, because $\text{Re}(f(s_c^-)) < \text{Re}(f(s_c^+))$ the contribution from s_c^- exponentially dominates the asymptotic.

For case (4) we no longer have a clear connection of the lines. We need to take $\tilde{\Gamma}$ to be the union of two disjoint curves: The line coming from $\infty e^{-i2\pi/3}$ passing through s_c^- and going to infinity parallel to the real line and the line coming from infinity parallel to the real line, passing through s_c^+ and going to $\infty e^{i2\pi/3}$. Again, the contribution from s_c^- exponentially dominates the asymptotic.

For case (5), when $\theta = \pi$, the path of steepest descent is similar to the one just taken. But a key difference is to be noted, now, as $\text{Re}(f(s_c^-)) = \text{Re}(f(s_c^+))$, contributions from both saddle points must be accounted in the asymptotic.

6.5.2 Hermite's Polynomials

This example will be a collage of many sources: Check [18] and [12] for the main references and notation. We already know the asymptotic for $x = o(n)$, done in 5.2.4. There are other two cases of interest that are related to methods explained in this Chapter. These are $x = o(1)$ and, more importantly, $x = o(\sqrt{n})$.

Regime $x = o(1)$

Proposition 6.5.2. *For $x \in \mathbb{R}$ fixed, we have the following asymptotic as $n \rightarrow \infty$:*

$$\text{He}_n(x) = \sqrt{2} \left(\frac{n}{2}\right)^{n/2} e^{x^2/4} \cos\left(\frac{n\pi}{2} - x\sqrt{n}\right)(1 + o(1))$$

Proof. Start with the integral formula given by Property 5.2.1 and write

$$\begin{aligned} \text{He}_n(x) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (x + it)^n e^{-t^2/2} dt = \frac{2}{\sqrt{2\pi}} \Re \left\{ \int_0^{\infty} (x + it)^n e^{-t^2/2} dt \right\} \\ &= \frac{2}{\sqrt{2\pi}} \Re \left\{ \int_{\Re(z)=x, \text{Im}(z)>0} e^{n \ln(z) + \frac{1}{2}(z-x)^2} \frac{dz}{i} \right\} \\ &= \frac{2}{\sqrt{2\pi}} \Re \left\{ \int_{\Re(z)=0, \text{Im}(z)>0} e^{n \ln(z) + \frac{1}{2}(z-x)^2} \frac{dz}{i} \right\} \end{aligned}$$

where you start with a line integral in $\mathbb{C}$ and move the contour over to the positive imaginary axis - justified by the Cauchy's Theorem and the exponential decay of the integrand near infinity. Now, we focus on $n \ln(z) + \frac{1}{2}(z-x)^2$ which has maximum at $z^* = i\sqrt{n} + o(\sqrt{n})$. The idea is to change variables by a factor of $\sqrt{z}$ so that the location of the maximum contribution stays $\mathcal{O}(1)$ as n gets large. Specifically, we do

$$z = i\sqrt{n}s, \quad \frac{dz}{i} = \sqrt{n}ds, \quad \ln(z) = \ln(\sqrt{n}) + \ln(s) + i\frac{\pi}{2}$$

which give us

$$\frac{1}{2}(z-x)^2 = -\frac{1}{2}ns^2 + \frac{1}{2}x^2 - i\sqrt{n}sx$$

so that we remain with

$$\begin{aligned} \text{He}_n(x) &= \frac{2}{\sqrt{2\pi}} \Re \left\{ \int_0^{\infty} e^{n \ln(\sqrt{n})} e^{n \ln(s)} e^{ni\frac{\pi}{2}} e^{\frac{1}{2}x^2} e^{-i\sqrt{n}sx} \sqrt{n}ds \right\} \\ &= \frac{2}{\sqrt{2\pi}} \sqrt{n} e^{n \ln(\sqrt{n})} e^{\frac{1}{2}x^2} \int_0^{\infty} e^{n(\ln(s) - \frac{1}{2}s^2)} \cos\left(n\frac{\pi}{2} - \sqrt{n}sx\right) ds, \end{aligned}$$

where we can apply Laplace's approximation discussed before. Noting that $\phi(s) = \ln(s) - \frac{1}{2}s^2$ has $\phi'(s) = \frac{1}{s} - s$ and $\phi''(s) = -\frac{1}{s^2} - 1$, with global maximum at $s^* = 1$ with $\phi(1) = \frac{1}{2}$ and $\phi''(1) = -2$, yielding

$$\text{He}_n(x) = \frac{2}{\sqrt{2\pi}} \sqrt{n} e^{n \ln(\sqrt{n})} e^{\frac{1}{2}x^2} \left(e^{n\phi(1)} \sqrt{\frac{2\pi}{n|\phi''(1)|}} \right) \cos\left(n\frac{\pi}{2} - \sqrt{n}x\right)(1 + o(1)),$$

which simplifies to the result. □

Regime $x = o(\sqrt{n})$

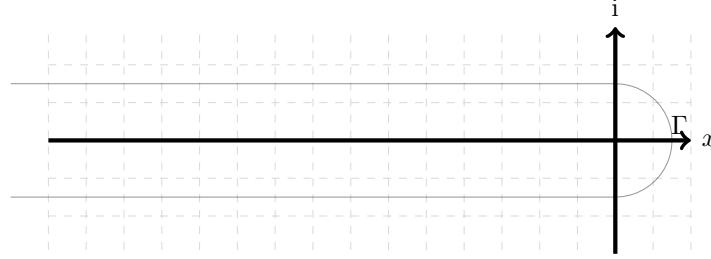
There are three asymptotics regimes for $\text{He}_n(x)$ when $x \rightarrow \infty$ depending on the value of $\lim_{N \rightarrow \infty} x/2\sqrt{N}$ for $N = n + 1$, each of which has qualitatively different behavior

- Small x values $-2\sqrt{N} < x < 2\sqrt{N}$: in this regime the Hermite Polynomials will be oscillating, and have a similar qualitative behavior as 6.5.2 with modification to phase and amplitude. These polynomials will have some connections to the *arcsine law* and *semi-circle law*.
- Large x values $|x| > 2\sqrt{N}$: In this regime the polynomial doesn't have more oscillations and is monotone increasing. As x grows larger, this approximation approaches 5.2.4.
- Edge x values $x = \pm 2\sqrt{N} + \mathcal{O}(N^{-\frac{1}{6}})$: in this regime the polynomials will be approximated by the Airy Function. This interpolates between the oscillatory small x regime and the monotone large x regimes.

We write the integral representation given by the Cauchy Integral applied to the Generating Function, this representation should be discussed in 5.2.5 and given by:

$$\begin{aligned}\text{He}_n(x) &:= \frac{n!}{2\pi i} \int_{\Gamma} z^{-n-1} e^{xz - \frac{z^2}{2}} dz \\ &= \frac{n!}{2\pi i} \int_{\Gamma} e^{xz - \frac{z^2}{2} - (n+1) \ln(z)} dz\end{aligned}$$

Where Γ is a contour that goes around the origin coming and going from minus infinity such as $\Gamma = \{\lambda - i\delta : \lambda < 0\} \cup \{\delta e^{i\theta} : -\pi/2 < \theta < \pi/2\} \cup \{\lambda + i\delta : \lambda < 0\}$. This is equivalent to using $\Gamma = \{e^{i\theta} : 0 \leq \theta < 2\pi\}$.



There are a few expressions we want to have. Namely we want to determine the formulas for He_n and He_{n-1} . Let's start working on the first case. To do this, set $N = n + 1$ and consider the scaling $x \mapsto N^a \alpha$ and $z = N^b \beta$

$$\begin{aligned}\text{He}_n(N^a \alpha) &= \frac{(N-1)!}{2\pi i} \int_{\Gamma} e^{(N^a \alpha)(N^b \beta) - \frac{1}{2}(N^b \beta)^2 - N \ln(N^b \beta)} d\beta \\ &\stackrel{(a=b=1/2)}{=} \frac{(N-1)! N^{\frac{1}{2}(1-N)}}{2\pi i} \int_{\Gamma} e^{-N(-\alpha\beta + \frac{1}{2}\beta^2 + \ln(\beta))} d\beta\end{aligned}$$

So that we have

$$\text{He}_n(\sqrt{N}\alpha) = C_n^0 \int_{\Gamma} e^{-N\phi_{\alpha}(\beta)} d\beta \quad (6.5.1)$$

with

$$\phi_{\alpha}(\beta) = \frac{1}{2}\beta^2 - \alpha\beta + \ln(\beta).$$

As we want to calculate the asymptotic by steepest descent, we need to determine the critical points of such a function and then the steepest descent paths passing through this points. We can easily determine the critical points as $\beta_c^{\pm} = \frac{\alpha \pm \sqrt{\alpha^2 - 4}}{2}$ - solutions to the quadratic formula given by solving the critical point of the function (the zeros of its derivatives). Now, the behavior of the critical points will depend on the interval of the real parameter α .

$$\beta_c^{\pm} = \frac{\alpha \pm \sqrt{\alpha^2 - 4}}{2} = \begin{cases} e^{\pm\gamma}, & \alpha = 2 \cosh(\gamma) > 2, \quad \gamma > 0; \\ e^{\pm i\gamma}, & \alpha = 2 \cos(\gamma) \in [0, 2], \quad \gamma \in (0, \pi/2]; \\ 1, & \alpha = 2. \end{cases}$$

Therefore, the following theorem follows.

Theorem 6.5.3. *The Hermite Polynomials $\text{He}_n(x)$ have the following three representation when $x = \mathcal{O}(\sqrt{n})$ and $x > 0$:*

1. *Small x values $-2\sqrt{N} < x < 2\sqrt{N}$: For fixed $\gamma \in (0, \pi/2)$ set $x = 2\sqrt{N} \cos(\gamma)$ and we obtain*

$$\text{He}_n(2\sqrt{N} \cos(\gamma)) = \sqrt{\frac{2}{\sin(\gamma)}} N^{\frac{1}{2}(N-1)} e^{N(\cos^2(\gamma) - \frac{1}{2})} \sin\left(\theta(\beta_c^+) - N\left(\frac{\sin(2\gamma)}{2} - \gamma\right)\right) (1 + \mathcal{O}(N^{-1}))$$

$$\text{He}_{n-1}(2\sqrt{N} \cos(\gamma)) = \sqrt{\frac{2}{\sin(\gamma)}} N^{\frac{1}{2}(N-2)} \frac{e^{N(\cos^2(\gamma) - \frac{1}{2})}}{e^{\cos^2(\gamma)}} \sin\left(\theta(\beta_c^+) - (N-1)\left(\frac{\sin(2\gamma)}{2} - \gamma\right)\right) (1 + \mathcal{O}(N^{-1}))$$

2. *Large x values $|x| > 2\sqrt{N}$: For fixed $\gamma > 1$, write $x = 2\sqrt{N} \cosh(\gamma)$. We obtain*

$$\text{He}_n(2\sqrt{N} \cosh(\gamma)) = e^{-N\left(\frac{e^{-2\gamma}}{2} + \gamma + 1\right)} \frac{N^{\frac{1}{2}(N-1)}}{\sqrt{|1 - e^{2\gamma}|}} (1 + \mathcal{O}(N^{-1})),$$

$$\text{He}_{n-1}(2\sqrt{N} \cosh(\gamma)) = \frac{e^{-N\left(\frac{e^{-2\gamma}}{2} + \gamma + 1\right)}}{e^{\left(\frac{e^{-2\gamma}}{2} + \gamma + \frac{3}{2}\right)}} \frac{N^{\frac{1}{2}(N-2)}}{\sqrt{|1 - e^{2\gamma}|}} (1 + \mathcal{O}(N^{-1})),$$

3. *Edge x values $x = \pm 2\sqrt{N} + \mathcal{O}(N^{-\frac{1}{6}})$: For fixed $u \in \mathbb{R}$, set $x = 2\sqrt{N} + N^{-1/6}\xi$, then we obtain*

$$\text{He}_n(2\sqrt{N} + N^{-1/6}\xi) = \sqrt{2\pi} e N^{-\frac{N}{2} + \frac{2}{3}} Ai(\xi) (1 + o(1)),$$

$$\text{He}_{n-1}(2\sqrt{N} + N^{-1/6}\xi) = \sqrt{2\pi} e N^{-\frac{N}{2} + \frac{7}{6}} Ai(\xi) (1 + o(1))$$

Case 1

For $-2\sqrt{N} < x < 2\sqrt{N}$, or, equivalently, $-2 < \alpha < 2$ we have, as Figure 6.1 suggests, both critical points as complex values, this can also be easily determined noting that if we set α constant with $0 < \alpha < \sqrt{2}$, we will have

$$\beta_c^\pm = \frac{\alpha}{2} \pm \frac{i}{2} \sqrt{4 - \alpha^2}.$$

And, if we parametrize $\alpha = 2 \cos(\gamma)$ for $\gamma \in (0, \pi/2]$ we can rewrite

$$\beta_c^\pm = e^{\pm i\gamma}.$$

If we are indeed interested in solving the paths where the imaginary parts are constant, as needed for the Laplace method, we will have to solve an equation such as

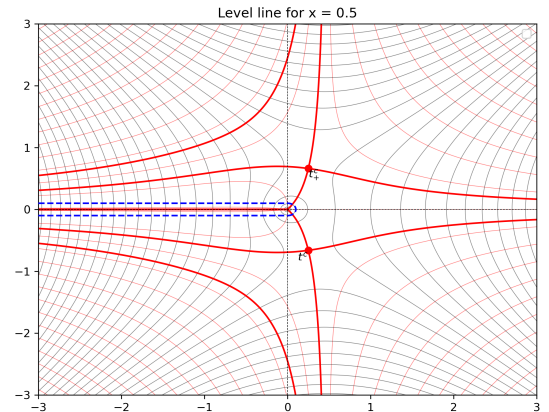
$$\text{Im}(\phi_\alpha(z) - \phi_\alpha(\beta_c^\pm)) = 0,$$

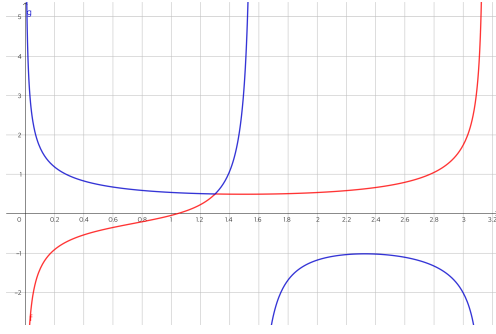
for some $z = r e^{i\theta}$. We first solve for the positive critical point - the negative will be analogous. This equation can be rewritten as a quadratic formulae

$$\frac{1}{2} r^2 \sin(2\theta) - r \cos(\gamma) \sin(\theta) + (\theta - \theta_0) = 0$$

where we have set $\theta_0 = \gamma - \frac{1}{2} \sin(2\gamma)$. Where we solve for the two solutions r_a^+ and r_d^- by meshing the two quadratic solutions given by the equations. This give us two smooth paths for the positive critical point

Figure 6.1: Figure for Case 1





$$r_d^+(\theta) = \begin{cases} \frac{\cos(\gamma) - \sqrt{\cos^2(\gamma) - (\theta - \theta_0)/\tan(\theta)}}{\cos(\theta)}; & \theta \in [\gamma, \pi) \\ \frac{\cos(\gamma) + \sqrt{\cos^2(\gamma) - (\theta - \theta_0)/\tan(\theta)}}{\cos(\theta)}; & \theta \in (0, \gamma) \end{cases}$$

$$r_a^+(\theta) = \begin{cases} \frac{\cos(\gamma) + \sqrt{\cos^2(\gamma) - (\theta - \theta_0)/\tan(\theta)}}{\cos(\theta)}; & \theta \in [\gamma, \pi/2) \\ \frac{\cos(\gamma) - \sqrt{\cos^2(\gamma) - (\theta - \theta_0)/\tan(\theta)}}{\cos(\theta)}; & \theta \in [\theta_0, \gamma) \end{cases}$$

With which we can construct both paths of constant imaginary part

$$\Gamma_d^+ = \{r_d^+(\theta) e^{i\theta} : \theta \in (0, \pi)\}; \quad \Gamma_a^+ = \{r_a^+(\theta) e^{i\theta} : \theta \in (0, \pi/2)\}$$

Analogously, for the negative part,

$$\Gamma_d^- = \{r_d^-(\theta) e^{i\theta} : \theta \in (0, \pi)\}; \quad \Gamma_a^- = \{r_a^-(\theta) e^{i\theta} : \theta \in (0, \pi/2)\}$$

Now, we only need to determine the steepest descent path. This should be easy as we know the asymptotic behavior $\phi_\alpha(t) \sim t^2$ for $t \rightarrow \infty$ and that $\phi_\alpha(t) \sim \ln(t)$ for $t \rightarrow 0$. From this, follows that Γ_d^+ and Γ_d^- are the paths we were looking for. We create therefore, a new path Γ_x that connects both Γ_d at the vertical line x . Then, doing $\Gamma \mapsto \Gamma_d^+ \cup \Gamma_d^- \cup \Gamma_x$ and noting that, as x goes to infinity, we have

$$\left| \int_{\Gamma_x} g(x) e^{-n\phi_\alpha(\beta)} d\beta \right| \leq \max_{y \in \Gamma_x} |g(x) e^{-n\phi_\alpha(\beta)}| \int_{\Gamma_x} |d\beta| \xrightarrow{t \rightarrow \infty} 0$$

where $g(t)$ can be both the inverse square root or inverse function. Therefore, we only need to consider the other two paths when considering the asymptotic. Let's briefly discuss how to get the tangent angle of the integration curve. We know that, locally, we will have $\beta_c^+ + \epsilon e^{i\theta}$. If we wish to find the angle θ from which the curve is tangent we must note that this will be the angle in which $\phi_\alpha(\beta_c^+ + \epsilon e^{i\theta})$ has constant imaginary part - as this is the way the curve is defined. Therefore, for some constant c ,

$$\begin{aligned} \text{Im} [\phi_\alpha(\beta_c^+ + \epsilon e^{i\theta})] &= c \\ \implies \text{Im} \left[\frac{\phi''(\beta_c)}{2} e^{2i\theta} \right] &= \sin \theta \sin(2\theta) = 0, \quad \forall \epsilon \\ \implies \theta &= k \frac{\pi}{2}, \quad \text{for } k = 1, 2, \dots \end{aligned}$$

Finally, retaking the expression for the Hermite Polynomial,

$$\begin{aligned}
\text{He}_n(2\sqrt{N} \cos(\gamma)) &= C_n^0 \int_{\Gamma} e^{-N\phi_{\alpha}(\beta)} d\beta = C_n^0 \left[\int_{\Gamma_d^-} e^{-N\phi_{\alpha}(\beta)} d\beta + \int_{\Gamma_d^+} e^{-N\phi_{\alpha}(\beta)} d\beta \right] \\
&\stackrel{(Steepest)}{=} C_n^0 \left(e^{-N(\phi_{\alpha}(\beta_c^+))} \left[\frac{\sqrt{2\pi}}{N^{\frac{1}{2}}} \frac{e^{i(\theta(\beta_c^+))}}{\sqrt{2 \sin(\gamma)}} + \mathcal{O}(N^{-\frac{3}{2}}) \right] + e^{-N(\overline{\phi_{\alpha}(\beta_c^+)})} \left[\frac{\sqrt{2\pi}}{N^{\frac{1}{2}}} \frac{e^{i(\overline{\theta(\beta_c^+)})}}{\sqrt{2 \sin(\gamma)}} + \mathcal{O}(N^{-\frac{3}{2}}) \right] \right) \\
&= C_n^0 \sqrt{\frac{2\pi}{2N \sin(\gamma)}} \left(e^{-N(\phi_{\alpha}(\beta_c^+))} \left[e^{i(\theta(\beta_c^+))} + \mathcal{O}(N^{-1}) \right] + e^{-N(\overline{\phi_{\alpha}(\beta_c^+)})} \left[e^{i(\overline{\theta(\beta_c^+)})} + \mathcal{O}(N^{-1}) \right] \right) \\
&= C_n^0 \sqrt{\frac{\pi}{N \sin(\gamma)}} \left(e^{-N(\phi_{\alpha}(\beta_c^+)) + i(\theta(\beta_c^+))} + e^{-N(\overline{\phi_{\alpha}(\beta_c^+)}) + i(\overline{\theta(\beta_c^+)})} \right) (1 + \mathcal{O}(N^{-1})) \\
&= C_n^0 \sqrt{\frac{\pi}{N \sin(\gamma)}} \left(e^{-i\theta(\beta_c^+) - N(-\alpha\beta_c^+ + \frac{1}{2}(\beta_c^+)^2 + \ln(\beta_c^+))} + e^{i\pi} e^{i\theta(\beta_c^+) - N(-\alpha\beta_c^+ + \frac{1}{2}(\beta_c^+)^2 + \ln(\beta_c^+))} \right) (1 + \mathcal{O}(N^{-1})) \\
&= C_n^0 \sqrt{\frac{\pi}{N \sin(\gamma)}} e^{N(\cos^2(\gamma) + \frac{1}{2})} \left(e^{-i(-N(\frac{\sin(2\gamma)}{2} - \gamma) + \theta(\beta_c^+))} - e^{i(-N(\frac{\sin(2\gamma)}{2} - \gamma) + \theta(\beta_c^+))} \right) (1 + \mathcal{O}(N^{-1})) \\
&= C_n^0 \sqrt{\frac{\pi}{N \sin(\gamma)}} e^{N(\cos^2(\gamma) + \frac{1}{2})} 2i \sin \left(\theta(\beta_c^+) - N \left(\frac{\sin(2\gamma)}{2} - \gamma \right) \right) (1 + \mathcal{O}(N^{-1})) \\
&= \frac{(N-1)! N^{\frac{1}{2}(1-N)}}{2\pi i} \sqrt{\frac{\pi}{N \sin(\gamma)}} e^{N(\cos^2(\gamma) + \frac{1}{2})} 2i \sin \left(\theta(\beta_c^+) - N \left(\frac{\sin(2\gamma)}{2} - \gamma \right) \right) (1 + \mathcal{O}(N^{-1})) \\
&= \sqrt{\frac{2}{\sin(\gamma)}} N^{\frac{1}{2}(N-1)} e^{N(\cos^2(\gamma) - \frac{1}{2})} \sin \left(\theta(\beta_c^+) - N \left(\frac{\sin(2\gamma)}{2} - \gamma \right) \right) (1 + \mathcal{O}(N^{-1}))
\end{aligned}$$

Where we used Stirling's Formula and the fact that we have

$$\begin{cases} \Re[(-\alpha\beta_c^+ + \frac{1}{2}(\beta_c^+)^2 + \ln(\beta_c^+))] = -\cos^2(\gamma) - \frac{1}{2}; & \text{Im}[(-\alpha\beta_c^+ + \frac{1}{2}(\beta_c^+)^2 + \ln(\beta_c^+))] = -\frac{\sin(2\gamma)}{2} + \gamma; \\ \Re[(-\alpha\beta_c^+ + \frac{1}{2}(\beta_c^+)^2 + \ln(\beta_c^+))] = -\cos^2(\gamma) - \frac{1}{2}; & \text{Im}[(-\alpha\beta_c^+ + \frac{1}{2}(\beta_c^+)^2 + \ln(\beta_c^+))] = \frac{\sin(2\gamma)}{2} - \gamma. \end{cases}$$

and

$$\begin{aligned}
(N-k)^{(aN-b)} &= N^{aN-b} \left(1 - \frac{k}{N}\right)^{aN-b} = N^{aN-b} \left(1 - \frac{k}{N}\right)^{aN} \left(1 - \frac{k}{N}\right)^{-b} \\
&= N^{aN-b} e^{aN \ln(1 - \frac{k}{N})} (1 + \mathcal{O}(N^{-1})) \\
&= N^{aN-b} e^{aN(-\frac{k}{N} + \mathcal{O}(N^{-2}))} (1 + \mathcal{O}(N^{-1})) \\
&= N^{aN-b} e^{-ak} (1 + \mathcal{O}(N^{-1}))
\end{aligned}$$

Of course, if we change for the other case we can simply rewrite

$$\text{He}_{n-1}(2\sqrt{N} \cos(\gamma)) = \sqrt{\frac{2}{\sin(\gamma)}} N^{\frac{1}{2}(N-2)} \frac{e^{N(\cos^2(\gamma) - \frac{1}{2})}}{e^{\cos^2(\gamma)}} \sin \left(\theta(\beta_c^+) - (N-1) \left(\frac{\sin(2\gamma)}{2} - \gamma \right) \right) (1 + \mathcal{O}(N^{-1}))$$

Case 2

Here, with $|x| > 2\sqrt{N}$, or equivalently $\alpha > 2$, we have - as the plot 6.2 suggests - both critical points as real values, this can be easily determined noting that if we set $\alpha > 2$, we will have

$$\beta_c^{\pm} = \frac{\alpha}{2} \pm \frac{1}{2} \sqrt{\alpha^2 - 4}.$$

And, if we parametrize $\alpha = 2 \cosh(\gamma)$ for $\gamma > 1$ we can rewrite

$$\beta_c^{\pm} = e^{\pm\gamma}.$$

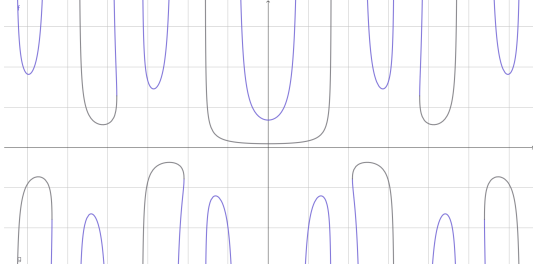
If we are indeed interested in solving the paths where the imaginary parts are constant, as needed for the Laplace method, we will have to solve an equation such as

$$\text{Im}(\phi_\alpha(z) - \phi_\alpha(\beta_c^\pm)) = 0,$$

for some $z = r e^{i\theta}$. We first solve for the positive critical point and the negative will be analogous. This can be rewritten as a quadratic equation

$$\frac{1}{2}r^2 \sin(2\theta) - r \cosh(\gamma) \sin(\theta) + \theta = 0$$

where we solve for the two solutions r_a^+ and r_d^- by meshing the two quadratic solutions given by the equations. This gives us two smooth paths for the positive critical point



$$r_d(\theta) = \frac{\cosh(\gamma) - \sqrt{\cosh^2(\gamma) - \frac{\cosh(\gamma)\theta}{\sin(\theta)}}}{\cos(\theta)}; \quad \theta \in [0, \pi)$$

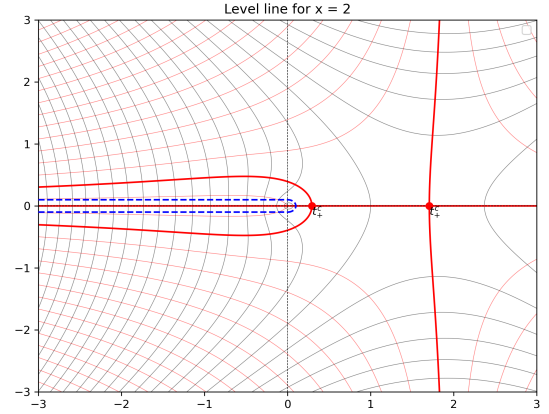
$$r_a(\theta) = \frac{\cosh(\gamma) + \sqrt{\cosh^2(\gamma) - \frac{\cosh(\gamma)\theta}{\sin(\theta)}}}{\cos(\theta)}; \quad \theta \in [0, \pi)$$

With which we can construct both paths of constant imaginary part

$$\Gamma_d = \{\tilde{r}_d(\theta) e^{i\theta} : \theta \in (0, 2\pi)\}; \quad \Gamma_a = \{\tilde{r}_a(\theta) e^{i\theta} : \theta \in (0, 2\pi)\}$$

We need now to determine the Steepest Descent Path. In this case it is more clear what the path should be: the one passing through the critical point $e^{-\gamma}$. An argument of symmetry allows us to show that the tangent angle of the path of steepest descent must be $2\pi/4$, as the curve - together with the real line - divides the plane into 4 parts. Then, repeating the process we did on Case 1 - changing

Figure 6.2: Figure for case 3



the path of integration - we can make the substitution

$$\begin{aligned}
\text{He}_n(2\sqrt{N} \cosh(\gamma)) &= C_n^0 \int_{\Gamma} e^{-N\phi_{\alpha}(\beta)} d\beta \\
&= C_n^0 \int_{\Gamma_d} e^{-N\phi_{\alpha}(\beta)} d\beta \\
&\stackrel{(Steepest)}{=} C_n^0 e^{N(\phi_{\alpha}(\beta_c^-))} \left[\frac{\sqrt{2\pi}}{N^{\frac{1}{2}}} \frac{e^{i(\theta(\beta_c^-))}}{\sqrt{|1 - e^{2\gamma}|}} + \mathcal{O}(N^{-\frac{3}{2}}) \right] \\
&= C_n^0 e^{-N(\phi_{\alpha}(\beta_c^-))} \frac{\sqrt{2\pi}}{N^{\frac{1}{2}}} \frac{e^{i(\theta(\beta_c^-))}}{\sqrt{|1 - e^{2\gamma}|}} (1 + \mathcal{O}(N^{-1})) \\
&= e^{-N(\phi_{\alpha}(\beta_c^-))} \frac{(N-1)! N^{\frac{1}{2}(1-N)}}{2\pi i} \frac{\sqrt{2\pi}}{N^{\frac{1}{2}}} \frac{i}{\sqrt{|1 - e^{2\gamma}|}} (1 + \mathcal{O}(N^{-1})) \\
&= e^{-N(\phi_{\alpha}(\beta_c^-)-1)+1} \frac{(N-1)^{N-\frac{1}{2}} N^{-\frac{N}{2}}}{\sqrt{|1 - e^{2\gamma}|}} (1 + \mathcal{O}(N^{-1})) \\
&= e^{-N(\phi_{\alpha}(\beta_c^-)-1)+1} \frac{(N-1)^{N-\frac{1}{2}} N^{-\frac{N}{2}}}{\sqrt{|1 - e^{2\gamma}|}} (1 + \mathcal{O}(N^{-1})) \\
&= e^{-N\left(\frac{e^{-2\gamma}}{2} + \gamma + 1\right) + 1} \frac{(N-1)^{N-\frac{1}{2}} N^{-\frac{N}{2}}}{\sqrt{|1 - e^{2\gamma}|}} (1 + \mathcal{O}(N^{-1})) \\
&= e^{-N\left(\frac{e^{-2\gamma}}{2} + \gamma + 1\right)} \frac{N^{\frac{1}{2}(N-1)}}{\sqrt{|1 - e^{2\gamma}|}} (1 + \mathcal{O}(N^{-1}))
\end{aligned}$$

Similarly, for the other case

$$\text{He}_{n-1}(2\sqrt{N} \cosh(\gamma)) = \frac{e^{-N\left(\frac{e^{-2\gamma}}{2} + \gamma + 1\right)}}{e^{\left(\frac{e^{-2\gamma}}{2} + \gamma + \frac{3}{2}\right)}} \frac{N^{\frac{1}{2}(N-2)}}{\sqrt{|1 - e^{2\gamma}|}} (1 + \mathcal{O}(N^{-1}))$$

Case 3

Let's start with a sketch of the proof for this behavior.

1. **Step One:** Write the integral for $N = n + 1$

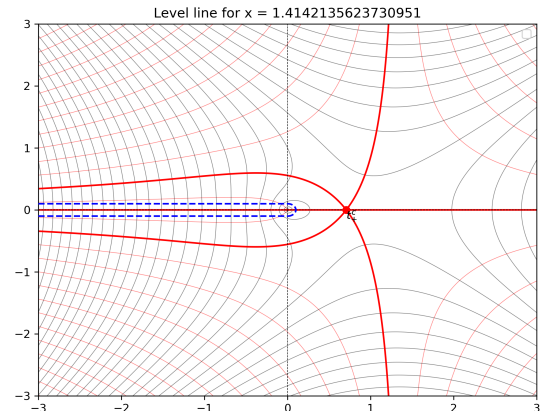
$$\text{He}_n(\sqrt{N}\alpha) = C_n^0 \int_{\Gamma} e^{-N\phi_{\alpha}(\beta)} d\beta$$

where $\phi_{\alpha}(\beta) = \beta^2 - 2\alpha\beta + \ln(\beta)$;

2. **Step two:** The function $\phi_{\alpha}(\beta)$ has a double critical point at 2. The steepest descent curve is defined by $\text{Im}(\phi_{\alpha}(z) - \phi_{\alpha}(2)) = 0$;
3. **Step three:** Deform the contour such that the only contribution for the integral comes from the critical point. Computing the Taylor approximation to the third order we have

$$\phi_{\alpha}(\beta) = \phi_{\alpha}(\sqrt{2}/2) + \phi_{\alpha}^{(3)}(\sqrt{2}/2)(\beta - \beta_c)^3 + \mathcal{O}(\beta^4)$$

Figure 6.3: Figure for case 3



and perform the change of variables $\lambda = N^{\frac{1}{3}}(\beta - \beta_c)$ to obtain

$$\text{He}_n(\sqrt{N}\alpha) \approx \tilde{C}_n \int_{\Gamma} e^{\lambda^3} d\lambda$$

4. **Step four** Recognize the Airy Function and write

$$\text{He}_n(\sqrt{N}\alpha) \propto \text{Ai}(0)$$

We start with the expression determined in (6.6.1), working on the edge, we will set $x = 2\sqrt{N} + \xi N^{-\frac{1}{6}}$, that is, $\alpha = 2 + \xi N^{-\frac{2}{3}}$. Remember the expression

$$\text{He}_n(\sqrt{N}\alpha) = C_n^0 \int_{\Gamma} e^{-N(\frac{1}{2}\beta^2 - \alpha\beta + \ln(\beta))} d\beta.$$

Applying the scale above, we have

$$\phi_{2+\xi N^{-\frac{2}{3}}}(\beta) = -(2 + \xi N^{-\frac{2}{3}})\beta + \frac{1}{2}\beta^2 + \ln(\beta) = \phi_2(\beta) - \xi N^{-\frac{2}{3}}\beta,$$

such that substituting it into the integral

$$\text{He}_n(\sqrt{N}(2 + \xi N^{-\frac{2}{3}})) = C_n^0 \int_{\Gamma} e^{-N(\phi_2(\beta) - \xi\beta N^{-\frac{2}{3}})} d\beta.$$

Now, it is clear from the previous cases that when we evaluate the integral, the contributions from the critical point $\beta_c = 1$ should be exponentially bigger - that is, the asymptotic of the integral can be deduced from the values around this point. We know that

$$\begin{aligned}\phi_2^{(0)}(1) &= -\frac{3}{2} \\ \phi_2^{(1)}(1) &= 0 \\ \phi_2^{(2)}(1) &= 0 \\ \phi_2^{(3)}(1) &= 2\end{aligned}$$

such that we can define for a neighborhood of the critical point the Taylor expansion of ϕ given by

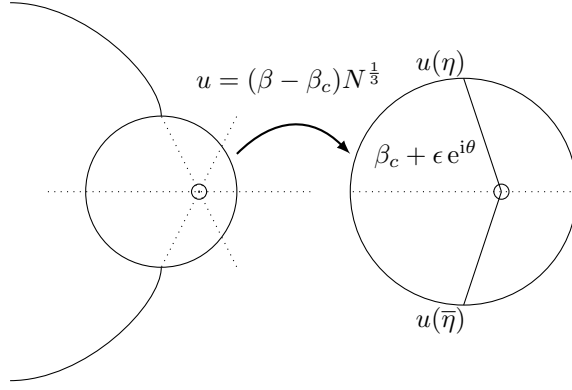
$$\phi_2(\beta) = \phi_2(1) + \frac{\phi_2^{(3)}(1)}{3!}(\beta - \beta_c)^3 + \mathcal{O}((\beta - \beta_c)^4).$$

This step is twofold: Noted that the evaluated of the integral comes from a neighborhood of the critical point, we change for a local variable (linear) of the problem as told before using $u = N^{\frac{1}{3}}(\beta - \beta_c)$ and take the two border points of the neighborhood given by $\eta = \beta_c + \epsilon e^{i\theta}$ for some small epsilon. With the change, check Figure ??, we will have the limits of the integration given by $u(\eta)$ and $u(\bar{\eta})$. Now, we only need to note that these lines - keeping the same angle as the original curve - are angled $e^{i\frac{2\pi}{3}}$ and $e^{i\frac{4\pi}{3}}$. this is because the original contour, by symmetric, divided the plane in six equal angles. Therefore, taking $\beta = \beta_c + uN^{-\frac{1}{3}}$, for u coming from $\infty e^{i\frac{4\pi}{3}}$ to $\infty e^{i\frac{2\pi}{3}}$, we write

$$\begin{aligned}\text{He}_n(2N^{\frac{1}{2}} + \xi N^{-\frac{1}{6}}) &= C_n^0 e^{\frac{3N}{2}} \int_{\Gamma} e^{-N(\frac{1}{3}(\beta - \beta_c)^3 - \xi\beta N^{-\frac{2}{3}} + \mathcal{O}((\beta - \beta_c)^4))} d\beta \\ &= C_n^0 e^{\frac{3N}{2}} \int_{u(\bar{\eta})}^{u(\eta)} e^{-N(\frac{1}{3}(uN^{-\frac{1}{3}})^3 - \xi(\beta_c + uN^{-\frac{1}{3}})N^{-\frac{2}{3}} + \mathcal{O}((uN^{-\frac{1}{3}})^4))} d(uN^{-\frac{1}{3}}) \\ &= C_n^0 N^{-\frac{1}{3}} e^{-2\xi\beta_c N^{\frac{1}{3}}} e^{\frac{3N}{2}} \int_{N^{\frac{1}{3}} e^{i\frac{2\pi}{3}}}^{-N^{\frac{1}{3}} e^{i\frac{4\pi}{3}}} e^{\frac{1}{3}u^3 - \xi u + \mathcal{O}(u^4 N^{-\frac{4}{3}})} du\end{aligned}$$

Then we will have asymptotically in N , remembering the definition of the Airy function 6.5.1,

$$\text{He}_n(2N^{\frac{1}{2}} + \xi N^{-\frac{1}{6}}) = \tilde{C}_N \int_{\gamma} e^{\left(\frac{u^3}{3} - \xi u\right)} du (1 + o(1)) = \tilde{C}_N \text{Ai}(\xi)(1 + o(1)),$$



for

$$\begin{aligned}\tilde{C}_N &= (N-1)! N^{\frac{1}{6} - \frac{N}{2}} e^{-2\xi\beta_c N^{\frac{1}{3}} + \frac{3N}{2}} = \sqrt{2\pi} (N-1)^{\frac{1}{2}} N^{1/6 - N/2} e^{(1/2 + N^{-1} - 2\xi\beta_c N^{-2/3})} (1 + \mathcal{O}(N^{-1})) \\ &= \sqrt{2\pi} e N^{-\frac{N}{2} + \frac{2}{3}} (1 + \mathcal{O}(N^{-1}))\end{aligned}$$

The result follow from the asymptotic for the Airy function given in 6.5.1. Again, we can change variables to express

$$\text{He}_{n-1}(2\sqrt{N} + N^{-1/6}\xi) = \sqrt{2\pi} e N^{-\frac{N}{2} + \frac{7}{6}} Ai(\xi)(1 + o(1))$$

6.6 Kernels

When dealing with the asymptotic of the j.p.d.f we will actually consider the Kernel. This is the object described in 3.3.4. However, we will actually work with the simplification given by 3.3.8. The first step is to consider the following expression

$$\text{He}_n(\alpha\sqrt{N} + yN^{-1/2}),$$

which will allow us to take the limit to get $K_n^{(2)}(x, x)$.

6.6.1 Sine Kernel

The computations don't change too much. Consider the expression

$$\begin{aligned}\text{He}_n(\sqrt{N}\alpha + N^{-1/2}y) &= \frac{(N-1)! N^{\frac{1}{2}(1-N)}}{2\pi i} \int_{\Gamma} e^{-N(-\alpha\beta + \frac{1}{2}\beta^2 + \ln(\beta)) + y\beta} d\beta \\ &= C_n^0 \int_{\Gamma} e^{-N\phi_{\alpha}(\beta) + y\beta} d\beta \\ &= C_n^0 \int_{\Gamma} e^{-N\phi_{\alpha}(\beta)} g_y(\beta) d\beta\end{aligned}\tag{6.6.1}$$

with

$$\phi_{\alpha}(\beta) = \frac{1}{2}\beta^2 - \alpha\beta + \ln(\beta); \quad g_y(\beta) = e^{y\beta}$$

The analysis remains the same - the difference now being that we have a function $g_y(\beta)$ that enters in the steepest descent expression. Setting our case for $-2\sqrt{N} < x < 2\sqrt{N}$, or, equivalently, $-2 < \alpha < 2$, we remember that

$\beta_c^\pm = e^{\pm i\gamma}$. Now, with this, and setting $\Delta = 2\sqrt{N} \cos(\gamma) + N^{-1/2}y$,

$$\begin{aligned}
\text{He}_n(\Delta) &= C_n^0 \int_{\Gamma} e^{-N\phi_\alpha(\beta)} g_y(\beta) d\beta = C_n^0 \left[\int_{\Gamma_d^-} e^{-N\phi_\alpha(\beta)} g_y(\beta) d\beta + \int_{\Gamma_d^+} e^{-N\phi_\alpha(\beta)} g_y(\beta) d\beta \right] \\
&\stackrel{(Steepest)}{=} C_n^0 \left(e^{-N(\phi_\alpha(\beta_c^+))} \left[\frac{\sqrt{2\pi}}{N^{\frac{1}{2}}} \frac{g_y(\beta_c^+) e^{i(\theta(\beta_c^+))}}{\sqrt{2\sin(\gamma)}} + \mathcal{O}(N^{-\frac{3}{2}}) \right] + e^{-N(\overline{\phi_\alpha(\beta_c^+)})} \left[\frac{\sqrt{2\pi}}{N^{\frac{1}{2}}} \frac{g_y(\beta_c^-) e^{i(\overline{\theta(\beta_c^+)})}}{\sqrt{2\sin(\gamma)}} + \mathcal{O}(N^{-\frac{3}{2}}) \right] \right) \\
&= C_n^0 \sqrt{\frac{2\pi}{2N\sin(\gamma)}} \left(e^{-N(\phi_\alpha(\beta_c^+))} \left[g_y(\beta_c^+) e^{i(\theta(\beta_c^+))} + \mathcal{O}(N^{-1}) \right] + e^{-N(\overline{\phi_\alpha(\beta_c^+)})} \left[g_y(\beta_c^-) e^{i(\overline{\theta(\beta_c^+)})} + \mathcal{O}(N^{-1}) \right] \right) \\
&= C_n^0 \sqrt{\frac{\pi}{N\sin(\gamma)}} e^{y \cos(\gamma)} \left(e^{-N(\phi_\alpha(\beta_c^+)) + i(\theta(\beta_c^+) + y \sin(\gamma))} + e^{-N(\overline{\phi_\alpha(\beta_c^+)}) + i(\overline{\theta(\beta_c^+)} - y \sin(\gamma))} \right) (1 + \mathcal{O}(N^{-1})) \\
&= C_n^0 \sqrt{\frac{\pi}{N\sin(\gamma)}} e^{y \cos(\gamma)} e^{N(\cos^2(\gamma) + \frac{1}{2})} 2i \sin \left(\theta(\beta_c^+) + y \sin(\gamma) - N \left(\frac{\sin(2\gamma)}{2} - \gamma \right) \right) (1 + \mathcal{O}(N^{-1})) \\
&= \frac{(N-1)! N^{\frac{1}{2}(1-N)}}{2\pi i} \sqrt{\frac{\pi}{N\sin(\gamma)}} e^{y \cos(\gamma)} e^{N(\cos^2(\gamma) + \frac{1}{2})} 2i \sin \left(\theta(\beta_c^+) + y \sin(\gamma) - N \left(\frac{\sin(2\gamma)}{2} - \gamma \right) \right) (1 + \mathcal{O}(N^{-1})) \\
&= \sqrt{\frac{2}{\sin(\gamma)}} N^{\frac{1}{2}(N-1)} e^{N(\cos^2(\gamma) - \frac{1}{2}) + y \cos(\gamma)} \sin \left(\theta(\beta_c^+) + y \sin(\gamma) - N \left(\frac{\sin(2\gamma)}{2} - \gamma \right) \right) (1 + \mathcal{O}(N^{-1}))
\end{aligned}$$

and for $N-1$,

$$\text{He}_{n-1}(\Delta) = \sqrt{\frac{2}{\sin(\gamma)}} N^{\frac{1}{2}(N-2)} \frac{e^{N(\cos^2(\gamma) - \frac{1}{2}) + y \cos(\gamma)}}{e^{\cos^2(\gamma)}} \sin \left(\theta(\beta_c^+) + y \sin(\gamma) - (N-1) \left(\frac{\sin(2\gamma)}{2} - \gamma \right) \right) (1 + \mathcal{O}(N^{-1}))$$

Remembering the expression for the Kernel, taking $\Delta_x = \sqrt{N}\alpha + N^{-1/2}x$ and $\Delta_y = \sqrt{N}\alpha + N^{-1/2}y$

$$\begin{aligned}
K_n^{(2)}(\Delta_x, \Delta_y) &= \frac{\sqrt{w(\Delta_x)w(\Delta_y)}}{r_{n-1}^{(2)}} \frac{H_n(\Delta_x)H_{n-1}(\Delta_y) - H_{n-1}(\Delta_x)H_n(\Delta_y)}{\Delta_x - \Delta_y} \\
&= W_{(N,x,y)} \frac{H_n(\Delta_x)H_{n-1}(\Delta_y) - H_{n-1}(\Delta_x)H_n(\Delta_y)}{(\sqrt{N}\alpha + N^{-1/2}x) - (\sqrt{N}\alpha + N^{-1/2}y)} \\
&= W_{(N,x,y)} N^{1/2} \frac{H_n(\Delta_x)H_{n-1}(\Delta_y) - H_{n-1}(\Delta_x)H_n(\Delta_y)}{x - y}
\end{aligned}$$

with

$$\begin{aligned}
W_{(N,x,y)} &= \frac{\sqrt{w(\sqrt{N}\alpha + N^{-1/2}x)w(\sqrt{N}\alpha + N^{-1/2}y)}}{r_{n-1}^{(2)}} \\
&= \frac{1}{(N-2)!} \left(\frac{\exp \left(-\frac{1}{2} \left(\sqrt{N}\alpha + N^{-1/2}x \right)^2 \right)}{\sqrt{2\pi}} \frac{\exp \left(-\frac{1}{2} \left(\sqrt{N}\alpha + N^{-1/2}y \right)^2 \right)}{\sqrt{2\pi}} \right)^{\frac{1}{2}} \\
&= \frac{1}{\sqrt{2\pi}(N-2)!} \left(\exp \left\{ -\frac{1}{2} (4N \cos^2(\gamma) + 4x \cos(\gamma) + x^2 N^{-1} + 4N \cos^2(\gamma) + 4y \cos(\gamma) + y^2 N^{-1}) \right\} \right)^{\frac{1}{2}} \\
&= \frac{1}{\sqrt{2\pi}(N-2)!} \left(\exp \left\{ -(2N \cos^2(\gamma) + 2x \cos(\gamma) + x^2 (2N)^{-1} + 2N \cos^2(\gamma) + 2y \cos(\gamma) + y^2 (2N)^{-1}) \right\} \right)^{\frac{1}{2}} \\
&= \frac{e^{-2N \cos^2(\gamma)}}{\sqrt{2\pi}(N-2)!} \exp \left\{ -\cos(\gamma)(x+y) - \frac{1}{4N}(x^2 + y^2) \right\}
\end{aligned}$$

Note that

$$H_n(\Delta_x)H_{n-1}(\Delta_y) \propto \sin \left(\theta(\beta_c^+) + x \sin(\gamma) - N \left(\frac{\sin(2\gamma)}{2} - \gamma \right) \right) \sin \left(\theta(\beta_c^+) + y \sin(\gamma) - (N-1) \left(\frac{\sin(2\gamma)}{2} - \gamma \right) \right)$$

and

$$H_{n-1}(\Delta_x)H_n(\Delta_y) \propto \sin \left(\theta(\beta_c^+) + y \sin(\gamma) - N \left(\frac{\sin(2\gamma)}{2} + \gamma \right) \right) \sin \left(\theta(\beta_c^+) + x \sin(\gamma) - (N-1) \left(\frac{\sin(2\gamma)}{2} - \gamma \right) \right)$$

therefore,

$$H_n(\Delta_x)H_{n-1}(\Delta_y) - H_{n-1}(\Delta_x)H_n(\Delta_y) \propto \sin \left(\frac{\sin(2\gamma)}{2} - \gamma \right) \sin(\sin(\gamma)(x-y))$$

Then, letting $x' = x \sin(\gamma)$ and $y' = y \sin(\gamma)$

$$K_n^{(2)}(\Delta_{x'}, \Delta_{y'}) = W_{(N, x', y')} N^{1/2} C e^{\cos(\gamma) \sin(\gamma)(x' + y')} \frac{\sin \left(\frac{\sin(2\gamma)}{2} - \gamma \right)}{\sin(\gamma)} \frac{\sin(x' - y')}{x' - y'}$$

where

$$\begin{aligned} C &= \left(\sqrt{\frac{2}{\sin(\gamma)}} N^{\frac{1}{2}(N-1)} e^{N(\cos^2(\gamma) - \frac{1}{2})} \right) \left(\sqrt{\frac{2}{\sin(\gamma)}} N^{\frac{1}{2}(N-2)} \frac{e^{N(\cos^2(\gamma) - \frac{1}{2})}}{e^{\cos^2(\gamma)}} \right) \\ &= \frac{2}{\sin(\gamma)} N^{N - \frac{3}{2}} \frac{e^{2N(\cos^2(\gamma) - \frac{1}{2})}}{e^{\cos^2(\gamma)}} \end{aligned}$$

and

$$W_{(N, x', y')} = \frac{e^{-2N \cos^2(\gamma)}}{\sqrt{2\pi}(N-2)!} \exp \left\{ -\sin(\gamma) \cos(\gamma)(x' + y') - \frac{\sin^2(\gamma)}{4N}(x'^2 + y'^2) \right\}$$

Here, we get to the point of interest.

Definition 6.6.1. *The Kernel*

$$\mathcal{S}(x, y) := \frac{\sin(x - y)}{x - y}$$

is called the Sine Kernel.

With this definition, we can now say that

Theorem 6.6.2. *For the conditions $-2\sqrt{N} < x < 2\sqrt{N}$, or, equivalently, $-2 < \alpha < 2$, we remember that $\beta_c^\pm = e^{\pm i\gamma}$. Set $\Delta_x = 2\sqrt{N} \cos(\gamma) + \frac{\sin(\gamma)}{\sqrt{N}}x$ and $\Delta_y = 2\sqrt{N} \cos(\gamma) + \frac{\sin(\gamma)}{\sqrt{N}}y$. Then, we have that*

$$K_n^{(2)}(\Delta_x, \Delta_y) = \frac{\sin \left(\frac{\sin(2\gamma)}{2} - \gamma \right)}{e^{\cos^2(\gamma)} \pi \sin^2(\gamma)} \mathcal{S}(x, y) (1 + \mathcal{O}(N^{-1}))$$

6.6.2 Airy Kernel

Chapter 7

Sung-Soo Paper

We are going to discuss in this chapter the ideas developed in the paper [2], where asymptotics were developed for a large class of random matrix ensembles using relatively straightforward asymptotic techniques taking advantage of the orthogonal polynomials present in these models. In the end, we hope to clarify the origin for the novel limit-behavior expression for the so-called *Partition Function* $\mathcal{Z}_N^\beta$. Namely, we get an expression of the type:

$$\log(\mathcal{Z}_{N,\beta=2}^Q) = -I^Q N^2 + \frac{1}{2} N \log N + E^Q N + G \log N + F^Q + o(1).$$

These functions perform the role of the normalization for the probability measure relative to the Coulomb Gas being worked on - a system of interacting particles confined by some potential. In this instance, we deal with complex plane gases, parameterized by some variable $\beta = 2$. This relates in RMT, to the famous *random normal matrix* and *planar symplectic ensembles*, depending on the symmetry imposed. If one takes, for example, the quadratic potential keeping the $2 - \beta$ regime, one gets ensembles correspondent to the *complex* and *planar* Ginibre ensembles, falling back into the classical models. For the results, another restriction is made, the confining potentials must be radially symmetric. For a large class of potentials Q , the first term I^Q was known for a long time, being given by the energy of the associated equilibrium measure. The coefficient $1/2$ of the order $N \log N$, as well as the entropy term E^Q , were known from recent work of Leblé and Serfaty [10], also for rather large classes of Q . The remaining terms in this expansion are new. The term G appears to depend only on whether the limiting spectrum is an annulus or a disc which, surprisingly, seems to not have been observed in the previous literature.

Although some recent work may indicate the direction to expand some of these results to larger classes of potentials [6], this is, so far, still a necessary condition for the techniques presented henceforth. Once the model is established, the techniques are involved, but should be reasonable: We express the partition function in terms of a summation of some orthogonal polynomials, for which we have an integral representation. Then, the integral can be estimated using the Laplace technique - separating the integral in a central main-contributing region and a bounded error-contributing complement region. Finally, the limit on the summation can be estimated with some quadrature techniques, namely the Euler-Maclaurin formula, giving us the final results after computations.

7.1 Some Complex Context

Before jumping into the asymptotics we lay some groundwork on the specific theoretical background needed for the main theorem, such as the RMT ensemble. To introduce this ensemble, we recover the idea of Coulomb Gases and try to connect its complex representative to the particular matrices we are interested in. This will lead us into constructing some Orthogonal Polynomials and, going through some asymptotic methods, we can get to the result.

7.1.1 Coulomb Gases in the Complex Plane

We start by describing a familiar distribution or physicists: The Boltzmann-Gibbs distribution. This is a probability distribution that represents the probability of a given state for a system of particles as a function of the state's "energy" ($\mathcal{H}$) and the "temperature" ($1/(\beta N^2)$) of the system. This is the distribution that maximizes the entropy: lower energy states have higher probability of being occupied than states with higher energy. Under the appropriate conditions, we name Coulomb Gas $\mathbf{p}_N$ the Boltzmann-Gibbs probability measure given in $(\mathbb{R}^d)^N$. The

measure $\mathbf{p}_N$ models an interacting gas of charged indistinguishable particles under some external field. Particles are interpreted to be at positions $x_1, x_2, \dots, x_N \in \mathbb{S}$ of dimension d embedded in some space $\mathbb{R}^n$. We write

$$d\mathbf{p}_N(x_1, x_2, \dots, x_N) = \frac{e^{-\beta N^2 \mathcal{H}_N(\vec{x})}}{\mathcal{Z}_N^{(beta)}} dx_1 dx_2 \dots dx_N$$

to be the measure of such system, where

$$\mathcal{H}_N(\vec{x}) = \frac{1}{N} \sum_{i=1}^N V(x_i) + \frac{1}{2N^2} \sum_{i \neq j}^N g(x_i - x_j)$$

is usually called the Hamiltonian, or more concretely, the energy of the system. The function $V: \mathbb{S} \mapsto \mathbb{R}$ is the external potential. If one wishes $\mathbf{p}_N$ to be a probability measure, then one must ensure that V is such that the normalizing constant $\mathcal{Z}_{N,\beta}$ is finite for all N and the measure support is compact. This is all to say that the potential should be confining: As the number of particles increases, it must counter the repulsion effect and keep limited the region where particles move around. The function $g: \mathbb{S} \mapsto (-\infty, \infty]$ is the *Coulomb Interacting Kernel*, solution to the Poisson equation given by $-\nabla g(\vec{x}) = c_n \delta_0$. The factor βN^2 is called the inverse temperature.

Recall the expression for invariant ensembles and note that, for an appropriate $V: \mathbb{R} \rightarrow \mathbb{R}$, we can take the appropriate dimensions for $d = 1$ and $n = 2$ to recover the, for example, Gaussian ensemble measure

$$\mathbf{p}_N(\vec{x}) = \frac{e^{-\beta N \mathcal{H}_N(\vec{x})}}{\mathcal{Z}_{N,\beta}}, \quad \mathcal{H}_N(\vec{x}) = \frac{1}{N} \sum_{i=1}^N V(x_i) + \frac{1}{N^2} \sum_{i < j}^N \log \frac{1}{|x_i - x_j|}.$$

In this case, we would be dealing with particles on the plane confined to move only on the real line. However, the gases measure accepts a natural extension for any potential that we call acceptable - that obeys some growth and smoothness conditions. Together with the dimensions, we can recover other random matrix models. On the complex plane one could write explicitly

$$d\mathbf{p}_N^{(\beta)}(z_1, \dots, z_N) := \frac{1}{\mathcal{Z}_N^{(beta)}} \prod_{j>k=1}^N |z_j - z_k|^\beta \prod_{j=1}^N e^{-\frac{\beta N}{2} Q(z_j)} dA(z_j) \quad (7.1.1)$$

where $dA(z) := d^2 z / \pi$ is the area measure. This measure would imply the normalization constant - or in our context, the *Partition Function*,

$$\mathcal{Z}_N^{(beta)} := \int_{\mathbb{C}^N} \prod_{j>k=1}^N |z_j - z_k|^\beta \prod_{j=1}^N e^{-\frac{\beta N}{2} Q(z_j)} dA(z_j).$$

This is the expression, if $\beta = 2$ for the measure of the **random normal matrix** ensemble. We still have the potential to tinker, for example, setting $Q(z) = |z|^2$ this would be the so-called **complex Ginibre ensemble**. We could also define the measure for the configurational canonical Coulomb gas ensemble in the upper-half plane - which has an additional complex conjugation symmetry and is governed by the law

$$d\tilde{\mathbf{p}}_N^{(\beta)}(z_1, \dots, z_N) := \frac{1}{\tilde{\mathcal{Z}}_N^{(beta)}} \prod_{j>k=1}^N |z_j - z_k|^\beta |z_j - \bar{z}_k|^\beta \prod_{j=1}^N |z_j - \bar{z}_j|^\beta e^{-\frac{\beta N}{2} Q(z_j)} dA(z_j)$$

and would imply the normalization constant

$$\tilde{\mathcal{Z}}_N^{(beta)} := \int_{\mathbb{C}^N} \prod_{j>k=1}^N |z_j - z_k|^\beta |z_j - \bar{z}_k|^\beta \prod_{j=1}^N |z_j - \bar{z}_j|^\beta e^{-\frac{\beta N}{2} Q(z_j)} dA(z_j)$$

and is related, for $\beta = 2$ to the **planar symplectic ensemble**. For $Q(z) = |z|^2$ this would be the so-called **symplectic Ginibre ensemble**. Virtually all results we saw for the random normal matrix ensemble are valid in the context of the planar symplectic ensemble with only a few adaptations. However, in the name of history telling and simplicity we will omit this measure: Adding this case to our chapter would cloud the point. For now on, we will only consider $\beta = 2$, this has some algebraic advantages that become clear by the Determinantal Point Processes (see Chapter 4 for details) - because of this, we omit the β in the variables ($\mathcal{Z}_N^{(\beta)} = \mathcal{Z}_N$).

7.1.2 Orthogonal Polynomials

For the models of random matrices we are interested in, we must consider $\{P_n(z) = \sum_{j=0}^n a_j z^j\}_n^{(w)}$ a family of orthogonal polynomials in respect to w where P_m has order m . We're dealing with weight functions of the type

$$w(z) = e^{-NQ(z)}$$

where $Q(z)$ is some complex potential that satisfies:

1. We have $Q(z) = q(|z|)$ for some $q : \mathbb{R} \mapsto \mathbb{R}$, i.e., Q is a radially symmetric potential;
2. The potential Q grows sufficiently fast, that is,

$$\liminf_{|z| \rightarrow \infty} \frac{Q(z)}{2 \log(|z|)} > 1;$$

3. Furthermore, assume Q to be smooth, subharmonic in $\mathbb{C}$, and strictly subharmonic in a neighborhood of the droplet.

There is still a question of what are these polynomials. We know by the assumption of orthogonality that

$$\langle P_n, P_m \rangle_w = \begin{cases} 0 & \text{if } m \neq n, \\ 1 & \text{if } m = n. \end{cases}$$

Let us take some arbitrary base of the polynomial space $\{z^j\}_j^{(w)}$. We can apply the Gram Schmidt scheme to get the proper orthogonality in respect to the weight w . If $P_n = x^n$, then the orthogonal element of the basis $\tilde{P}_n$ should be

$$\tilde{P}_n = P_n - \sum_{j=0}^{n-1} \frac{\langle P_n, P_j \rangle_w}{\langle P_j, P_j \rangle_w} P_j$$

but notice that

$$\begin{aligned} \langle P_n, P_j \rangle_w &= \int_{\mathbb{C}} z^{n-j} e^{-NQ(z)} dA(z), \\ &= \int_0^\infty \int_0^{2\pi} r^{n-j} e^{i\theta(n-j)} e^{-Nq(r)} r dr d\theta, \\ &= \int_0^{2\pi} e^{i\theta(n-j)} d\theta \int_0^\infty r^{n-j+1} e^{-Nq(r)} dr, \\ &= 0, \quad \text{since } n \neq j. \end{aligned}$$

Therefore, we have that $\{z^j\}_j^{(w)}$ is the proper orthogonal family. More than that, if we impose the normality,

$$\begin{aligned} \int_{\mathbb{C}} a_n^2 z^{n-n} e^{-NQ(z)} dA(z) &= \int_{\mathbb{C}} a_n^2 e^{-NQ(z)} dA(z) = 1 \\ \implies a_n &= \sqrt{\frac{1}{\int_{\mathbb{C}} w(z) dA(z)}}. \end{aligned}$$

Note that we will have the same family if the internal product is defined in respect to some arbitrary $\rho \neq 0$, that is, if

$$\langle P_n, P_m \rangle_w = \int_{D(\rho)} z^{n-m} e^{-NQ(z)} dA(z),$$

then

$$\begin{aligned} \langle P_n, P_j \rangle_w &= \int_{D(\rho)} z^{n-j} e^{-NQ(z)} dA(z), \\ &= \int_0^\rho \int_0^{2\pi} r^{n-j} e^{i\theta(n-j)} e^{-Nq(r)} r dr d\theta, \\ &= \int_0^{2\pi} e^{i\theta(n-j)} d\theta \int_0^\rho r^{n-j+1} e^{-Nq(r)} dr, \\ &= 0 \quad \text{since } n \neq j. \end{aligned}$$

One can say that the choice of basis was weirdly good. Not to worry, one could also suppose arbitrary polynomials P_m and P_n to see that as

$$P_n(z)\bar{P}_m(z) = \sum_{k=0}^n \sum_{j=0}^m a_k b_j z^{k-j},$$

then the inner product I can be written as

$$\begin{aligned} I &= \sum_{k=0}^n \sum_{j=0}^m a_k b_j \int_0^\infty \int_0^{2\pi} r^{k-j+1} e^{i(k-j)\theta} e^{-Nq(r)} dr d\theta, \\ &= \sum_{k=0}^n \sum_{j=0}^m a_k b_j \int_0^{2\pi} e^{i(k-j)\theta} d\theta, \int_0^\infty r^{k-j+1} e^{-Nq(r)} dr \\ &= 2\pi \sum_{j=0}^n a_j b_j \int_0^\infty r e^{-Nq(r)} dr, \end{aligned}$$

where the same argument as before, in the angular integral, was used to annihilate the off-diagonal terms of the summation. Now, note that we need

$$\sum_{j=0}^n a_j b_j = 0 \quad \text{for } m \neq n.$$

This is easily satisfied by the family of monomials as $b_j = 0 \ \forall j < m$.

7.2 Droplet Results

We will use some potential $Q(z) : \mathbb{C} \rightarrow \mathbb{R}$ which satisfies some nice growth and smoothness conditions, as specified in (3). Because of the radial symmetry, one can take some function $q : \mathbb{R} \mapsto \mathbb{R}$ such that $q(|z|) = Q(z)$; then we can say that the previous conditions on the potential Q implies that $rq'(r)$ is increasing in $\mathbb{C}$ and strictly increasing in the droplet. Using (3.3.9), one can write the free energy

$$\ln \mathcal{Z}_N = \ln N! - 2 \sum_{j=0}^{N-1} \log h_j, \quad h_k := \int_{\mathbb{C}} |z|^{2j} e^{-NQ(z)} dA(z). \quad (7.2.1)$$

where h_k is the norming constant for the monomial family of orthogonal polynomials in respect to w . One could also write, for the planar symplectic ensemble

$$\ln \tilde{\mathcal{Z}}_N = \ln N! - 2 \sum_{j=0}^{N-1} \log 2\tilde{h}_{2j+1} \quad \tilde{h}_k := \int_{\mathbb{C}} |z|^{2j} e^{-2NQ(z)} dA(z).$$

The expressions derived for the normal ensemble would be easily adapted for this model and it is done in [2], however we will not follow that computation in name of clarity and storytelling.

In order to derive the asymptotic formulas for the free energy we need to derive the asymptotics for the orthogonal polynomials and its norming constants. The first step in the asymptotics for the normal matrices is to prepare for the Laplace method. Our goal is to identify a region that gives the main contribution, in order, for value of the integral - and consequently to identify a error-contributing region of the integration region. Let's reformulate our integral to identify the integral as usual in the Laplace method, such as in Section 6.3. We know $Q(z) = q(|z|)$ a radially symmetric potential such that, defining $z := r e^{i\theta}$, that is, $|z| = r$, we would have

$$\begin{aligned} h_j &= \int_{\Theta} \int_{\mathbb{R}_+} r^{2j} e^{-Nq(r)} \frac{r}{\pi} dr d\theta \\ &= \frac{2\pi}{\pi} \int_{\mathbb{R}_+} r^{2j+1} e^{-Nq(r)} dr \\ &= \int_{\mathbb{R}_+} 2r e^{-N(q(r) - \frac{2j}{N} \log(r))} dr \end{aligned}$$

We first redefine our potential such that the integrand has an expression better suited to apply the Laplace method. Using $V_{\tau(j)} = q(r) - 2\tau(j) \log(r)$, where $\tau(j) := j/N \in [0, 1]$, one can rewrite the integral as

$$h_j = \int_{\mathbb{R}_+} 2r e^{-NV_{\tau(j)}(r)} dr$$

Where we have set the stage for the Laplace technique. Note that now τ will perform the function of the previous index j and relates to which coefficient we're calculating. Taking a look at the first derivative of the newly defined potential we find for the critical point

$$V'_\tau(r) = q'(r) - \frac{2\tau}{r} \implies r_c q'(r_c) = 2\tau.$$

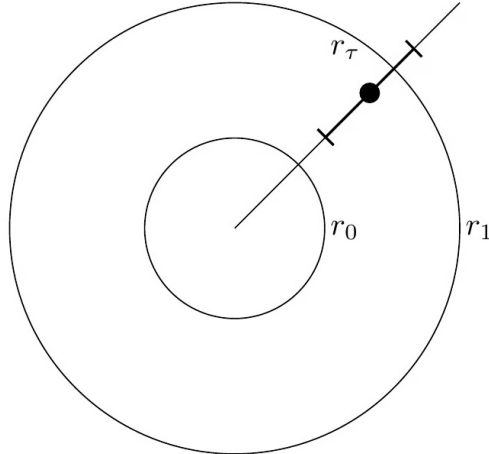
This is: We can find a critical point of the potential r_c , parameterized by the degree of the polynomial we are calculating, or more concretely, by τ . Of course, because we know the range of τ , it is worth defining the limit points

$$\begin{cases} r_0, & \text{the largest solution of } r_0 q'(r_0) = 0 \\ r_1, & \text{the smallest solution of } r_1 q'(r_1) = 2 \end{cases}$$

Our main goal here is to evaluate the integral using the surroundings of the critical point. If this point is unique, then we need no more than to calculate the integral in a single small interval around $r_c(\tau)$. This is, of course, true: We have mentioned that the properties on the potential directly implies that $r q'(r)$ is a strictly increasing function on $[r_0, r_1]$, then the solution must be unique. Now, as outside the droplet the growth is not strict, the definitions of r_0 and r_1 had to be as mentioned before: Respectively the largest and smallest solution to the proper instance of $r q'(r) = 2\tau$. An important distinction must be made between the cases when $r_0 = 0$ and $r_0 > 0$, corresponding to Subsections 7.2.2 and 7.2.1, respectively. We begin noticing that $r_\tau | r_\tau q'(r_\tau) = 2\tau$ is an increasing function on τ and that $r_0 \leq r_\tau \leq r_1 \quad \forall \tau \in [0, 2]$.

7.2.1 Asymptotics - Annulus

We start by considering, in this regime, $r_0 > 0$. We know the measure (7.1.1) to be the law for the distribution of the particles in this problem. When we say that $r_0 > 0$, we are saying that the region of space $[r_0, r_1]$ where this measure is non-zero for $N \rightarrow \infty$ has the annulus shape. If we think of the equivalent Coulomb gas, this means the particles will concentrate in this shape when we perform an asymptotic on N . This is the support for the limit measure.



Asymptotic for the orthogonal polynomial

We begin as is usual when applying asymptotic techniques in integrals; separating a central part where the integral value is concentrated and another part where we can control the order of contribution. This is of course possible because the integrand is proportional to some exponential, with negative exponent, with a unique critical

point, around which, the value of the expression concentrates. Define $\delta_N = \log(N)/\sqrt{N}$ - this will be our the cut-size for the interest region and will be justified during the calculations -, then

$$h_j = \int_{|r-r_j|>\delta_N} 2r e^{-NV_{\tau(j)}(r)} dr + \int_{|r-r_j|<\delta_N} 2r e^{-NV_{\tau(j)}(r)} dr = (1) + (2).$$

As mentioned, the first integral must be somehow controlled and the second estimated as the main value for the result.

Theorem 7.2.1. *We have the following asymptotic for h_j ,*

$$h_j = \frac{e^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} \left(\frac{8\pi r_{\tau(j)}^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{1}{2}} \left[1 + \frac{1}{N} \mathfrak{B}_1(r_{\tau(j)}) + \mathcal{O}(N^{-2}) \right]. \quad (7.2.2)$$

where

$$\mathfrak{B}_1(r_{\tau(j)}) = \frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^2} \frac{1}{4} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^{\frac{5}{2}}} \frac{5}{12} + \frac{1}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \frac{2}{3r_{\tau(j)}}$$

Proof. Let's first show that we can control the first part of the integral, for that, rewrite

$$\begin{aligned} (1) &= e^{-NV_{\tau(j)}(r_{\tau(j)})} \int_{|r-r_j|>\delta_N} 2r e^{-N(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)}))} dr, \\ &= e^{-NV_{\tau(j)}(r_{\tau(j)})} \int_{|r-r_j|>\delta_N} 2r e^{-N(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)}))} e^{-(M-M)(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)}))} dr, \\ &= e^{-NV_{\tau(j)}(r_{\tau(j)})} \int_{|r-r_j|>\delta_N} 2r e^{-M(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)}))} e^{-(N-M)(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)}))} dr. \end{aligned}$$

Now, we need some way to estimate the difference $(V_{\tau(j)}(r) - V_{\tau(j)}(r_{\tau(j)}))$. If we are able to say that this difference is at least $\mathcal{O}(N^{-1})$, we could control the growth of the integral choosing M to be large enough such that

$$\int_{|r-r_j|>\delta_N} \left(e^{-q(r)} r^{2\tau(j)} \right)^M r dr$$

converges. Using the definition of the function V and the mean value theorem, we start by evaluating the difference $V_{\tau(j+1)}(r_{\tau(j+1)}) - V_{\tau(j)}(r_{\tau(j)})$, in the following manner

$$\begin{aligned} V_{\tau(j+1)}(r_{\tau(j+1)}) - V_{\tau(j)}(r_{\tau(j)}) &= q(r_{\tau(j+1)}) - q(r_{\tau(j)}) - 2(\tau(j+1) \log(r_{\tau(j+1)}) - \tau(j) \log(r_{\tau(j)})), \\ &= q'(c)(r_{\tau(j+1)} - r_{\tau(j)}) - \frac{2}{N} \left(\log(r_{\tau(j+1)}) + j \log\left(\frac{r_{\tau(j+1)}}{r_{\tau(j)}}\right) \right), \\ &= q'(c)r'(c)(\tau(j+1) - \tau(j)) - \frac{2}{N} (\log(r_{\tau(j+1)}) + j\mathcal{O}(N^{-1})), \\ &= q'(c)r'(c)\mathcal{O}(N^{-1}) - \mathcal{O}(N^{-1}), \\ &= \mathcal{O}(N^{-1}). \end{aligned}$$

The last equality holds as both functions are continuous and c is some value that is limit by some compact disk on the plane. Then, we can add zero and use the estimation

$$V_{\tau(j+1)}(r_{\tau(j+1)}) - V_{\tau(j+1)}(r) + V_{\tau(j+1)}(r) - V_{\tau(j)}(r_{\tau(j)}) = \mathcal{O}(N^{-1}).$$

to derive the result we are after. It is enough to estimate $V_{\tau(j+1)}(r) - V_{\tau(j)}(r_{\tau(j)})$. For that, because $rq'(r)$ is increasing, for all $r > r_{\tau(j)} + \delta_N$ we can say that

$$V_{\tau(j+1)}(r) \geq V_{\tau(j+1)}(r_{\tau(j)} + \delta_N) = V_{\tau(j+1)}(r_{\tau(j+1)}) + \frac{1}{2} V_{\tau(j+1)}''(r_{\tau(j+1)})(r_{\tau(j)} + \delta_N - r_{\tau(j+1)})^2 + \mathcal{O}(\delta_N^3),$$

and therefore can write that

$$\begin{aligned} V_{\tau(j+1)}(r) - V_{\tau(j)}(r_{\tau(j)}) &= \frac{1}{2} V''_{\tau(j+1)}(r_{\tau(j+1)})(r_{\tau(j)} + \delta_N - r_{\tau(j+1)})^2 + \mathcal{O}(\delta_N^3) \\ &= \delta^2 c + \mathcal{O}(N^{-1}) + \mathcal{O}(\delta_N^3) \\ &\geq \delta^2 c \end{aligned}$$

Finally, using this one can write

$$V_{\tau(j+1)}(r) - V_{\tau(j+1)}(r_{\tau(j+1)}) \geq c\delta_N^2.$$

Getting this inequality back to the integral expression we can write that

$$\begin{aligned} (1) &= e^{-NV_{\tau(j)}(r_{\tau(j)})} \int_{|r-r_j|>\delta_N} 2r e^{-M(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)}))} e^{-(N-M)(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)}))} dr \\ &= e^{-NV_{\tau(j)}(r_{\tau(j)})} e^{-c\delta_N^2(N-M)} \int_{|r-r_j|>\delta_N} 2r e^{-M(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)}))} dr \end{aligned}$$

where you can check that

$$\begin{aligned} \int_{|r-r_j|>\delta_N} 2r e^{-M(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)}))} dr &= \int_{|r-r_j|>\delta_N} 2r e^{-M((q(r)-2\tau(j)\log(r))-(q(r_{\tau(j)})-2\tau(j)\log(r_{\tau(j)})))} dr \\ &= 2 \left(e^{-q(r_{\tau(j)})} r_{\tau(j)}^{-2\tau(j)} \right)^M \int_{|r-r_j|>\delta_N} \left(e^{-q(r)} r^{2\tau(j)} \right)^M r dr. \end{aligned}$$

We chose M to be a big enough number such that the integral above converges. If that is true,

$$(1) = \int_{|r-r_j|>\delta_N} 2r e^{-NV_{\tau(j)}(r)} dr = e^{-NV_{\tau(j)}(r_{\tau(j)})} \epsilon_N$$

with $\epsilon_N = \mathcal{O}(e^{-c(\log(N))^2})$.

Now we must deal with the second part of the integral. We begin by setting $x = r - r_{\tau(j)}$, then

$$\begin{aligned} (2) &= \int_{-\delta_N}^{\delta_N} 2(x + r_{\tau(j)}) e^{-NV_{\tau(j)}(x+r_{\tau(j)})} dx \\ &= e^{-NV_{\tau(j)}(r_{\tau(j)})} \int_{-\delta_N}^{\delta_N} 2(x + r_{\tau(j)}) e^{-N[V_{\tau(j)}(x+r_{\tau(j)})-V_{\tau(j)}(r_{\tau(j)})]} dx \end{aligned}$$

By re-scaling the integral using $y = \sqrt{N}x$ we get yet

$$(2) = \frac{e^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} \int_{-\log(N)}^{\log(N)} 2 \left(\frac{y}{\sqrt{N}} + r_{\tau(j)} \right) e^{-N[V_{\tau(j)}(\frac{y}{\sqrt{N}}+r_{\tau(j)})-V_{\tau(j)}(r_{\tau(j)})]} dy$$

Now, the idea here is quite direct: to use the Taylor expansion for $V_{\tau(j)}\left(\frac{y}{\sqrt{N}} + r_{\tau(j)}\right)$ near the critical point $r_{\tau(j)}$. That is,

$$\begin{aligned} (2) &= \frac{e^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} \int_{-\log(N)}^{\log(N)} 2 \left(\frac{y}{\sqrt{N}} + r_{\tau(j)} \right) e^{-N \left(\frac{y^2}{2N} V_{\tau(j)}^{(2)}(r_{\tau(j)}) + \frac{y^3}{3!N^{\frac{3}{2}}} V_{\tau(j)}^{(3)}(r_{\tau(j)}) + \frac{y^4}{4!N^2} V_{\tau(j)}^{(4)}(r_{\tau(j)}) + \mathcal{O}\left(\left(\frac{y}{\sqrt{N}}\right)^5\right) \right)} dy \\ &= 2r_{\tau(j)} \frac{e^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} \int_{-\log(N)}^{\log(N)} \left(\frac{y}{r_{\tau(j)}\sqrt{N}} + 1 \right) e^{-\frac{y^2}{2} V_{\tau(j)}^{(2)}(r_{\tau(j)}) - \frac{y^3}{3!\sqrt{N}} V_{\tau(j)}^{(3)}(r_{\tau(j)}) - \frac{y^4}{4!N} V_{\tau(j)}^{(4)}(r_{\tau(j)}) + \mathcal{O}\left(\frac{y^5}{N^{\frac{3}{2}}}\right)} dy \\ &= 2r_{\tau(j)} \frac{e^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} \int_{-\log(N)}^{\log(N)} \left(\frac{y}{r_{\tau(j)}\sqrt{N}} + 1 \right) e^{-\frac{y^2}{2} V_{\tau(j)}^{(2)}(r_{\tau(j)})} \left(e^{-\frac{y^3}{3!\sqrt{N}} V_{\tau(j)}^{(3)}(r_{\tau(j)}) - \frac{y^4}{4!N} V_{\tau(j)}^{(4)}(r_{\tau(j)}) + \mathcal{O}\left(\frac{y^5}{N^{\frac{3}{2}}}\right)} \right) dy \end{aligned}$$

Now we take the Taylor Series for

$$\begin{aligned} e^{-\frac{y^3}{3!\sqrt{N}} V_{\tau(j)}^{(3)}(r_{\tau(j)}) - \frac{y^4}{4!N} V_{\tau(j)}^{(4)}(r_{\tau(j)}) + \mathcal{O}\left(\frac{y^5}{N^{\frac{3}{2}}}\right)} &= 1 - \left[\frac{y^3}{3!\sqrt{N}} V_{\tau(j)}^{(3)}(r_{\tau(j)}) + \frac{y^4}{4!N} V_{\tau(j)}^{(4)}(r_{\tau(j)}) + \mathcal{O}\left(\frac{y^5}{N^{\frac{3}{2}}}\right) \right] \\ &\quad - \frac{1}{2} \left[\frac{y^6}{N} \left(\frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})}{6} \right)^2 + \frac{y^8}{N^2} \left(\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{24} \right)^2 + \frac{y^7}{N^3} \left(\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{6} \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})}{24} \right) + \mathcal{O}\left(\frac{y^8}{N^2}\right) \right] + \mathcal{O}\left(\frac{y^a}{N^b}\right) \end{aligned}$$

to get the asymptotic expansion (since odd terms vanish) we reorganize

$$(2) = 2r_{\tau(j)} \frac{e^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} \int_{-\log(N)}^{\log(N)} e^{-\frac{y^2}{2} V_{\tau(j)}^{(2)}(r_{\tau(j)})} \left[1 - \frac{y^4}{3!r_{\tau(j)}N} V_{\tau(j)}^{(3)}(r_{\tau(j)}) - \frac{y^4}{4!N} V_{\tau(j)}^{(4)}(r_{\tau(j)}) - \frac{y^6}{2 \cdot (3!)^2 N} V_{\tau(j)}^{(3)}(r_{\tau(j)})^2 \right] dy$$

then

$$(2) = 2r_{\tau(j)} \frac{e^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} \left[\left(-\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{24N} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})}{6r_{\tau(j)}N} \right) \int_{-\log(N)}^{\log(N)} e^{-\frac{y^2}{2} V_{\tau(j)}^{(2)}(r_{\tau(j)})} y^4 \right. \\ \left. - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2}{72N} \int_{-\log(N)}^{\log(N)} e^{-\frac{y^2}{2} V_{\tau(j)}^{(2)}(r_{\tau(j)})} y^6 + \int_{-\log(N)}^{\log(N)} e^{-\frac{y^2}{2} V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right]$$

the result can be directly evaluated using the Gaussian integrals

$$\int_{\mathbb{R}} e^{-2ar^2} dr = \sqrt{\frac{\pi}{2}} \frac{1}{a^{1/2}} \implies \int_{\mathbb{R}} e^{-2\left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4}\right)y^2} y^2 = \sqrt{\frac{\pi}{2}} \left(\frac{4}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{1}{2}}$$

$$\int_{\mathbb{R}} e^{-2ar^2} r^4 dr = \sqrt{\frac{\pi}{2}} \frac{3}{16} \frac{1}{a^{5/2}} \implies \int_{\mathbb{R}} e^{-2\left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4}\right)y^2} y^4 = \sqrt{\frac{\pi}{2}} \frac{3}{16} \left(\frac{4}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{5}{2}}$$

$$\int_{\mathbb{R}} e^{-2ar^2} r^6 dr = \sqrt{\frac{\pi}{2}} \frac{15}{64} \frac{1}{a^{7/2}} \implies \int_{\mathbb{R}} e^{-2\left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4}\right)y^2} y^6 = \sqrt{\frac{\pi}{2}} \frac{15}{64} \left(\frac{4}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{7}{2}}$$

we can finally calculate

$$(2) = \sqrt{2\pi} r_{\tau(j)} \frac{e^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} \left[\left(-\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{24N} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})}{6r_{\tau(j)}N} \right) \left(\frac{4}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{1}{2}} \right. \\ \left. - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2}{72N} \frac{3}{16} \left(\frac{4}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{5}{2}} + \frac{15}{64} \left(\frac{4}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{7}{2}} \right]$$

and rearranging the terms

$$(2) = \frac{e^{-NV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{N}} \left(\frac{8\pi r_{\tau(j)}^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{1}{2}} \left[1 + \frac{1}{N} B_1(r_{\tau(j)}) + \mathcal{O}(N^{-2}) \right]$$

where

$$\mathfrak{B}_1(r_{\tau(j)}) = \frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^2} \frac{1}{4} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^{\frac{5}{2}}} \frac{5}{12} + \frac{1}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \frac{2}{3r_{\tau(j)}}$$

Notice that the error term of (1) is absorbed into the error order of 7.2.1. Therefore we have the asymptotic expression for h_j given by (7.2.2). □

Asymptotics on the Summation

So now we need to return to (7.2.1) and calculate

$$\sum_{j=0}^{N-1} \log h_j = \sum_{j=0}^{N-1} -NV_{\tau(j)}(r_{\tau(j)}) + \frac{1}{2} \left(\log(8\pi r_{\tau(j)}^2) - \log N - \log V_{\tau(j)}^{(2)}(r_{\tau(j)}) \right) + \frac{1}{N} \mathfrak{B}_1(r_{\tau(j)}) + \mathcal{O}(N^{-2}) \quad (7.2.3)$$

Theorem 7.2.2.

$$-2N^2 I_Q[\mu_Q] + N(3 \log(2\pi) - E_Q[\mu_Q] - 1) + \frac{1}{2} \log(N) + 2 \int_S \mathfrak{B}_1 d\mu_Q + \mathcal{O}(N^{-1})$$

where we have set

$$\mathfrak{B}_1(r_{\tau(j)}) = \frac{V_{\tau_j}^{(4)}(r_{\tau(j)})}{V_{\tau_j}^{(2)}(r_{\tau(j)})^2} \frac{1}{4} - \frac{V_{\tau_j}^{(3)}(r_{\tau(j)})^2}{V_{\tau_j}^{(2)}(r_{\tau(j)})^{\frac{5}{2}}} \frac{5}{12} + \frac{1}{V_{\tau_j}^{(2)}(r_{\tau(j)})} \frac{2}{3r_{\tau(j)}},$$

$$I_Q[\mu_Q] = \left[\frac{1}{4} \int_{r_0}^{r_1} q(s) s V_{\tau_j}^{(2)}(s) ds - \log(r_1) + \frac{1}{2} q(r_1) \right]$$

and

$$E_Q[\mu_Q] = \frac{1}{2} \int_{r_0}^{r_1} \log(V^{(2)}(s)) (sq'(s))' ds.$$

Proof. For each term depending on j we need an summation expression, we first look at the first term using the Euler-Maclaurin formula

$$\sum_{j=0}^{N-1} V_{\tau(j)}(r_{\tau(j)}) = \int_0^N V_{\tau(j)}(r_{\tau(j)}) dt - \frac{1}{2} (V_{\tau(j)}(r_{\tau(j)}) - V_{\tau(0)}(r_{\tau(0)})) + \frac{1}{12} [\partial_t V_{\tau(j)}(r_{\tau(j)})|_{t=N} - \partial_t V_{\tau(j)}(r_{\tau(j)})|_{t=0}] + \mathcal{O}(N^{-3}). \quad (7.2.4)$$

The first term in (7.2.4) is solved doing a integral by parts. We remember the expression for the potential $V_{\tau(j)}(r_{\tau(j)}) = q(r_{\tau(j)}) - 2\tau(j) \log r$, with $\tau(j) = j/N$. Therefore the differential can be written as

$$\begin{aligned} d(r_{\tau(j)} q'(r_{\tau(j)})) &= 2d\tau(j) \\ dr q'(r) + dr r q''(r) &= 2d\tau(j) \\ dr(q'(r) + r q''(r)) &= 2d\tau(j). \end{aligned}$$

This implies that the derivative for r is

$$\frac{dr}{d\tau} = \frac{2}{r_{\tau(j)} V_{\tau_j}^{(2)}(r_{\tau(j)})} = \frac{2}{(r q'(r))'}.$$

Note that in the critical point $r_{\tau(j)}$, we should have $r_{\tau(j)} q'(r_{\tau(j)}) = 2\tau$. We then are able to rewrite, setting $s = r_{\tau(j)}$,

$$\begin{aligned} \int_0^N V_{\tau(t)}(r_{\tau(t)}) dt &= \int_0^N (q(r_{\tau(t)}) - 2\tau(t) \log(r_{\tau(t)})) dt \\ &= \frac{1}{2} \int_{r_0}^{r_1} (q(s) - s q'(s) \log(s)) s V_{\tau_j}^{(2)}(s) N ds \\ &= N \left[\frac{1}{2} \int_{r_0}^{r_1} q(s) s V_{\tau_j}^{(2)}(s) ds - \frac{1}{2} \int_{r_0}^{r_1} s^2 q'(s) \log(s) V_{\tau_j}^{(2)}(s) ds \right] \end{aligned}$$

and finally using that $r_1 q'(r_1) = 2$ and $r_0 q'(r_0) = 0$ we can apply the integration by parts on the second term and

get

$$\begin{aligned}
\frac{1}{2} \int_{r_0}^{r_1} s^2 q'(s) \log(s) V_{\tau_j}^{(2)}(s) ds &= \frac{1}{2} \int_{r_0}^{r_1} s^2 q'(s) \log(s) \frac{1}{s} (sq'(s))' ds \\
&= \frac{1}{2} \int_{r_0}^{r_1} sq'(s) \log(s) (sq'(s))' ds \\
&= \frac{1}{4} \int_{r_0}^{r_1} ((sq'(s))^2)' \log(s) ds \\
&= \frac{1}{4} (sq'(s))^2 \log(s) \Big|_{r_0}^{r_1} + \frac{1}{4} \int_{r_0}^{r_1} (sq'(s))^2 \frac{1}{s} ds \\
&= \log(r_1) + \frac{1}{4} \int_{r_0}^{r_1} sq'(s)^2 ds \\
&= \log(r_1) + \frac{1}{2} q(r_1) - \frac{1}{4} \int_{r_0}^{r_1} q(s) (sq'(s))' ds
\end{aligned}$$

Leaving us with

$$\begin{aligned}
\int_0^N V_{\tau(j)}(r_{\tau(j)}) dt &= N \left[\frac{1}{2} \int_{r_0}^{r_1} q(s) s V_{\tau_j}^{(2)}(s) ds - \log(r_1) + \frac{1}{2} q(r_1) - \frac{1}{4} \int_{r_0}^{r_1} q(s) (sq'(s))' ds \right] \\
&= N \left[\frac{1}{4} \int_{r_0}^{r_1} q(s) s V_{\tau_j}^{(2)}(s) ds - \log(r_1) + \frac{1}{2} q(r_1) \right] \\
&:= NI_Q[\mu_Q]
\end{aligned} \tag{7.2.5}$$

We also need an expression for $V_{\tau(j)}(r_{\tau(j)}) - V_{\tau(0)}(r_{\tau(0)})$. This is direct noting that

$$V_{\tau(j)}(r_{\tau(j)}) - V_{\tau(0)}(r_{\tau(0)}) = V_1(r_1) - V_0(r_0) = q(r_1) - q(r_0) - 2 \log(r_1) =: 2U_{\mu_Q}(0) \tag{7.2.6}$$

And finally we can the remaining term , we use the Leibniz rule and obtain

$$\begin{aligned}
\partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=N} &= \partial_t (q(r_{\tau(t)}) - 2\tau(t) \log(r_{\tau(t)}))|_{t=N} \\
&= \partial_\tau (q(r_{\tau(t)}) - 2\tau(t) \log(r_{\tau(t)}))|_{\tau=1} \partial_t (\tau(t))|_{t=N} \\
&= \frac{1}{N} (\partial_\tau (q(r_\tau)) - 2(\log(r_\tau) + \tau \partial_\tau (\log(r_\tau)))|_{\tau=1}) \\
&= \frac{1}{N} [\partial_t (q(r_{\tau(t)}) - 2 \log(r_{\tau(t)}))|_{\tau=1} - 2 \log(r_1)]
\end{aligned}$$

and

$$\partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=0} = \frac{1}{N} [\partial_t (q(r_{\tau(t)}) - 2 \log(r_{\tau(t)}))|_{\tau=0} - 2 \log(r_0)].$$

And since we can calculate, from $r_1 q'(r_1) = 2$ and $r_0 q'(r_0) = 0$

$$\begin{aligned}
\partial_\tau q(r_\tau)|_{\tau=0} &= \left(\frac{2q'(r)}{(rq'(r))'} \right)_{r_0} = 0, \\
\partial_\tau q(r_\tau)|_{\tau=1} &= \left(\frac{2q'(r)}{(rq'(r))'} \right)_{r_1} = \frac{4}{r_1(q'(r_1) + r_1 q''(r_1))}, \\
\partial_\tau (2 \log(r_\tau))|_{\tau=1} &= \left(\frac{4}{r(rq'(r))'} \right)_{r_1} = \frac{4}{r_1(q'(r_1) + r_1 q''(r_1))},
\end{aligned}$$

we can also express the difference as

$$\partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=N} - \partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=0} = \frac{1}{12N} [2(\log(r_0) - \log(r_1))] = \frac{1}{6N} \log\left(\frac{r_0}{r_1}\right). \tag{7.2.7}$$

And the proof for the summation $\sum_{j=0}^{N-1} V_{\tau(j)}(r_{\tau(j)})$ is complete if one gathers the terms (7.2.5), (7.2.7) and (7.2.6),

$$-N \sum_{j=0}^{N-1} V_{\tau(j)}(r_{\tau(j)}) = -N^2 I_Q[\mu_Q] - 2NU_{\mu_Q}(0) - \frac{1}{6} \log\left(\frac{r_0}{r_1}\right) + \mathcal{O}(N^{-2}) \tag{7.2.8}$$

We move on to the next term from where we need to determine both terms in $\sum_{j=0}^{N-1} \left(\log(2\pi r_{\tau(j)}^2) - \log V_{\tau_j}^{(2)}(r_{\tau(j)}) \right)$. Lets start now with the second summation, to which we apply Euler-Maclaurin again

$$\sum_{j=0}^{N-1} \log V_{\tau_j}^{(2)}(r_{\tau(j)}) = \int_0^N \log V_{\tau_t}^{(2)}(r_{\tau(t)}) dt - \frac{1}{2} \log \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) + \frac{1}{12} \left[\partial_t \log V_{\tau_t}^{(2)}(r_{\tau(t)})|_{t=1} - \partial_t \log V_{\tau_t}^{(2)}(r_{\tau(t)})|_{t=0} \right] + \mathcal{O}(N^{-3})$$

We first tackle the integral on the right hand side, changing variables as $s = r_{\tau(t)}$ we would have

$$\int_0^N \log V_{\tau_t}^{(2)}(r_{\tau(t)}) dt = \frac{N}{2} \int_{r_0}^{r_1} \log(V^{(2)}(s))(sq'(s))' ds$$

And, noticing that

$$\partial_t \log V_{\tau_t}^{(2)}(r_{\tau(t)}) = \partial_V \left(\log V_{\tau_t}^{(2)}(r_{\tau(t)}) \right) \partial_r \left(V_{\tau_t}^{(2)}(r_{\tau(t)}) \right) \partial_r (r_{\tau(t)}) = \frac{(\partial_r V^{(2)})(r_{\tau(t)})}{r_{\tau(t)} (V^{(2)}(r_{\tau(t)}))^2}$$

We have for the third term on the right hand side,

$$\partial_t \log V_{\tau_t}^{(2)}(r_{\tau(t)})|_{t=1} - \partial_t \log V_{\tau_t}^{(2)}(r_{\tau(t)})|_{t=0} = \frac{1}{2N} \left(\frac{(\partial_r V^{(2)})(r_1)}{r_1 (V^{(2)}(r_1))^2} - \frac{(\partial_r V^{(2)})(r_0)}{r_0 (V^{(2)}(r_0))^2} \right) = \mathcal{O}(N^{-1})$$

then,

$$\sum_{j=0}^{N-1} \log V_{\tau_j}^{(2)}(r_{\tau(j)}) = \frac{N}{2} \int_{r_0}^{r_1} \log(V^{(2)}(s))(sq'(s))' ds - \frac{1}{2} \log \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) + \mathcal{O}(N^{-1}) \quad (7.2.9)$$

Similarly, for the first summation, we apply Euler-Maclaurin again

$$\sum_{j=0}^{N-1} \log(8\pi r_{\tau(j)}^2) = \log(8\pi) + 2 \left(\int_0^N \log(r_{\tau(t)}) dt - \frac{1}{2} \log \left(\frac{r_1}{r_0} \right) + \frac{1}{12} \left[\partial_t \log r_{\tau(t)}|_{t=1} - \partial_t \log r_{\tau(t)}|_{t=0} \right] + \mathcal{O}(N^{-3}) \right)$$

We first tackle the integral on the right hand side, changing variables as $s = r_{\tau(t)}$ we would have

$$\int_0^N \log r_{\tau(t)} dt = \frac{N}{2} \int_{r_0}^{r_1} \log(s)(sq'(s))' ds$$

And, noticing that

$$\partial_t \log r_{\tau(t)} = \partial_r (\log(r_{\tau(t)})) \partial_r (r_{\tau(t)}) = \frac{2}{V^{(2)}(r_{\tau(t)}) (r_{\tau(t)})^2}$$

We have for the third term on the right hand side,

$$\partial_t \log r_{\tau(t)}|_{t=1} - \partial_t \log r_{\tau(t)}|_{t=0} = \frac{1}{2N} \left(\frac{2}{V^{(2)}(r_1) (r_1)^2} - \frac{2}{V^{(2)}(r_0) (r_0)^2} \right) = \mathcal{O}(N^{-1})$$

then

$$\sum_{j=0}^{N-1} \log(8\pi r_{\tau(j)}^2) = N \log(8\pi) + 2 \left(\frac{N}{2} \int_{r_0}^{r_1} \log(s)(sq'(s))' ds - \frac{1}{2} \log \left(\frac{r_1}{r_0} \right) + \mathcal{O}(N^{-1}) \right) \quad (7.2.10)$$

With all of this, gathering the terms in (7.2.3) with the expressions in (7.2.8), (7.2.9) and (7.2.10), one can now finally express

$$\sum_{j=0}^{N-1} \log(h_j) = -N^2 I_Q[\mu_Q] - \frac{N}{2} \log N + N \left(\frac{3 \log(2\pi)}{2} - \frac{1}{2} E_Q[\mu_Q] \right) - \frac{1}{2} \log \left(\frac{r_1}{r_0} \right) + \frac{1}{4} \log \left(\frac{V^{(2)}(r_1)}{V^{(2)}(r_0)} \right) + \int_S \mathfrak{B}_1 d\mu_Q + \mathcal{O}(N^{-1})$$

where

$$E_Q[\mu_Q] = \frac{1}{2} \int_{r_0}^{r_1} \log(V^{(2)}(s))(sq'(s))' ds$$

from which we can get the asymptotic expansion using that

$$\log(N!) = N \log(N) - N + \frac{1}{2} \log(N) + \frac{1}{2} \log(2\pi) + \mathcal{O}(N^{-1})$$

that is,

$$\begin{aligned} \ln Z_N &= \ln N! - 2 \sum_{j=0}^{N-1} \log h_j \\ &= N \log(N) - N + \frac{1}{2} \log(N) + \frac{1}{2} \log(2\pi) + 2 \left(-N^2 I_Q[\mu_Q] - \frac{N}{2} \log N + N \left(\frac{3 \log(2\pi)}{2} - \frac{1}{2} E_Q[\mu_Q] \right) - \frac{1}{2} \log \left(\frac{r_1}{r_0} \right) + \frac{1}{4} \log \right. \\ &= -2N^2 I_Q[\mu_Q] + N (3 \log(2\pi) - E_Q[\mu_Q] - 1) + \frac{1}{2} \log(N) + 2 \int_S \mathfrak{B}_1 d\mu_Q + \mathcal{O}(N^{-1}) \end{aligned}$$

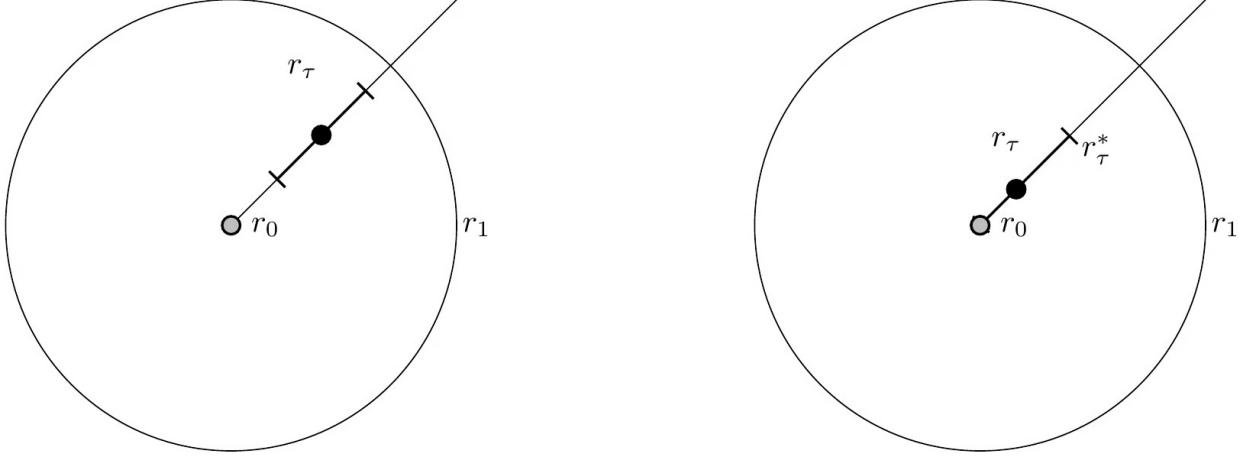
□

where we have set

$$F_q[A] = \frac{1}{2} \log(2\pi) + \frac{1}{2} \log \left(\frac{V^{(2)}(r_1)}{V^{(2)}(r_0)} \right) - \log \left(\frac{r_1}{r_0} \right)$$

7.2.2 Asymptotics - Disk

In this case, we consider $r_0 = 0$.



Asymptotic for the orthogonal polynomial

In the asymptotic expansions of h_j , one should distinguish the following two cases depending on a small constant $\epsilon > 0$.

- Case 1: $r_{j/N} \gg N^{-\epsilon}$. For the annular droplet case where $r_0 > 0$, this case covers all $j = 0, 1, \dots, N-1$. On the other hand, for the disc droplet case where $r_0 = 0$, this case covers only $j = m_N, m_N + 1, \dots, N-1$ for $m_N = N^\epsilon$ with some $\epsilon > 0$;
- $r_{j/N} \ll N^{-\epsilon}$. This covers the remaining disc droplet case with $j = 0, 1, \dots, m_N - 1$. Notably, the asymptotic expansion involves gamma functions in this case.

We separate this in two proofs, the first one for $j = m_N, m_N + 1, \dots, N-1$ is very similar to our previous proof. To actually expand on the details, we need some useful facts. **First, recall that r_τ satisfies $r_\tau = \mathcal{O}(\tau^{\frac{1}{2}})$ as $\tau \rightarrow 0$.** Note also that for $d \geq 1$, we can choose C_1 a constant uniformly for all τ and write

$$|V_\tau^{(d)}(r_\tau)| = |q^{(d)}(r_\tau) + (-1)^d 2\tau(d-1)! r_\tau^{-d}| \leq C_1(1 + \tau^{-\frac{d}{2}+1}).$$

We then start as usual dividing the integral into two parts

$$\begin{aligned} h_j &= \int_0^\infty 2r^{2j+1} e^{-Nq(r)} dr = \int_0^\infty 2r e^{-NV_{\tau(j)}(r)} dr \\ &= \int_{r_{\tau(j)} - \delta_N}^{r_{\tau(j)} + \delta_N} 2r e^{-NV_{\tau(j)}(r)} dr + \int_{|r - r_{\tau(j)}| > \delta_N} 2r e^{-NV_{\tau(j)}(r)} dr \end{aligned}$$

and now we have two integrals to solve. Let's start with the outer integral, which we should be able to control the growth. For $m_N \leq j < N$, we have

$$r_{\tau(j+1/2)} - r_{\tau(j)} = \mathcal{O}((jN)^{-\frac{1}{2}}).$$

Since the potential is increasing in (r_τ, ∞) , the Taylor series expansion for V_τ is for all $r > r_{\tau(j)} + \delta_N$

$$\begin{aligned} V_{\tau(j+1/2)}(r) &\geq V_{\tau(j+1/2)}(r_{\tau(j+1/2)}) \\ &= V_{\tau(j+1/2)}(r_{\tau(j+1/2)}) + \frac{1}{2} V_{\tau(j+1/2)}^{(2)}(r_{\tau(j+1/2)})(r_{\tau(j+1/2)} - r_{\tau(j)})^2 + \mathcal{O}(\tau(j)^{-1/2} \delta_N^3). \end{aligned}$$

where $\mathcal{O}(\tau(j)^{-1/2} \delta_N^3) = \mathcal{O}(N^{-1} j^{-1/2} (\log(N))^3)$. Similarly, since V_τ is decreasing in $(0, r_\tau)$, we have for all $r < r_{\tau(j)} - \delta_N$

$$\begin{aligned} V_{\tau(j+1/2)}(r) &\geq V_{\tau(j+1/2)}(r_{\tau(j+1/2)}) \\ &= V_{\tau(j+1/2)}(r_{\tau(j+1/2)}) + \frac{1}{2} V_{\tau(j+1/2)}^{(2)}(r_{\tau(j+1/2)})(r_{\tau(j+1/2)} - r_{\tau(j)})^2 + \mathcal{O}(\tau(j)^{-1/2} \delta_N^3). \end{aligned}$$

Using the Taylor series for V_τ again, we have

$$\begin{aligned} V_{\tau(j+1/2)}(r_{\tau(j+1/2)}) - V_{\tau(j)}(r_{\tau(j)}) &= V_{\tau(j)}(r_{\tau(j+1/2)}) - V_{\tau(j)}(r_{\tau(j)}) - \frac{1}{N} \log r_{\tau(j+1/2)} \\ &= \mathcal{O}((jN)^{-1}) - \frac{1}{N} \log r_{\tau(j+1/2)} \end{aligned}$$

Thus, it follows from the last two inequalities that

$$\begin{aligned} e^{NV_{\tau(j)}(r_{\tau(j)})} \int_{|r - r_{\tau(j)}| > \delta_N} 2r^{2j+1} e^{-Nq(r)} dr &= \int_{|r - r_{\tau(j)}| > \delta_N} 2e^{-N(V_{\tau(j+1/2)}(r) - V_{\tau(j)}(r_{\tau(j)}))} dr \\ &\leq e^{-c_1 (\log(N))^2} e^{\frac{N-M}{N} \log(r_{\tau(j+1/2)})} \int_{|r - r_{\tau(j)}| > \delta_N} 2e^{-M(V_{\tau(j+1/2)}(r) - V_{\tau(j)}(r_{\tau(j)}))} dr = \mathcal{O}(e^{c_2}) \end{aligned}$$

for some constants $c_1, c_2 > 0$. We now need to examine the integral near the critical point. That is,

$$\begin{aligned} &\int_{r_{\tau(j)} - \delta_N}^{r_{\tau(j)} + \delta_N} 2r e^{-NV_{\tau(j)}(r)} dr \\ &= e^{-NV_{\tau(j)}(r_{\tau(j)})} \int_{-\delta_N}^{\delta_N} 2(r_{\tau(j)} + t) e^{-N\left(\frac{1}{2} V_{\tau(j)}^{(2)}(r_{\tau(j)})(t)^2 + \frac{1}{3!} V_{\tau(j)}^{(3)}(r_{\tau(j)})(t)^3 + \frac{1}{4!} V_{\tau(j)}^{(4)}(r_{\tau(j)})(t)^4 + \mathcal{O}(\tau(j)^{-\frac{3}{2}} t^5)\right)} dt \\ &= e^{-NV_{\tau(j)}(r_{\tau(j)})} \int_{-\sqrt{N}\delta_N}^{\sqrt{N}\delta_N} 2e^{-2\Delta Q(r_{\tau(j)})t^2} \left(r_{\tau(j)} + \frac{t}{\sqrt{N}}\right) \left(1 - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})(t)^3}{6\sqrt{N}} - \frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})(t)^4}{24N} + \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2(t)^6}{72N} + \epsilon'_{N,1}\right) dt \end{aligned}$$

where $\epsilon_{N,1} = \mathcal{O}(j^{-\frac{3}{2}} (\log N)^\alpha)$ for some $\alpha > 0$.

Now, for $j \leq m_N$, we need to define some $r_\tau^* := r_\tau \log(N)$ from which we separate the integral in an important part close to the origin and another part which has a lesser order. Consider

$$h_j = \int_0^\infty 2r^{2j+1} e^{-Nq(r)} dr = \int_0^{r_\tau^*} 2r^{2j+1} e^{-Nq(r)} dr + \int_{r_\tau^*}^\infty 2r^{2j+1} e^{-Nq(r)} dr$$

Due to strict subharmonicity of Q in a neighborhood of the droplet, r_τ satisfies $r_\tau = \mathcal{O}(\tau^{\frac{1}{2}})$ as $\tau \rightarrow 0$. Since V_τ has a global minimum at $r = r_\tau$ and increases for larger values, for all $r > r_{\tau(j+1/2)}^*$ we have

$$\begin{aligned} V_{\tau(j+1/2)}(r) &\geq V_{\tau(j+1/2)}(r)(r_{\tau(j+1/2)}^*) = q(r_{\tau(j+1/2)}^*) - \frac{2j+1}{N} \log(r_{\tau(j+1/2)}^*) \\ &= q(0) + \frac{1}{2}q^{(2)}(0)(r_{\tau(j+1/2)}^*)^2 - \frac{2j+1}{N} \log(r_{\tau(j+1/2)}^*) + \mathcal{O}(r_{\tau(j+1/2)}^*)^3. \end{aligned}$$

Remember that $q'(0) = 0$ and $\Delta Q(z) \in (0, \infty)$ near the origin. Using this inequality,

$$\begin{aligned} e^{Nq(0)} \int_{r_{\tau(j+\frac{1}{2})}^*}^{\infty} 2r^{2j+1} e^{-Nq(r)} dr &= \int_{r_{\tau(j+\frac{1}{2})}^*}^{\infty} 2e^{-N(V_{\tau(j+1/2)}(r)q(0))} dr \\ &\leq e^{-c_1(N-M)(r_{\tau(j+\frac{1}{2})}^*)^2} (r_{\tau(j+\frac{1}{2})}^*)^{(2j+1)\frac{N-M}{N}} \int_{r_{\tau(j+\frac{1}{2})}^*}^{\infty} 2e^{-M(V_{\tau(j+1/2)}(r)q(0))} dr := \epsilon_N(j) \end{aligned}$$

for some $c_1 > 0$ and M as given in the other proof, such that the appropriate integral converges. Since $r_\tau = \mathcal{O}(\tau^{\frac{1}{2}})$, there is a constant $c_2 > 0$ such that

$$\epsilon_N(j) = \mathcal{O}\left((r_{\tau(j+\frac{1}{2})}^*)^{(2j+1)\frac{N-M}{N}} e^{-c_2(j+\frac{1}{2})(\log N)^2}\right) = \mathcal{O}\left(\left(\frac{2j+1}{2N}(\log(N))^2\right)^{(2j+1)\frac{N-M}{N}} e^{-c_2(j+\frac{1}{2})(\log N)^2}\right)$$

Therefore we obtain

$$\begin{aligned} h_j &= \int_0^{\infty} 2r^{2j+1} e^{-Nq(r)} dr = \int_0^{r_{\tau(j+\frac{1}{2})}^*} 2r^{2j+1} e^{-Nq(r)} dr + e^{-Nq(0)} \epsilon_N(j) \\ &= \int_0^{r_{\tau(j+\frac{1}{2})}^*} 2r^{2j+1} e^{-N(q(0) + \frac{1}{2}q^{(2)}(0)r^2) + \mathcal{O}(N(r_{\tau(j+\frac{1}{2})}^*)^3)} dr + e^{-Nq(0)} \epsilon_N(j) \\ &= e^{-Nq(0)} \left[N - (j+1) \int_0^{r_{\tau(j+\frac{1}{2})}^*} 2r^{2j+1} e^{-\frac{1}{2}q^{(2)}(0)r^2} dr \left(1 + \mathcal{O}(N(r_{\tau(j+\frac{1}{2})}^*)^3)\right) + \epsilon_N(j) \right] \\ &= e^{-Nq(0)} \left[N - (j+1)\Gamma(j+1)\left(\frac{1}{2}q^{(2)}(0)\right)^{-(j+1)} \left(1 + \mathcal{O}(N^{-\frac{1}{2}}(j+\frac{1}{2})^{\frac{3}{2}}(\log N)^3)\right) + \epsilon_N(j) \right] \end{aligned}$$

and this completes this part of the proof.

Asymptotics on the Summation

Now we complete the case calculating, using the results above,

$$\sum_{j=0}^N \log(h_j) = \sum_{j=0}^{m_N-1} \log(h_j) + \sum_{j=m_N}^{N-1} \log(h_j)$$

The first term in the right is easily determined, we need only to write

$$\begin{aligned} \sum_{j=0}^{m_N-1} \log(h_j) &= -m_N Nq(0) - \frac{m_N(m_N+1)}{2} \left(\log(N) + \log\left(\frac{1}{2}q^{(2)}(0)\right) \right) + \log(G(m_N+1)) + \mathcal{O}(N^{-\frac{1}{2}(1-5\epsilon)}(\log(N))^3) \\ &= -m_N Nq(0) - \frac{m_N(m_N+1)}{2} \left(\log(N) + \log\left(\frac{1}{2}q^{(2)}(0)\right) \right) + \frac{m_N^2 \log(m_N)}{2} - \frac{3}{4}m_N^2 + \frac{\log(2\pi)m_N}{2} - \frac{\log(m_N)}{12} + \gamma' \end{aligned}$$

where we have used the asymptotic for the G-Barnes function

$$\log(G(z+1)) = \frac{z^2 \log(z)}{2} - \frac{3}{4}z^2 + \frac{\log(2\pi)z}{2} - \frac{\log(z)}{12} + \gamma'(-1) + \mathcal{O}\left(\frac{1}{z^2}\right)$$

The second summation is a bit more detailed, we use the previously defined

$$\log(h_j) = -NV_{\tau(j)}(r_{\tau(j)}) + \frac{1}{2} \log \left(\frac{2\pi r_{\tau(j)}^2}{N\Delta Q(r_{\tau(j)})} \right) + \frac{1}{N} \mathfrak{B}_1(r_{\tau(j)}) + \mathcal{O}(j^{-\frac{3}{2}}(\log(N))^\alpha)$$

and need to determine three things in order to get the summation,

$$\begin{aligned} & \sum_{j=m_N}^{N-1} V_{\tau(j)}(r_{\tau(j)}), \\ & \sum_{j=m_N}^{N-1} \log \Delta Q(r_{\tau(j)}), \\ & \sum_{j=m_N}^{N-1} \log(r_{\tau(j)}). \end{aligned}$$

For the first we proceed as before and perform the Euler-Maclaurin formula.

$$\sum_{j=m_N}^{N-1} V_{\tau(j)}(r_{\tau(j)}) = \int_{m_N}^N V_{\tau(j)}(r_{\tau(j)}) dt - \frac{1}{2} (V_{\tau(N)}(r_{\tau(N)}) - V_{\tau(m_N)}(r_{\tau(m_N)})) + \frac{1}{12} [\partial_t V_{\tau(j)}(r_{\tau(j)})|_{t=N} - \partial_t V_{\tau(j)}(r_{\tau(j)})|_{t=m_N}] + \mathcal{O}(N^{-3})$$

where we used $\partial_t^3(V_{\tau(t)}(r_{\tau(t)}))|_{t=m_N} = \mathcal{O}(N^{-3}(\tau(m_N))^{-2}) = \mathcal{O}(m_N^{-2}N^{-1})$. The first term can be solved in a very similar way as before.

$$\begin{aligned} \int_{m_N}^N V_{\tau(j)}(r_{\tau(j)}) &= 2N \int_{r_{\tau(m_N)}}^{r_1} (q(s) - sq'(s) \log s) s \Delta Q(s) ds \\ &= N \left(\frac{1}{2} \int_{S \setminus S_{\tau(m_N)}} Q \Delta Q dA - \log(r_1) + (\tau(m_N))^2 \log(r_{\tau(m_N)}) + \frac{1}{2} (q(r_1) - \tau(m_N)q(r_{\tau(m_N)})) \right) \\ &= NI_Q[\mu_Q] - \frac{N}{2} \int_{S_{\tau(m_N)}} Q \Delta Q dA + \frac{m_N^2}{N} \log(r_{\tau(m_N)}) - \frac{m_N}{2} q(r_{\tau(m_N)}) \end{aligned}$$

where, **because**

$$r_\tau = \left(\frac{2\tau}{q^{(2)}(0)} \right)^{\frac{1}{2}} + \mathcal{O}(\tau) = \left(\frac{\tau}{\Delta Q(0)} \right)^{\frac{1}{2}} + \mathcal{O}(\tau), \quad \text{as } \tau \rightarrow 0$$

we have that

$$\log(r_{\tau(m_N)}) = \frac{1}{2} \log \left(\frac{\tau(m_N)}{\Delta Q(0)} \right) + \mathcal{O}(\tau(m_N)^{\frac{1}{2}}) = \frac{1}{2} \log \left(\frac{\tau(m_N)}{\Delta Q(0)} \right) + \mathcal{O}(N^{-\frac{1}{2}(1-\epsilon)})$$

and

$$q(r_{\tau(m_N)}) = q(0) + \frac{1}{2} q^{(2)}(0) (r_{\tau(m_N)})^2 + \mathcal{O}((\tau(m_N))^{\frac{3}{2}}) = q(0) + \tau(m_N) + \mathcal{O}(N^{-\frac{3}{2}(1-\epsilon)})$$

7.3 Examples

Chapter 8

External Field Model

Some random matrix ensembles are in terms of multiple orthogonal polynomials. Consider now that we couple a deterministic source matrix $\hat{A}$ to the random matrix $\hat{M}$, in a way that the weight

$$d\mu_n(M) = \frac{1}{Z_A} \exp \left\{ -n \left(\text{Tr} \left(V(\hat{M}) - \hat{A}\hat{M} \right) \right) \right\} dM.$$

describes the model distribution. We take $\hat{A}$ to be an Hermitian matrix and dM to be the entry-wise Lebesgue measure. The eigenvalues of $\hat{M}$ represent the energy levels of a system without time-reversal invariance. When the external source $\hat{A}$ is nonzero and $V(\hat{M}) = \hat{M}^2/2$, this measure arises in the study of Hamiltonian that can be written as the sum of a random matrix and a deterministic source matrix. In this definition,

$$Z_A = \int \exp \left\{ -n \left(\text{Tr} \left(V(\hat{M}) - \hat{A}\hat{M} \right) \right) \right\} dM$$

Without any loss of generality, we can assume $\hat{A}$ to be diagonal.

Of course, when $\hat{A} = 0$, that is, no external source, and $V(\hat{M}) = \hat{M}^2/2$, this measure describes the Gaussian Unitary Ensemble - GUE. In this case, we can say that the average characteristic polynomial is

$$\mathbb{E} \left[x \hat{I}_n - \hat{M}_n \right] := \text{He}_{\bar{n}}(x)$$

where $\text{He}_{\bar{n}}(x)$ is a (type II) multiple Hermite polynomial.

For reasonable choices of $V(z)$ the spectrum tends to accumulate on closed intervals of the real axis. Introducing an external source $\hat{A}$ can have the effect of perturbing the expected position of the spectrum. Some study has been done on the case for the matrix $\hat{A}$ with two eigenvalues $\pm a$, each of multiplicity $n/2$. When a is sufficiently small, the eigenvalues of $\hat{M}$ accumulate with probability one on a single interval, just as when $\hat{A} = 0$. As a increases the interval splits into two disconnected intervals. There is a transitional value of a between these two cases where the local eigenvalue density is described by the so-called *Pearcey process*. This is precisely the case which we will explore asymptotically for. Define the monic polynomial

$$P_n(z) = \int \det(z - \hat{M}) d\mu_n(M)$$

then, we have the following proposition

Proposition 8.0.1. Suppose $\hat{A}$ has p distinct eigenvalues a_i with respective multiplicity n_i , so that $n_1 + \dots + n_p = n$. Let $n^{(i)} = n_1 + \dots + n_i$ and $n^{(0)} = 0$. Define

$$w_j(x) = x^{d_j-1} e^{-(V(x)-a_j x)}, \quad j = 1, \dots, n.$$

where $i = i_j$ is such that $n^{(i-1)} < j \leq n^{(i)}$ and $d_j = j - n^{(i-1)}$. Then the following hold

- There is a constant $\tilde{Z}_n$, with $d\lambda = d\lambda_1 \dots d\lambda_n$, such that

$$P_n(z) = \frac{1}{\tilde{Z}} \int \prod_{j=1}^n (z - \lambda_j) \prod_{j=1}^n w_j(\lambda_j) \prod_{i>j} (\lambda_i - \lambda_j) d\lambda$$

- Let

$$m_{j,k} = \int_{-\infty}^{\infty} x^k w_j(x) dx$$

Then we have the determinantal formula

$$P_n(z) = \frac{1}{\tilde{Z}_n} = \begin{vmatrix} m_{1,0} & m_{1,1} & \dots & m_{1,n} \\ \vdots & \vdots & \ddots & \vdots \\ m_{n,0} & m_{n,1} & \dots & m_{n,n} \\ 1 & z & \dots & z^n \end{vmatrix}$$

- For $j = 1, \dots, p$,

$$\int_{-\infty}^{\infty} P_n(x) x^j e^{V(x)-a_j x} dx = 0, \quad j = 0, \dots, n_i - 1$$

and these equations uniquely determine $P_n(x)$.

The relations can be viewed as multiple orthogonality conditions for the polynomial P_n . There are p weights $e^{-(V(x)-a_j x)}$, $j = 1, \dots, p$, and for each weight there are a number of orthogonality conditions, so that the total number of them is n . This point of view is especially useful in the case when $\hat{A}$ has only a small number of distinct eigenvalues. Let $\sum_n$ be the collection of functions

$$\sum_n := \{x_j e^{a_i x} \mid i = 1, \dots, p, j = 0, \dots, n_i - 1\}$$

We start with a lemma.

Lema 8.0.2. There exists a unique function Q_{n-1} in the linear span of $\sum_n$ such that

$$\int_{-\infty}^{\infty} x^j Q_{n-1}(x) e^{-V(x)} dx = 0, \quad j = 0, \dots, n-2,$$

and

$$\int_{-\infty}^{\infty} x^{n-1} Q_{n-1}(x) e^{-V(x)} dx = 1.$$

Consider $P_0, \dots, P_n$, a sequence of multiple orthogonal polynomials such that $\deg P_k = k$, with an increasing sequence of the multiplicity vectors, so that $k_i \leq l_i$, $i = 1, \dots, p$, when $k \leq l$. Consider the biorthogonal system of functions, $\{Q_k(x), k = 0, \dots, n-1\}$,

$$\int_{-\infty}^{\infty} P_j(x) Q_k(x) e^{-V(x)} dx = \delta_{jk}, \quad \text{for } j, k = 0, \dots, n-1.$$

Define the kernel

$$K_n(x, y) = e^{-\frac{1}{2}(V(x)+V(y))} \sum_{k=0}^{n-1} P_k(x) Q_k(y).$$

This kernel, as in the simplest case, will enable us to determine asymptotic on the equilibrium measure.

Chapter 9

Large Deviation Theory

Chapter 10

Miscellaneous

10.1 Laplace Transform

10.2 Fourier Transform

We start by defining what we call a Fourier Series. Take $f(x)$ to be a a -periodic function, i.e., $f(x+a) = f(x)$, $\forall x \in \mathbb{R}$. How to characterize such a function? We may try to specify the values it takes on the interval but that's uncountable big. We can also try to approximate it somehow, one such approach is to use geometric periodic functions. We write:

$$f(x) = \sum_{n=1}^{\infty} \alpha_n \sin\left(\frac{2\pi nx}{a}\right) + \sum_{m=1}^{\infty} \beta_m \cos\left(\frac{2\pi mx}{a}\right) = \sum_{-\infty}^{\infty} f_n e^{2\pi i n \frac{x}{a}}$$

Note that such a summation must be a -periodic as all functions used are a -periodic. Of course, not all functions are guarantee to be expressed in such a way. A know class that works, and usually enough for our purposes, are square-integrable functions, for which

$$\int_{-a/2}^{a/2} |f(x)|^2 dx$$

exists and is finite. The concept we have is useful but limited if used as such. We want to extend such a concept for functions that are not a -periodic - but how? Define a family of functions $\{f_a(x) | a \in \mathbb{R}^+\}$ such that $f_a(x)$ is a -periodic and $\lim_{a \rightarrow \infty} f_a(x) = f(x)$. We have that every function f_a in the family can be written as a Fourier Series as it is periodic, so we write:

$$f_a(x) = \sum_{-\infty}^{\infty} f_{an} e^{ik_n x} : k_n = n\Delta k; \Delta k = \frac{2\pi}{a}.$$

And as we know about the limit, we can write

$$\begin{aligned} f(x) &= \lim_{a \rightarrow \infty} \left[\sum_{-\infty}^{\infty} f_{an} e^{ik_n x} \right] = \lim_{a \rightarrow \infty} \left[\sum_{-\infty}^{\infty} \frac{\Delta k}{2\pi} e^{ik_n x} \left(\frac{2\pi f_{an}}{\Delta k} \right) \right] \\ &= \int_{-\infty}^{\infty} \frac{dk}{2\pi} e^{ikx} F(k); \text{ where } F(k) = \lim_{a \rightarrow \infty} \left[\frac{2\pi f_{an}}{\Delta k} \right]. \end{aligned}$$

where we call $F(x)$ the Fourier Tranform of the function $f(x)$. We can also define the inverse transform

$$\begin{aligned} F(k) &= \lim_{a \rightarrow \infty} \left[\frac{2\pi}{\Delta k} f_{an}(x) \right] \\ &= \lim_{a \rightarrow \infty} \left[\frac{2\pi}{\frac{2\pi}{a}} \frac{1}{a} \int_{-a/2}^{a/2} dx e^{-ik_n x} \right] \\ &= \int_{-\infty}^{\infty} dx e^{-ikx} f(x) \end{aligned}$$

With this definition we can make explicit some properties of the transform, namely:

1. Linearity: Given $f(x)$ and $g(x)$, with its respective $F(k)$ and $G(k)$ Fourier Transforms we have that

$$af(x) + bg(x) \rightarrow_{FT} aF(k) + bG(k);$$

2. Translation: Given $f(x)$ translated by b we write

$$f(x + b) \rightarrow_{FT} e^{ikb} F(k);$$

3. Derivative: Given $f(x)$ we have that the derivative is written

$$\frac{d}{dx} f(x) \rightarrow_{FT} ikF(k)$$

10.2.1 On Differential Equations

Take the damped harmonic oscillator subjected to an additional force $f(t)$. The equation of motion is given by

$$\frac{d^2 x(t)}{dt^2} + 2\gamma \frac{dx(t)}{dt} + \omega_0^2 x(t) = \frac{f(t)}{m}.$$

To solve for $x(t)$ we first take the Fourier Transform

$$-\omega^2 X(\omega) - 2i\gamma\omega X(\omega) + \omega_0^2 X(\omega) = \frac{F(\omega)}{m}$$

where now we can reorganize to solve for $X(\omega)$.

$$X(\omega) = \frac{\frac{F(\omega)}{m}}{-\omega^2 - 2i\gamma\omega + \omega_0^2}.$$

Now it is a simple case of taking the inverse of the Fourier Transform

$$x(t) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \frac{e^{-i\omega t} F(\omega)}{-\omega^2 - 2i\gamma\omega + \omega_0^2}; \quad \text{where } F(\omega) = \int_{-\infty}^{\infty} dt e^{i\omega t} f(t)$$

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